

Technology Brief 7: Resistive Sensors

An **electrical sensor** is a device capable of responding to an applied **stimulus** by generating an electrical signal whose voltage, current, or some other attribute is related to the intensity of the stimulus.

► The family of possible stimuli encompasses a wide array of physical, chemical, and biological quantities, including temperature, pressure, position, distance, motion, velocity, acceleration, concentration (of a gas or liquid), blood flow, etc. ◀

The sensing process relies on measuring resistance, capacitance, inductance, induced electromotive force (emf), oscillation frequency or time delay, among others. Sensors are integral to the operation of just about every instrument that uses electronic systems, from automobiles and airplanes to computers and cell phones (Fig. TF7-1). This Technology Brief covers resistive sensors. **Capacitive**, **inductive**, and **emf sensors** are covered separately (here and in later chapters).

Piezoresistivity

According to Eq. (4.70), the resistance of a cylindrical resistor or wire conductor is given by $R = l / \sigma A$, where l is the cylinder's length, A is its cross-sectional area, and σ is the conductivity of its material. Stretching the wire by an applied external force causes l to increase and A to decrease. Consequently, R increases (Fig. TF7-2). Conversely, compressing the wire causes R to decrease. The Greek word **piezein** means to press, from which the term piezoresistivity is derived. *This should not be confused with piezoelectricity, which is an emf effect. (See EMF Sensors in Technology Brief 12.)*

The relationship between the resistance R of a piezoresistor and the **applied force** F can be modeled by the approximate linear equation

$$R = R_0 \left(1 + \frac{\alpha F}{A_0} \right),$$

where R_0 is the unstressed resistance (@ $F = 0$), A_0 is the unstressed cross-sectional area of the resistor, and α is the **piezoresistive coefficient** of the resistor material. The force F is positive if it is causing the resistor to stretch and negative if it is compressing it.

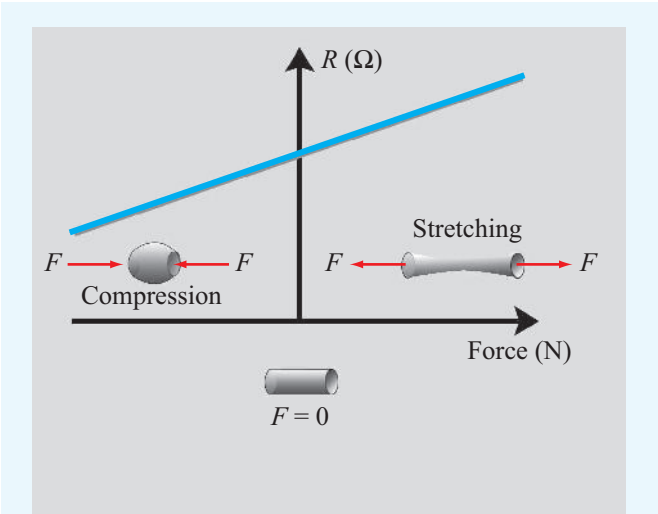
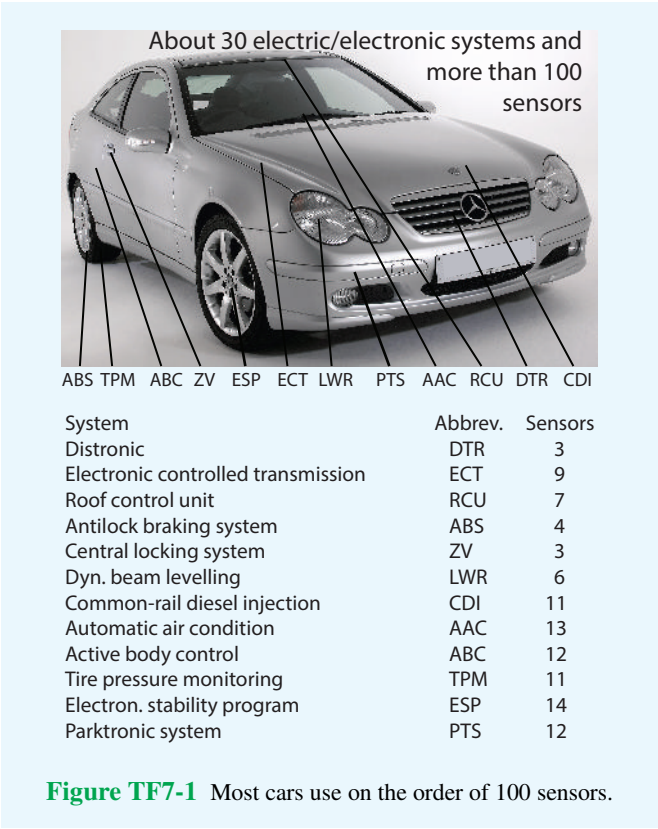


Figure TF7-2 Piezoresistance varies with applied force.

技术简介 7：电阻传感器

电传感器是一种能够通过生成电信号来响应施加的刺激的设备，该电信号的电压、电流或一些其他属性与刺激的强度相关。

► 可能的刺激家族涵盖了广泛的物理、化学和生物量，包括温度、压力、位置、距离、运动、速度、加速度、（气体或液体）浓度、血流等。◀

传感过程依赖于测量电阻、电容、电感、感应电动势 (emf)、振荡频率或时间延迟等。从汽车和飞机到计算机和手机，几乎所有使用电子系统的仪器的操作都离不开传感器（图 TF7-1）。本技术简介涵盖电阻传感器。电容式、电感式和电动势传感器分别介绍（此处和后面的章节）。

压阻率

根据方程。根据方程 (4.70)，圆柱形电阻器或导线的电阻由 $R = l / \sigma A$ 给出，其中 l 是圆柱体的长度， A 是其横截面积， σ 是其材料的电导率。通过施加外力拉伸导线会导致 l 增加而 A 减少。因此， R 增加（图 TF7-2）。相反，压缩导线会导致 R 减小。希腊语“piezein”的意思是“按压”，“压阻率”一词就是由此而来。这不应与压电相混淆，压电是一种电动势效应。（请参阅技术简介 12 中的 EMF 传感器。）

压敏电阻的电阻 R 与施加的力 F 之间的关系可以通过近似线性方程来建模

$$R = R_0 \left(1 + \frac{\alpha F}{A_0} \right),$$

其中 R_0 是无应力电阻 (@ $F = 0$)， A_0 是电阻器的无应力横截面积， α 是电阻器材料的压阻系数。如果力 F 导致电阻器拉伸，则力 F 为正；如果力 F 压缩电阻器，则力 F 为负。

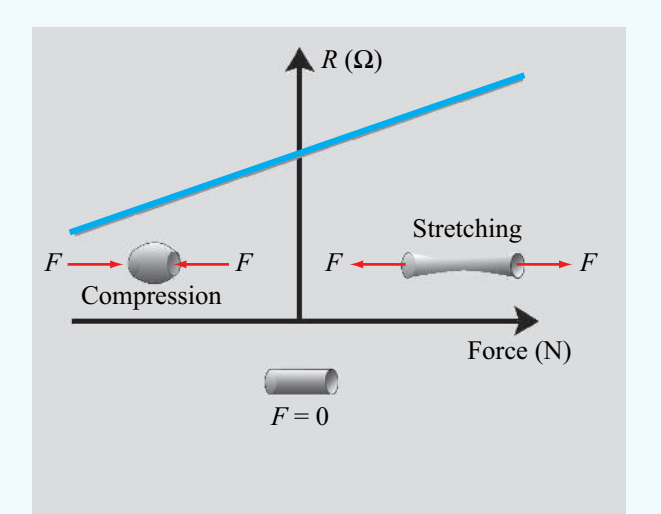
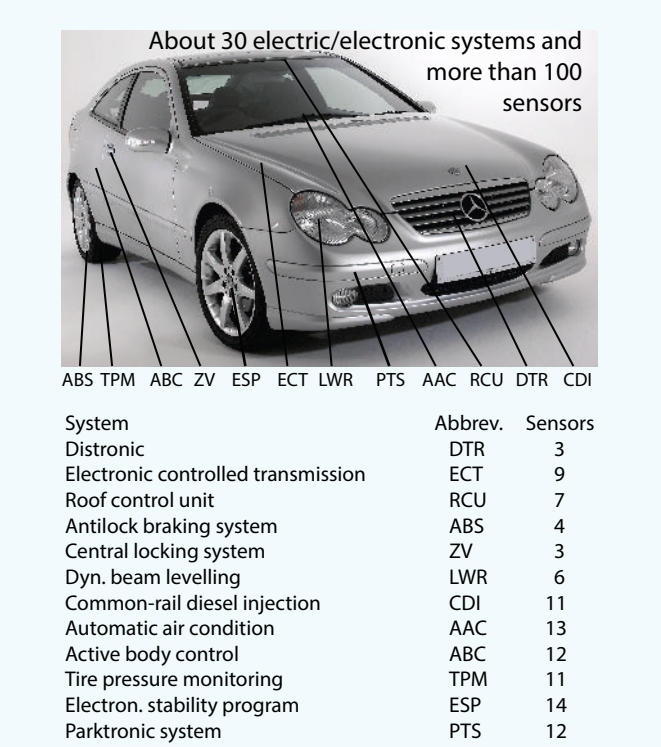
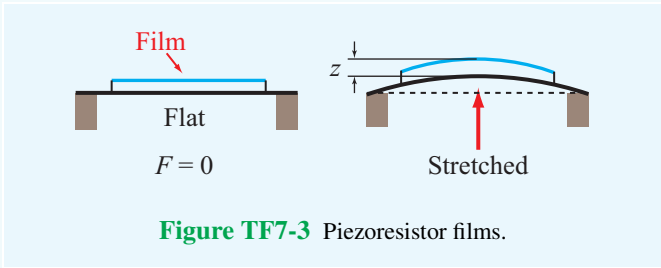


图 TF7-2 压阻随施加的力而变化。



► By connecting the piezoresistor to a Wheatstone bridge circuit (Fig. TF7-5) where the other three resistors are all identical in value and equal to R_0 (the resistance of the piezoresistor when no external force is present), the voltage output becomes directly proportional to the normalized resistance change: $\Delta R/R_0$.

An elastic resistive sensor is well suited for measuring the deformation z of a surface (Fig. TF7-3), which can be related to the pressure applied to the surface; and if z is recorded as a function of time, it is possible to derive the velocity and acceleration of the surface's motion. To realize high longitudinal piezoresistive sensitivity (the ratio of the normalized change in resistance, $\Delta R/R_0$, to the corresponding change in length, $\Delta l/l_0$, caused by the applied force), the piezoresistor is often designed as a serpentine-shaped wire (Fig. TF7-4(a)) bonded on a flexible plastic substrate and glued onto the surface whose deformation is to be monitored. Copper and nickel alloys are commonly used for making the sensor wires, although in some applications silicon is used instead (Fig. TF7-4(b)) because it has a very high piezoresistive sensitivity.

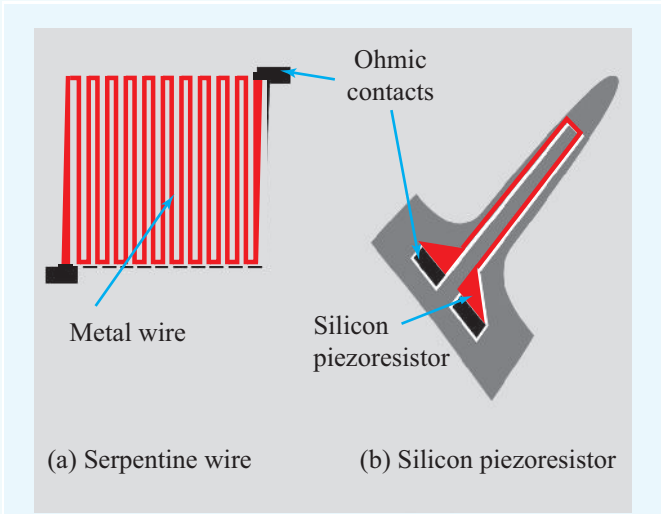


Figure TF7-4 Metal and silicon piezoresistors.

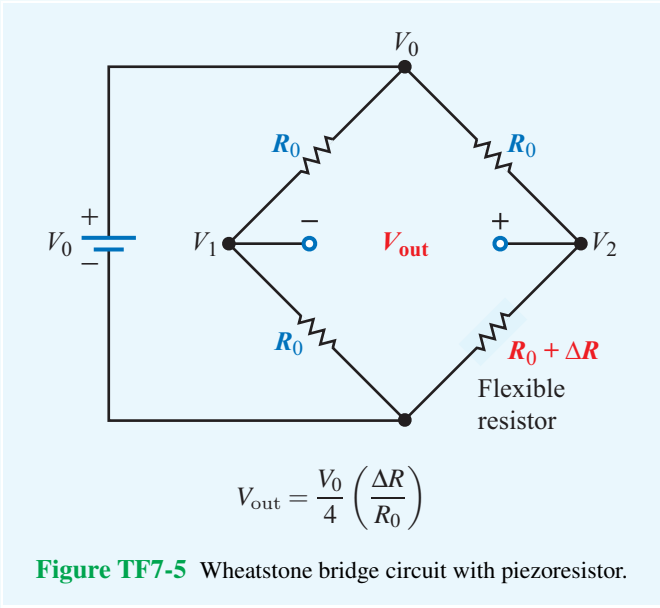


Figure TF7-5 Wheatstone bridge circuit with piezoresistor.

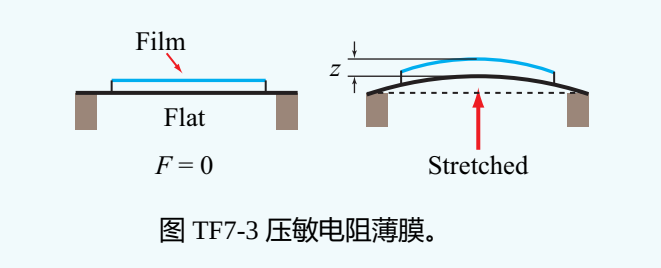


图 TF7-3 压敏电阻薄膜。

弹性电阻传感器非常适合测量表面的变形 z (图 TF7-3)，这可能与施加到表面的压力有关；如果将 z 记录为时间的函数，则可以推导出表面运动的速度和加速度。为了实现高纵向压阻灵敏度（归一化电阻变化 $\Delta R/R_0$ 与由施加的力引起的相应长度变化 $\Delta l/l_0$ 的比率），压敏电阻通常设计为蛇形线 (图 TF7-4(a))，粘合在柔性塑料基板上，并粘合到要监测变形的表面上。铜和镍合金通常用于制造传感器线，尽管在某些应用中使用硅代替 (图 TF7-4 (b))，因为它具有非常高的压阻灵敏度。

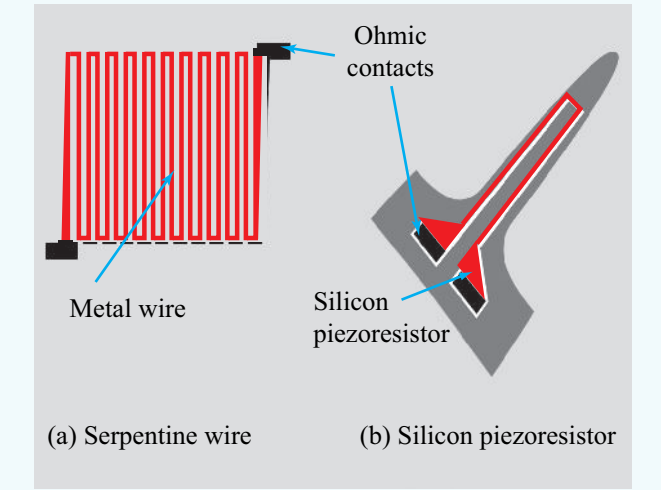


图 TF7-4 金属和硅压敏电阻。

► 通过将压敏电阻连接到惠斯通电桥电路 (图 TF7-5)，其中其他三个电阻的阻值都相同，并且等于 R_0 (无外力时压敏电阻的电阻)，电压输出与归一化电阻变化成正比: $\Delta R/R_0$ 。

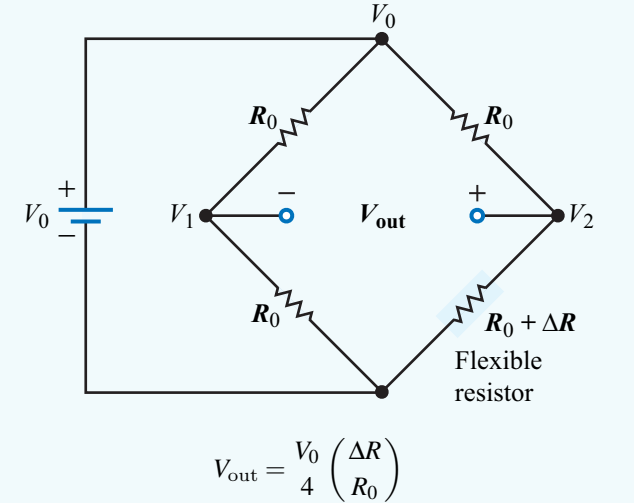


图 TF7-5 带压敏电阻的惠斯登电桥电路。

4-5.2 Electric Potential Due to Point Charges

The electric field due to a point charge q located at the origin is given by Eq. (4.32) as

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \quad (\text{V/m}). \quad (4.54)$$

The field is radially directed and decays quadratically with the distance R from the observer to the charge.

As was stated earlier, the choice of integration path between the end points in Eq. (4.43) is arbitrary. Hence, we can conveniently choose the path to be along the radial direction $\hat{\mathbf{R}}$, in which case $d\mathbf{l} = \hat{\mathbf{R}} dR$ and

$$V = - \int_{\infty}^R \left(\hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \right) \cdot \hat{\mathbf{R}} dR = \frac{q}{4\pi\epsilon R} \quad (\text{V}). \quad (4.55)$$

If the charge q is at a location other than the origin, say at position vector $\mathbf{R}_1$, then V at observation position vector $\mathbf{R}$ becomes

$$V = \frac{q}{4\pi\epsilon |\mathbf{R} - \mathbf{R}_1|} \quad (\text{V}), \quad (4.56)$$

where $|\mathbf{R} - \mathbf{R}_1|$ is the distance between the observation point and the location of the charge q . The principle of superposition applied previously to the electric field $\mathbf{E}$ also applies to the electric potential V . Hence, for N discrete point charges $q_1, q_2, \dots, q_N$ residing at position vectors $\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_N$, the electric potential is

$$V = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i}{|\mathbf{R} - \mathbf{R}_i|} \quad (\text{V}). \quad (4.57)$$

4-5.3 Electric Potential Due to Continuous Distributions

To obtain expressions for the electric potential V due to continuous charge distributions over a volume $\mathcal{V}'$, across a surface S' , or along a line l' , we (1) replace q_i in Eq. (4.57) with $\rho_v d\mathcal{V}'$, $\rho_s ds'$, and $\rho_\ell dl'$, respectively; (2) convert the summation into an integration; and (3) define $R' = |\mathbf{R} - \mathbf{R}_i|$ as the distance between the integration point and the observation point. These steps lead to the following expressions:

$$V = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v}{R'} d\mathcal{V}' \quad \textbf{(volume distribution)}, \quad (4.58a)$$

$$V = \frac{1}{4\pi\epsilon} \int_{S'} \frac{\rho_s}{R'} ds' \quad \textbf{(surface distribution)}, \quad (4.58b)$$

$$V = \frac{1}{4\pi\epsilon} \int_{l'} \frac{\rho_\ell}{R'} dl' \quad \textbf{(line distribution)}. \quad (4.58c)$$

4-5.4 Electric Field as a Function of Electric Potential

In Section 4-5.1, we expressed V in terms of a line integral over $\mathbf{E}$. Now we explore the inverse relationship by reexamining Eq. (4.37):

$$dV = -\mathbf{E} \cdot d\mathbf{l}. \quad (4.59)$$

For a scalar function V , Eq. (3.73) gives

$$dV = \nabla V \cdot d\mathbf{l}, \quad (4.60)$$

where ∇V is the gradient of V . Comparison of Eq. (4.59) with Eq. (4.60) leads to

$$\mathbf{E} = -\nabla V. \quad (4.61)$$

▶ This differential relationship between V and $\mathbf{E}$ allows us to determine $\mathbf{E}$ for any charge distribution by first calculating V and then taking the negative gradient of V to find $\mathbf{E}$. ◀

The expressions for V , given by Eqs. (4.57) to (4.58c), involve scalar sums and scalar integrals, and as such are usually much easier to evaluate than the vector sums and integrals in the expressions for $\mathbf{E}$ derived in Section 4-3 on the basis of Coulomb’s law. Thus, even though the electric potential approach for finding $\mathbf{E}$ is a two-step process, it is conceptually and computationally simpler to apply than the direct method based on Coulomb’s law.

Example 4-9: Electric Field of an Electric Dipole

An *electric dipole* consists of two point charges of equal magnitude but opposite polarity separated by a distance d (**Fig. 4-15(a)**). Determine V and $\mathbf{E}$ at any point P given that P is at a distance $R \gg d$ from the dipole center and the dipole resides in free space.

Solution: To simplify the derivation, we align the dipole along the z axis and center it at the origin (**Fig. 4-15(a)**). For the two charges shown in **Fig. 4-15(a)**, application of Eq. (4.57) gives

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_1} + \frac{-q}{R_2} \right) = \frac{q}{4\pi\epsilon_0} \left(\frac{R_2 - R_1}{R_1 R_2} \right).$$

Since $d \ll R$, the lines labeled R_1 and R_2 in **Fig. 4-15(a)** are approximately parallel to each other, in which case the following approximations apply:

$$R_2 - R_1 \approx d \cos \theta, \quad R_1 R_2 \approx R^2.$$

4-5.2 点电荷产生的电势

由位于原点的点电荷 q 产生的电场由方程 1 给出。 (4.32) 作为

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \quad (\text{V/m}). \quad (4.54)$$

该场呈放射状，并随观察者到电荷的距离 R 呈二次方衰减。

如前所述，方程中端点之间积分路径的选择。 (4.43) 是任意的。因此，我们可以方便地选择沿径向方向 $\hat{\mathbf{R}}$ 的路径，在这种情况下 $d\mathbf{l} = \hat{\mathbf{R}} dR$ 和

$$V = - \int_{\infty}^R \left(\hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \right) \cdot \hat{\mathbf{R}} dR = \frac{q}{4\pi\epsilon R} \quad (\text{V}). \quad (4.55)$$

如果电荷 q 位于原点以外的位置，例如位置矢量 $\mathbf{R}_1$ ，则观测位置矢量 $\mathbf{R}$ 处的 V 变为

$$V = \frac{q}{4\pi\epsilon |\mathbf{R} - \mathbf{R}_1|} \quad (\text{V}), \quad (4.56)$$

其中 $|\mathbf{R} - \mathbf{R}_1|$ 是观察点与电荷 q 位置之间的距离。先前应用于电场 $\mathbf{E}$ 的叠加原理也适用于电势 V 。因此，对于位于位置向量 $\mathbf{R}_1$ 、 $\mathbf{R}_2$ 、 $\dots$ 、 $\mathbf{R}_N$ 处的 N 个离散点电荷 q_1 、 q_2 、 $\dots$ 、 q_N ，电势为

$$V = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i}{|\mathbf{R} - \mathbf{R}_i|} \quad (\text{V}). \quad (4.57)$$

4-5.3 连续分布产生的电势

为了获得由于体积 $\mathcal{V}'$ 、表面 S' 或沿线 l' 的连续电荷分布而产生的电势 V 的表达式，我们 (1) 替换方程 1 中的 q_i 。 (4.57) 分别为 $\rho_v d\mathcal{V}'$ 、 $\rho_s ds'$ 和 $\rho_\ell dl'$ ； (2) 将求和转化为积分； (3) 定义 $R' = |\mathbf{R} - \mathbf{R}_i|$ 为积分点与观测点之间的距离。这些步骤导致以下表达式：

$$V = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v}{R'} d\mathcal{V}' \quad \textbf{(volume distribution)}, \quad (4.58a)$$

$$V = \frac{1}{4\pi\epsilon} \int_{S'} \frac{\rho_s}{R'} ds' \quad \textbf{(surface distribution)}, \quad (4.58b)$$

$$V = \frac{1}{4\pi\epsilon} \int_{l'} \frac{\rho_\ell}{R'} dl' \quad \textbf{(line distribution)}. \quad (4.58c)$$

4-5.4 电场作为电势的函数

在第 4-5.1 节中，我们用 $\mathbf{E}$ 上的线积分来表示 V 。现在我们通过重新检查方程来探索逆关系。 (4.37)：

$$dV = -\mathbf{E} \cdot d\mathbf{l}. \quad (4.59)$$

对于标量函数 V ，方程： (3.73) 给出

$$dV = \nabla V \cdot d\mathbf{l}, \quad (4.60)$$

其中 ∇V 是 V 的梯度。等式的比较(4.59) 与等式。 (4.60) 导致

$$\mathbf{E} = -\nabla V. \quad (4.61)$$

▶ V 和 $\mathbf{E}$ 之间的这种微分关系使我们能够通过首先计算 V ，然后采用 V 的负梯度来找到 $\mathbf{E}$ ，从而确定任何电荷分布的 $\mathbf{E}$ 。 ◀

V 的表达式由等式给出。 (4.57) 到 (4.58c) 涉及标量和和标量积分，因此通常比第 4-3 节中根据库仑定律导出的 $\mathbf{E}$ 表达式中的向量和和积分更容易计算。因此，尽管寻找 $\mathbf{E}$ 的电势方法是一个两步过程，但它在概念上和计算上都比基于库仑定律的直接方法更容易应用。

例 4-9：电偶极子的电场

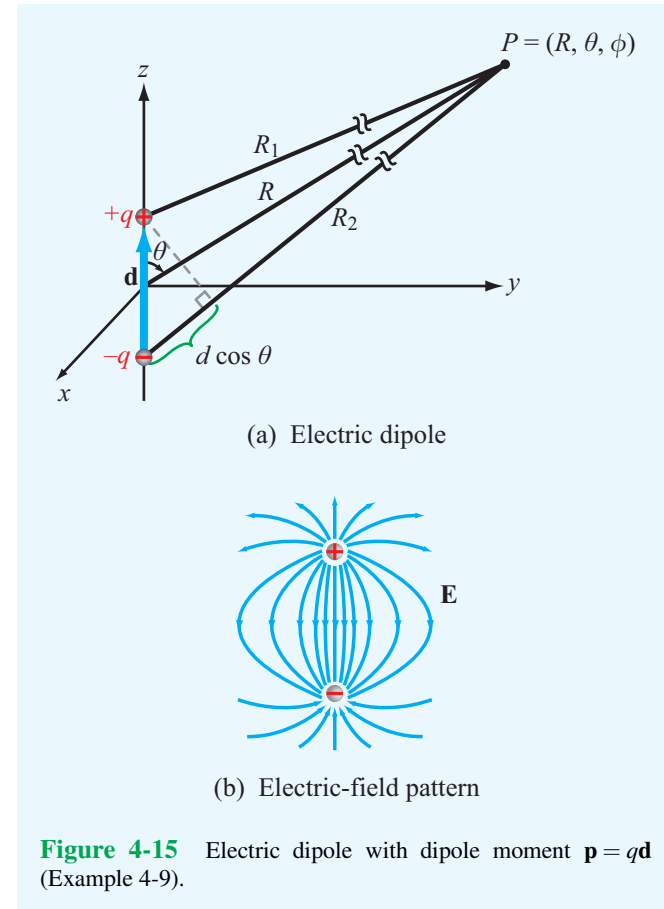
电偶极子由两个大小相等但极性相反、间隔距离 d 的点电荷组成（图 4-15(a)）。假设 P 距偶极子中心的距离为 $R \gg d$ 并且偶极子位于自由空间中，请确定任意点 P 处的 V 和 $\mathbf{E}$ 。

解：为了简化推导，我们将偶极子沿 z 轴对齐，并以原点为中心（图4-15 (a) ）。对于图 4-15(a) 所示的两种电荷，应用等式： (4.57) 给出

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_1} + \frac{-q}{R_2} \right) = \frac{q}{4\pi\epsilon_0} \left(\frac{R_2 - R_1}{R_1 R_2} \right).$$

由于 $d \ll R$ ，图 4-15(a) 中标记为 R_1 和 R_2 的线近似彼此平行，在这种情况下，适用以下近似：

$$R_2 - R_1 \approx d \cos \theta, \quad R_1 R_2 \approx R^2.$$



Hence,

$$V = \frac{qd \cos \theta}{4\pi\epsilon_0 R^2}. \quad (4.62)$$

To generalize this result to an arbitrarily oriented dipole, note that the numerator of Eq. (4.62) can be expressed as the dot product of $q\mathbf{d}$ (where $\mathbf{d}$ is the distance vector from $-q$ to $+q$) and the unit vector $\hat{\mathbf{R}}$ pointing from the center of the dipole toward the observation point P . That is,

$$qd \cos \theta = q\mathbf{d} \cdot \hat{\mathbf{R}} = \mathbf{p} \cdot \hat{\mathbf{R}}, \quad (4.63)$$

where $\mathbf{p} = q\mathbf{d}$ is called the **dipole moment**. Using Eq. (4.63) in Eq. (4.62) then gives

$$V = \frac{\mathbf{p} \cdot \hat{\mathbf{R}}}{4\pi\epsilon_0 R^2} \quad (\text{electric dipole}). \quad (4.64)$$

In spherical coordinates, Eq. (4.61) is given by

$$\mathbf{E} = -\nabla V = -\left(\hat{\mathbf{R}} \frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}} \frac{1}{R} \frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}\right), \quad (4.65)$$

where we have used the expression for ∇V in spherical coordinates given in Appendix C. Upon taking the derivatives of the expression for V given by Eq. (4.62) with respect to R and θ and then substituting the results in Eq. (4.65), we obtain

$$\mathbf{E} = \frac{qd}{4\pi\epsilon_0 R^3} (\hat{\mathbf{R}} 2 \cos \theta + \hat{\boldsymbol{\theta}} \sin \theta) \quad (\text{V/m}). \quad (4.66)$$

We stress that the expressions for V and $\mathbf{E}$ given by Eqs. (4.64) and (4.66) apply only when $R \gg d$. To compute V and $\mathbf{E}$ at points in the vicinity of the two dipole charges, it is necessary to perform all calculations without resorting to the far-distance approximations that led to Eq. (4.62). Such an exact calculation for $\mathbf{E}$ leads to the field pattern shown in **Fig. 4-15(b)**.

4-5.5 Poisson's Equation

With $\mathbf{D} = \epsilon\mathbf{E}$, the differential form of Gauss's law given by Eq. (4.26) may be cast as

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon}. \quad (4.67)$$

Inserting Eq. (4.61) in Eq. (4.67) gives

$$\nabla \cdot (\nabla V) = -\frac{\rho_v}{\epsilon}. \quad (4.68)$$

Given Eq. (3.110) for the Laplacian of a scalar function V ,

$$\nabla^2 V = \nabla \cdot (\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}, \quad (4.69)$$

Eq. (4.68) can be cast in the abbreviated form

$$\nabla^2 V = -\frac{\rho_v}{\epsilon} \quad (\text{Poisson's equation}). \quad (4.70)$$

This is known as **Poisson's equation**. For a volume $\mathcal{V}'$ containing a volume charge density distribution ρ_v , the solution for V derived previously and expressed by Eq. (4.58a) as

$$V = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v}{R'} d\mathcal{V}' \quad (4.71)$$

satisfies Eq. (4.70). If the medium under consideration contains no charges, Eq. (4.70) reduces to

$$\nabla^2 V = 0 \quad (\text{Laplace's equation}), \quad (4.72)$$

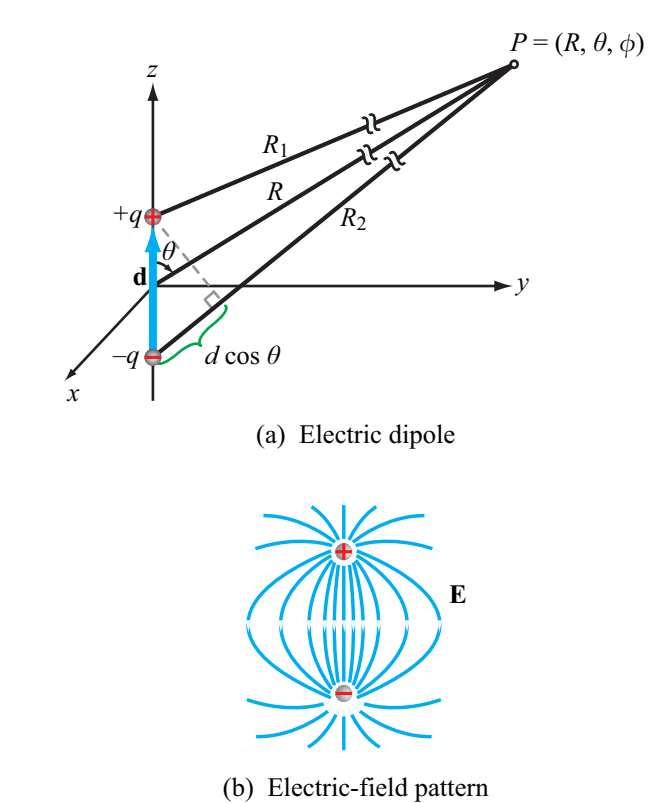


图 4-15 偶极矩为 $\mathbf{p} = q\mathbf{d}$ 的电偶极子 (示例 4-9)。

因此,

$$V = \frac{qd \cos \theta}{4\pi\epsilon_0 R^2}. \quad (4.62)$$

为了将此结果推广到任意方向的偶极子, 请注意方程的分子。(4.62) 可以表示为 qd (其中 d 是从 $-q$ 到 $+q$ 的距离向量) 与从偶极子中心指向观测点 P 的单位向量 $\hat{\mathbf{R}}$ 的点积。即

$$qd \cos \theta = q\mathbf{d} \cdot \hat{\mathbf{R}} = \mathbf{p} \cdot \hat{\mathbf{R}}, \quad (4.63)$$

其中 $\mathbf{p} = q\mathbf{d}$ 称为偶极矩。使用方程式方程 (4.63) (4.62) 然后给出

$$V = \frac{\mathbf{p} \cdot \hat{\mathbf{R}}}{4\pi\epsilon_0 R^2} \quad (\text{electric dipole}). \quad (4.64)$$

在球坐标系中, 方程(4.61) 由下式给出

$$\mathbf{E} = -\nabla V = -\left(\hat{\mathbf{R}} \frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}} \frac{1}{R} \frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}\right), \quad (4.65)$$

其中我们使用了附录 C 中给出的球坐标中的 ∇V 表达式。(4.62) 关于 R 和 θ , 然后将结果代入式 (4.62)。(4.65), 我们得到

$$\mathbf{E} = \frac{qd}{4\pi\epsilon_0 R^3} (\hat{\mathbf{R}} 2 \cos \theta + \hat{\boldsymbol{\theta}} \sin \theta) \quad (\text{V/m}). \quad (4.66)$$

我们强调 V 和 $\mathbf{E}$ 的表达式由等式给出。(4.64) 和 (4.66) 仅当 $R \gg d$ 时适用。为了计算两个偶极子电荷附近点处的 V 和 $\mathbf{E}$, 有必要执行所有计算, 而无需诉诸导致方程 1 的远距离近似。(4.62)。对 $\mathbf{E}$ 的这种精确计算得出如图 4-15(b) 所示的场模式。

4-5.5 泊松方程

对于 $\mathbf{D} = \epsilon\mathbf{E}$, 高斯定律的微分形式由方程给出。(4.26) 可以表示为

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon}. \quad (4.67)$$

插入方程式方程 (4.61) (4.67) 给出

$$\nabla \cdot (\nabla V) = -\frac{\rho_v}{\epsilon}. \quad (4.68)$$

给定方程。(3.110) 对于标量函数 V 的拉普拉斯算子,

$$\nabla^2 V = \nabla \cdot (\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}, \quad (4.69)$$

等式。(4.68) 可以缩写为形式

$$\nabla^2 V = -\frac{\rho_v}{\epsilon} \quad (\text{Poisson's equation}). \quad (4.70)$$

这就是所谓的泊松方程。对于包含体积电荷密度分布 ρ_v 的体积 $\mathcal{V}'$, 之前导出的 V 的解由方程 1 表示。(4.58a) 为

$$V = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v}{R'} d\mathcal{V}' \quad (4.71)$$

满足方程。(4.70)。如果所考虑的介质不包含电荷, 则方程: (4.70) 减少为

$$\nabla^2 V = 0 \quad (\text{Laplace's equation}), \quad (4.72)$$

and it is then referred to as ***Laplace’s equation***. Poisson’s and Laplace’s equations are useful for determining the electrostatic potential V in regions with boundaries on which V is known, such as the region between the plates of a capacitor with a specified voltage difference across it.

Concept Question 4-9: What is a conservative field?

Concept Question 4-10: Why is the electric potential at a point in space always defined relative to the potential at some reference point?

Concept Question 4-11: Explain why Eq. (4.40) is a mathematical statement of Kirchhoff’s voltage law.

Concept Question 4-12: Why is it usually easier to compute V for a given charge distribution and then find $\mathbf{E}$ using $\mathbf{E} = -\nabla V$ than to compute $\mathbf{E}$ directly by applying Coulomb’s law?

Concept Question 4-13: What is an electric dipole?

Exercise 4-10: Determine the electric potential at the origin due to four $20\text{ }\mu\text{C}$ charges residing in free space at the corners of a $2\text{ m} \times 2\text{ m}$ square centered about the origin in the x – y plane.

Answer: $V = \sqrt{2} \times 10^{-5}/(\pi\epsilon_0)\text{ (V)}$. (See [EM](#).)

Exercise 4-11: A spherical shell of radius a has a uniform surface charge density ρ_s . Determine (a) the electric potential and (b) the electric field with both at the center of the shell.

Answer: (a) $V = \rho_s a/\epsilon\text{ (V)}$, (b) $\mathbf{E} = 0$. (See [EM](#).)

4-6 Conductors

The electromagnetic ***constitutive parameters*** of a material medium are its electrical permittivity ϵ , magnetic permeability μ , and conductivity σ . A material is said to be ***homogeneous*** if its constitutive parameters do not vary from point to point, and ***isotropic*** if they are independent of direction. Most materials are isotropic, but some crystals are not. Throughout this book, all materials are assumed to be homogeneous and isotropic. This section is concerned with σ , Section 4-7 examines ϵ , and discussion of μ is deferred to Chapter 5.

► The ***conductivity*** of a material is a measure of how easily electrons can travel through the material under the influence of an externally applied electric field. ◀

Materials are classified as ***conductors*** (metals) or ***dielectrics*** (insulators) according to the magnitudes of their conductivities. A conductor has a large number of loosely attached electrons in the outermost shells of its atoms. In the absence of an external electric field, these free electrons move in random directions and with varying speeds. Their random motion produces zero average current through the conductor. Upon applying an external electric field, however, the electrons migrate from one atom to the next in the direction opposite that of the external field. Their movement gives rise to a ***conduction current density***

$$\mathbf{J} = \sigma \mathbf{E} \quad (\text{A/m}^2) \quad \textbf{(Ohm’s law)}, \quad (4.73)$$

where σ is the material’s conductivity with units of siemen per meter (S/m).

In yet other materials, called dielectrics, the electrons are tightly bound to the atoms—so much so that it is very difficult to detach them under the influence of an electric field. Consequently, no significant conduction current can flow through them.

► A ***perfect dielectric*** is a material with $\sigma = 0$. In contrast, a ***perfect conductor*** is a material with $\sigma = \infty$. Some materials, called superconductors, exhibit such a behavior. ◀

The conductivity σ of most metals is in the range from 10^6 to 10^7 S/m when compared with 10^{-10} to 10^{-17} S/m for good insulators ([Table 4-1](#) on p. 194). A class of materials called ***semiconductors*** allow for conduction currents even though their conductivities are much smaller than those of metals. The conductivity of pure germanium, for example, is 2.2 S/m . Tabulated values of σ at room temperature (20°C) are given in Appendix B for some common materials, and a subset is reproduced in [Table 4-1](#).

► The conductivity of a material depends on several factors, including temperature and the presence of impurities. In general, σ of metals increases with decreasing temperature. Most superconductors operate in the neighborhood of absolute zero. ◀

然后它被称为拉普拉斯方程。泊松方程和拉普拉斯方程可用于确定具有 V 已知边界的区域中的静电电势 V ，例如具有指定电压差的电容器极板之间的区域。

概念问题 4-9：什么是保守域？

概念问题 4-10：为什么空间中某一点的电势总是相对于某个参考点的电势来定义？

概念问题 4-11：解释为什么(4.40) 是基尔霍夫电压定律的数学表述。

概念问题 4-12：为什么计算给定电荷分布的 V ，然后使用 $\mathbf{E} = -\nabla V$ 求 $\mathbf{E}$ ，通常比应用库仑定律直接计算 $\mathbf{E}$ 更容易？

概念问题 4-13：什么是电偶极子？

练习 4-10：确定由于位于 x - y 平面中以原点为中心的 $2\text{ m} \times 2\text{ m}$ 正方形的角上的自由空间中的四个 $20\text{ }\mu\text{C}$ 电荷而在原点处的电势。

Answer: $V = \sqrt{2} \times 10^{-5}/(\pi\epsilon_0)\text{ (V)}$. (See [EM](#).)

练习4-11：半径为a的球壳具有均匀的表面电荷密度 ρ_s 。
。确定 (a) 电势和 (b) 电场 （两者均位于壳的中心）
。

答案：(a) $V = \rho_s a/\epsilon\text{ (V)}$ ，(b) $\mathbf{E} = 0$ 。（参见^{EM}。）

4-6 导体

材料介质的电磁本构参数是其介电常数 ϵ 、磁导率 μ 和电导率 σ 。如果一种材料的本构参数在点与点之间不发生变化，则该材料被称为均质材料；如果材料的本构参数与方向无关，则该材料被称为各向同性材料。大多数材料是各向同性的，但有些晶体不是。在本书中，所有材料都被假设为均匀且各向同性的。本节关注 σ ，第 4-7 节研究 ϵ ，对 μ 的讨论推迟到第 5 章。

► 材料的电导率是电子在外部施加的电场的影响下穿过材料的容易程度的量度。 ◀

根据其电导率的大小，材料被分为导体（金属）或电介质（绝缘体）。导体的原子最外层有大量松散连接的电子。在没有外部电场的情况下，这些自由电子以随机方向 and 不同速度移动。它们的随机运动产生通过导体的平均电流为零。然而，一旦施加外部电场，电子就会以与外部电场相反的方向从一个原子迁移到下一个原子。它们的运动产生传导电流密度

$$\mathbf{J} = \sigma \mathbf{E} \quad (\text{A/m}^2) \quad \textbf{(Ohm’s law)}, \quad (4.73)$$

其中 σ 是材料的电导率，单位为西门子每米 (S/m)。

在另一些称为电介质的材料中，电子与原子紧密结合，以至于在电场的影响下很难将它们分离。因此，没有明显的传导电流可以流过它们。

► 理想的电介质是 $\sigma = 0$ 的材料。相反，理想的导体是 $\sigma = \infty$ 的材料。一些称为超导体的材料表现出这种行为。 ◀

大多数金属的电导率 σ 范围为 10^6 至 10^7 S/m ，而良好绝缘体的电导率为 10^{-10} 至 10^{-17} S/m （第 194 页的表 4-1）。一类称为半导体的材料允许传导电流，尽管它们的电导率比金属小得多。例如，纯锗的电导率为 2.2 S/m 。附录 B 中给出了一些常见材料在室温 (20°C) 下的 σ 表格值，表 4-1 中再现了一个子集。

► 材料的电导率取决于多种因素，包括温度和杂质的存在。一般来说，金属的 σ 随着温度的降低而增加。大多数超导体在绝对零附近运行。 ◀

Technology Brief 8:
Supercapacitors as Batteries

As recent additions to the language of electronics, the names **supercapacitor**, **ultracapacitor**, and **nanocapacitor** suggest that they represent devices that are somehow different from or superior to traditional capacitors. Are these just fancy names attached to traditional capacitors by manufacturers, or are we talking about a really different type of capacitor?

► The three aforementioned names refer to variations on an energy storage device (Fig. TF8-1) known by the technical name **electrochemical double-layer capacitor (EDLC)**, in which *energy storage is realized by a hybrid process that incorporates features from both the traditional electrostatic capacitor and the electrochemical voltaic battery*. ◀

For the purposes of this Technology Brief, we refer to this relatively new device as a supercapacitor. The battery is far superior to the traditional capacitor with regard to energy storage, but a capacitor can be charged and discharged much more rapidly than a battery. As a hybrid technology, the supercapacitor offers features that are intermediate between those of the battery and the traditional capacitor. The supercapacitor is now used to support a wide range of applications, from motor startups in large engines (trucks, locomotives, submarines, etc.) to flash lights in digital cameras, and its use is rapidly

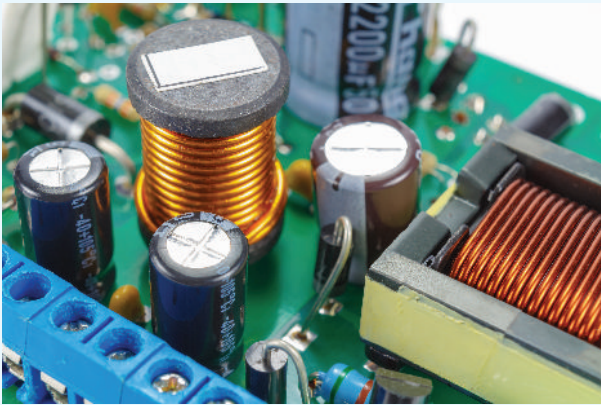


Figure TF8-1 Examples of supercapacitors. (Courtesy of Vladimir Zhupanenko/Shutterstock.)

extending into consumer electronics (cell phones, MP3 players, laptop computers) and electric cars (Fig. TF8-2).

Capacitor Energy Storage Limitations

Energy density W' is often measured in watts-hours per kg (Wh/kg), with $1 \text{ Wh} = 3.6 \times 10^3 \text{ joules}$. Thus, the energy capacity of a device is normalized to its mass. For batteries, W' extends between about 30 Wh/kg for a lead-acid battery to as high as 150 Wh/kg for a lithium-ion battery. In contrast, W' rarely exceeds 0.02 Wh/kg for a traditional capacitor. Let us examine what limits the value of W' for the capacitor by considering a small parallel-plate capacitor with plate area A and separation between plates d . For simplicity, we assign the capacitor a voltage rating of 1 V (maximum anticipated voltage across the capacitor). Our goal is to maximize the energy density W' . For a parallel-plate capacitor $C = \epsilon A/d$, where ϵ is the permittivity of the insulating material. Using Eq. (4.121) leads to

$$W' = \frac{W}{m} = \frac{1}{2m} CV^2 = \frac{\epsilon AV^2}{2md} \quad (\text{J/kg}),$$

where m is the mass of the conducting plates and the insulating material contained in the capacitor. To keep the analysis simple, we assume that the plates can be made so thin as to ignore their masses relative to the mass of the insulating material. If the material's density



Figure TF8-2 Examples of systems that use supercapacitors.

技术简介 8：超级电容器作为
电池

作为电子语言的最新补充，超级电容器、超级电容器和纳米电容器等名称表明它们代表了与传统电容器不同或优于传统电容器的设备。这些只是制造商给传统电容器起的奇特名称，还是我们谈论的是一种真正不同类型的电容器？

► 上述三个名称指的是能量存储设备的变体（图TF8-1），其技术名称为电化学双层电容器（EDLC），其中能量存储是通过结合传统静电电容器和电化学伏打电池特征的混合过程实现的。◀

就本技术简介而言，我们将这种相对较新的设备称为超级电容器。电池在储能方面远远优于传统电容器，但电容器的充电和放电速度比电池快得多。作为一种混合技术，超级电容器的功能介于电池和传统电容器之间。超级电容器现在用于支持广泛的应用，从大型发动机（卡车、机车、潜艇等）中的电机启动到数码相机中的闪光灯，并且其使用正在迅速发展

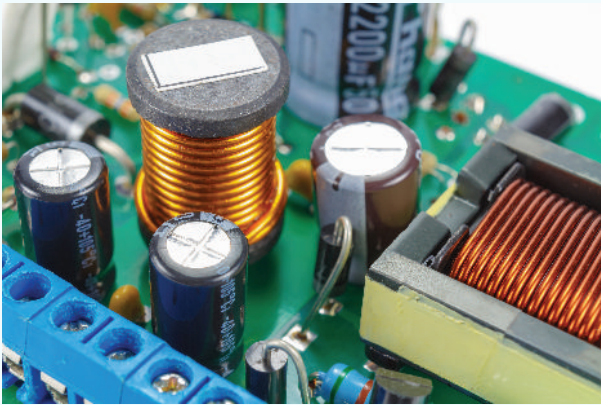


图 TF8-1 超级电容器示例。（由 Vladimir Zhupanenko/Shutterstock 提供。）

扩展到消费电子产品（手机、MP3播放器、笔记本电脑）和电动汽车（图TF8-2）。

电容器储能的局限性

能量密度 W' 通常以瓦时每千克 (Wh/kg) 为单位进行测量，其中 $1 \text{ Wh} = 3.6 \times 10^3 \text{ 焦耳}$ 。因此，设备的能量容量被归一化为其质量。对于电池， W' 范围从铅酸电池的约 30 Wh/kg 到锂离子电池的高达 150 Wh/kg。相比之下，传统电容器的 W' 很少超过 0.02 Wh/kg。让我们通过考虑一个具有板面积 A 和板间距 d 的小型平行板电容器来检查限制电容器 W' 值的因素。为简单起见，我们为电容器指定 1 V 的额定电压（电容器两端的最大预期电压）。我们的目标是最大化能量密度 W' 。对于平行板电容器 $C = \epsilon A/d$ ，其中 ϵ 是绝缘材料的介电常数。使用方程式(4.121) 导致

$$W' = \frac{W}{m} = \frac{1}{2m} CV^2 = \frac{\epsilon AV^2}{2md} \quad (\text{J/kg}),$$

其中 m 是电容器中包含的导电板和绝缘材料的质量。为了使分析简单，我们假设板可以做得很薄，以忽略它们相对于绝缘材料质量的质量。如果材料的密度



图 TF8-2 使用超级电容器的系统示例。

is ρ (kg/m³), then $m = \rho Ad$ and

$$W' = \frac{\varepsilon V^2}{2\rho d^2} \quad (\text{J/kg}).$$

To maximize W' , we need to select d to be the smallest possible, but we also have to be aware of the constraint associated with dielectric breakdown. To avoid sparking between the capacitor's two plates, the electric field strength should not exceed E_{ds} , which is the dielectric strength of the insulating material. Among the various types of materials commonly used in capacitors, mica has one of the highest values of E_{ds} , nearly 2×10^8 V/m. Breakdown voltage V_{br} is related to E_{ds} by $V_{\text{br}} = E_{\text{ds}}d$, so given that the capacitor is to have a voltage rating of 1 V, let us choose V_{br} to be 2 V, thereby allowing a 50% safety margin. With $V_{\text{br}} = 2$ V and $E_{\text{ds}} = 2 \times 10^8$ V/m, it follows that d should not be smaller than 10^{-8} m, or 10 nm. For mica, $\varepsilon \approx 6\varepsilon_0$ and $\rho = 3 \times 10^3$ kg/m³. Ignoring for the moment the practical issues associated with building a capacitor with a spacing of only 10 nm between conductors, the expression for energy density leads to $W' \approx 90$ J/kg. Converting W' to Wh/kg (by dividing by 3.6×10^3 J/Wh) gives

$$W'(\text{max}) = 2.5 \times 10^{-2} \quad (\text{Wh/kg}),$$

for a traditional capacitor at a voltage rating of 1 V.

► The energy storage capacity of a traditional capacitor is about four orders of magnitude smaller than the energy density capability of a lithium-ion battery. ◀

Energy Storage Comparison

The table in the upper part of **Fig. TF8-3** displays typical values or ranges of values for each of five attributes commonly used to characterize the performance of energy storage devices. In addition to the energy density W' , they include the power density P' , the charge and discharge rates, and the number of charge/discharge cycles that the device can withstand before deteriorating in performance. For most energy storage devices, the discharge rate usually is shorter than the charge rate, but for the purpose of the present discussion, we treat them as equal. As a first-order approximation, the **discharge rate** is related to P' and W' by

$$T = \frac{W'}{P'}.$$

► Supercapacitors are capable of storing 100 to 1000 times more energy than a traditional capacitor but 10 times less than a battery (**Fig. TF8-3**). On the other hand, *supercapacitors can discharge their stored energy in a matter of seconds when compared with hours for a battery.* ◀

Moreover, the supercapacitor's cycle life is on the order of 1 million when compared with only 1000 for a rechargeable battery. Because of these features, the supercapacitor has greatly expanded the scope and use of capacitors in electronic circuits and systems.

Future Developments

The upper right-hand corner of the plot in **Fig. TF8-3** represents the ideal energy storage device with $W' \approx 100\text{--}1000$ Wh/kg and $P' \approx 10^3\text{--}10^4$ W/kg. The corresponding discharge rate is $T \approx 10\text{--}100$ ms. Current research aims to extend the capabilities of batteries and supercapacitors in the direction of this prized domain of the **energy-power space**.

是 ρ (kg/m³), 则 $m = \rho Ad$ 和

$$W' = \frac{\varepsilon V^2}{2\rho d^2} \quad (\text{J/kg}).$$

为了最大化 W' , 我们需要选择尽可能小的 d , 但我们也必须意识到与介电击穿相关的约束。为了避免电容器两块极板之间产生火花, 电场强度不应超过 E_{ds} , 即绝缘材料的介电强度。在电容器常用的各类材料中, 云母是 E_{ds} 值最高的之一, 接近 2×10^8 V/m。击穿电压 V_{br} 与 E_{ds} 的关系为 $V_{\text{br}} = E_{\text{ds}}d$, 因此考虑到电容器的额定电压为 1 V, 我们选择 V_{br} 为 2 V, 从而允许 50% 的安全裕度。对于 $V_{\text{br}} = 2$ V 和 $E_{\text{ds}} = 2 \times 10^8$ V/m, 得出 d 不应小于 10^{-8} m 或 10 nm。对于云母, $\varepsilon \approx 6\varepsilon_0$ 和 $\rho = 3 \times 10^3$ kg/m³。暂时忽略与构建导体间距仅为 10 nm 的电容器相关的实际问题, 能量密度的表达式得出 $W' \approx 90$ J/kg。将 W' 转换为 Wh/kg (除以 3.6×10^3 J/Wh) 得出

$$W'(\text{max}) = 2.5 \times 10^{-2} \quad (\text{Wh/kg}),$$

对于额定电压为 1 V 的传统电容器。

► 传统电容器的能量存储容量比锂离子电池的能量密度小约四个数量级。 ◀

储能比较

图 TF8-3 上方的表格列出储能器件的五类典型性能参数: 比能量 W' 、比功率 P' 、充电时间、放电时间, 以及性能衰退前可承受的充放电循环次数。多数器件的放电时间短于充电时间; 此处为便于比较, 将二者视为相等。在一阶近似下, 放电时间 T 与 W' 、 P' 的关系为:

$$T = \frac{W'}{P'}.$$

► 超级电容器能够存储比传统电容器多 100 到 1000 倍的能量, 但比电池少 10 倍 (图 TF8-3)。另一方面, 超级电容器可以在几秒钟内释放其存储的能量, 而电池则需要数小时。 ◀

此外, 超级电容器的循环寿命约为 100 万次, 而充电电池的循环寿命仅为 1000 次。由于这些特性, 超级电容器极大地扩展了电容器在电子电路和系统中的范围和用途。

未来发展

图 TF8-3 的右上区域代表高比能量、高比功率的理想储能器件: $W' \approx 100\text{--}1000$ Wh/kg, $P' \approx 10^3\text{--}10^4$ W/kg。研究旨在使电池与超级电容器的性能向该区域发展。
校注: $T = W'/P'$ 的单位为小时; 上述范围对应 0.01–1 h (36–3600 s)。原文 “10–100 ms” 与所列数值不符。“rate” 在此按时间量译为 “时间”。

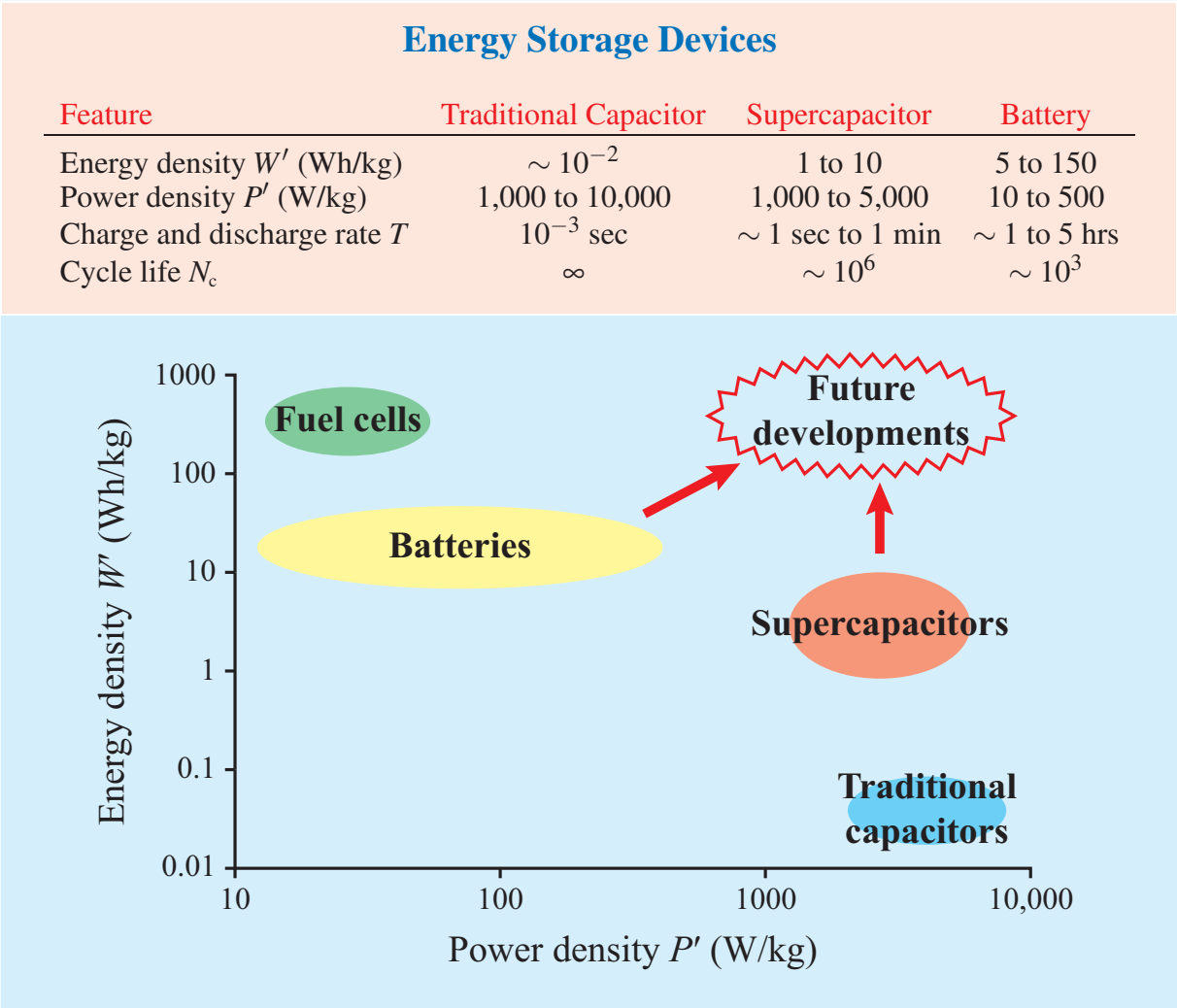
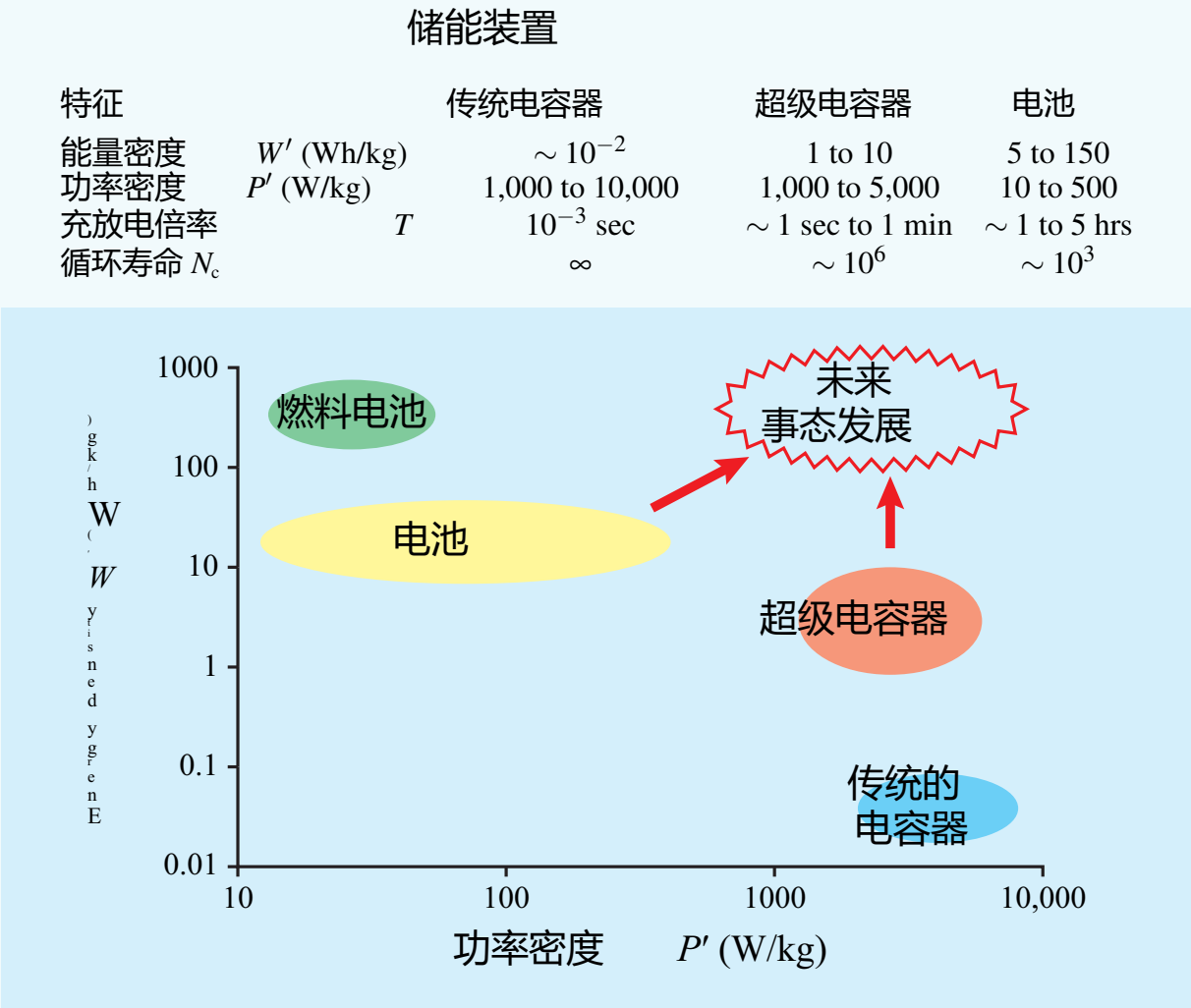


Figure TF8-3 Comparison of energy storage devices.



图TF8-3 储能装置对比。

Table 4-1 Conductivity of some common materials at 20 °C.

Material	Conductivity, σ (S/m)
<i>Conductors</i>	
Silver	6.2×10^7
Copper	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Iron	10^7
Mercury	10^6
Carbon	3×10^4
<i>Semiconductors</i>	
Pure germanium	2.2
Pure silicon	4.4×10^{-4}
<i>Insulators</i>	
Glass	10^{-12}
Paraffin	10^{-15}
Mica	10^{-15}
Fused quartz	10^{-17}

Concept Question 4-14: What are the electromagnetic constitutive parameters of a material?

Concept Question 4-15: What classifies a material as a conductor, a semiconductor, or a dielectric? What is a superconductor?

Concept Question 4-16: What is the conductivity of a perfect dielectric?

4-6.1 Drift Velocity

The drift velocity $\mathbf{u}_e$ of electrons in a conducting material is related to the externally applied electric field $\mathbf{E}$ through

$$\mathbf{u}_e = -\mu_e \mathbf{E} \quad (\text{m/s}), \quad (4.74a)$$

where μ_e is a material property call the *electron mobility* with units of (m²/V·s). In a semiconductor, current flow is due to the movement of both electrons and holes, and since holes are positive-charge carriers, the *hole drift velocity* $\mathbf{u}_h$ is in the same direction as $\mathbf{E}$,

$$\mathbf{u}_h = \mu_h \mathbf{E} \quad (\text{m/s}), \quad (4.74b)$$

where μ_h is the *hole mobility*. The mobility accounts for the effective mass of a charged particle and the average distance

over which the applied electric field can accelerate it before it is stopped by colliding with an atom and then starts accelerating all over again. From Eq. (4.11), the current density in a medium containing a volume density ρ_v of charges moving with velocity $\mathbf{u}$ is $\mathbf{J} = \rho_v \mathbf{u}$. In the most general case, the current density consists of a component $\mathbf{J}_e$ due to electrons and a component $\mathbf{J}_h$ due to holes. Thus, the total *conduction current density* is

$$\mathbf{J} = \mathbf{J}_e + \mathbf{J}_h = \rho_{ve} \mathbf{u}_e + \rho_{vh} \mathbf{u}_h \quad (\text{A/m}^2), \quad (4.75)$$

where $\rho_{ve} = -N_e e$ and $\rho_{vh} = N_h e$ with N_e and N_h being the number of free electrons and the number of free holes per unit volume and $e = 1.6 \times 10^{-19}$ C is the absolute charge of a single hole or electron. Use of Eqs. (4.74a) and (4.74b) gives

$$\mathbf{J} = (-\rho_{ve} \mu_e + \rho_{vh} \mu_h) \mathbf{E} = \sigma \mathbf{E}, \quad (4.76)$$

where the quantity inside the parentheses is defined as the *conductivity* of the material, σ . Thus,

$$\sigma = -\rho_{ve} \mu_e + \rho_{vh} \mu_h = (N_e \mu_e + N_h \mu_h) e \quad (\text{S/m}), \quad (4.77a)$$

(semiconductor)

and its unit is siemens per meter (S/m). For a good conductor, $N_h \mu_h \ll N_e \mu_e$, and Eq. (4.77a) reduces to

$$\sigma = -\rho_{ve} \mu_e = N_e \mu_e e \quad (\text{S/m}). \quad (4.77b)$$

(good conductor)

► In view of Eq. (4.76), in a *perfect dielectric* with $\sigma = 0$, $\mathbf{J} = 0$ regardless of $\mathbf{E}$. Similarly, in a *perfect conductor* with $\sigma = \infty$, $\mathbf{E} = \mathbf{J}/\sigma = 0$ regardless of $\mathbf{J}$. ◀

That is,

Perfect dielectric:	$\mathbf{J} = 0,$
Perfect conductor:	$\mathbf{E} = 0.$

Because σ is on the order of 10⁶ S/m for most metals, such as silver, copper, gold, and aluminum (**Table 4-1**), it is common practice to treat them as perfect conductors and to set $\mathbf{E} = 0$ inside them.

A perfect conductor is an *equipotential* medium, meaning that the electric potential is the same at every point in the conductor. This property follows from the fact that V_{21} , which is the voltage difference between two points in the conductor,

表 4-1 一些常见材料在 20 °C 时的电导率。

Material	Conductivity, σ (S/m)
<i>Conductors</i>	
Silver	6.2×10^7
Copper	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Iron	10^7
Mercury	10^6
Carbon	3×10^4
<i>Semiconductors</i>	
Pure germanium	2.2
Pure silicon	4.4×10^{-4}
<i>Insulators</i>	
Glass	10^{-12}
Paraffin	10^{-15}
Mica	10^{-15}
Fused quartz	10^{-17}

概念问题 4-14：什么是材料的电磁本构参数？

概念问题 4-15：如何将材料分类为导体、半导体或电介质？什么是超导体？

概念问题 4-16：理想电介质的电导率是多少？

4-6.1 漂移速度

导电材料中电子的漂移速度 $\mathbf{u}_e$ 与外部施加的电场 $\mathbf{E}$ 相关：

$$\mathbf{u}_e = -\mu_e \mathbf{E} \quad (\text{m/s}), \quad (4.74a)$$

其中 μ_e 是称为电子迁移率的材料属性，单位为 (m²/V·s)。在半导体中，电流是由电子和空穴的运动引起的，并且由于空穴是正电荷载流子，因此空穴漂移速度 $\mathbf{u}_h$ 与 $\mathbf{E}$ 方向相同，

$$\mathbf{u}_h = \mu_h \mathbf{E} \quad (\text{m/s}), \quad (4.74b)$$

其中 μ_h 是空穴迁移率。迁移率考虑了带电粒子的有效质量和平均距离

在它与原子碰撞而停止之前，所施加的电场可以对其进行加速，然后再次开始加速。从方程。 (4.11) 中，含有以速度 $\mathbf{u}$ 运动的体积密度 ρ_v 的电荷的介质中的电流密度为 $\mathbf{J} = \rho_v \mathbf{u}$ 。在最一般的情况下，电流密度由电子引起的分量 $\mathbf{J}_e$ 和空穴引起的分量 $\mathbf{J}_h$ 组成。因此，总传导电流密度为

$$\mathbf{J} = \mathbf{J}_e + \mathbf{J}_h = \rho_{ve} \mathbf{u}_e + \rho_{vh} \mathbf{u}_h \quad (\text{上午}^2), \quad (4.75)$$

其中 $\rho_{ve} = -N_e e$ 和 $\rho_{vh} = N_h e$ ，其中 N_e 和 N_h 是每单位体积的自由电子数量和自由空穴数量， $e = 1.6 \times 10^{-19}$ C 是单个空穴或电子的绝对电荷。使用等式。 (4.74a) 和 (4.74 b) 给出

$$\mathbf{J} = (-\rho_{ve} \mu_e + \rho_{vh} \mu_h) \mathbf{E} = \sigma \mathbf{E}, \quad (4.76)$$

其中括号内的数量定义为材料的电导率， σ 。因此，

$$\sigma = -\rho_{ve} \mu_e + \rho_{vh} \mu_h = (N_e \mu_e + N_h \mu_h) e \quad (\text{S/m}), \quad (4.77a)$$

(semiconductor)

单位为西门子每米 (S/m)。对于良导体， $N_h \mu_h \ll N_e \mu_e$ 和方程： (4.77a) 减少为

$$\sigma = -\rho_{ve} \mu_e = N_e \mu_e e \quad (\text{S/m}). \quad (4.77b)$$

(good conductor)

► 鉴于方程。 (4.76)，在具有 $\sigma = 0$ 的理想电介质中，无论 $\mathbf{E}$ 为何， $\mathbf{J} = 0$ 。类似地，在具有 $\sigma = \infty$ 的理想导体中，无论 $\mathbf{J}$ 为何， $\mathbf{E} = \mathbf{J}/\sigma = 0$ 。 ◀

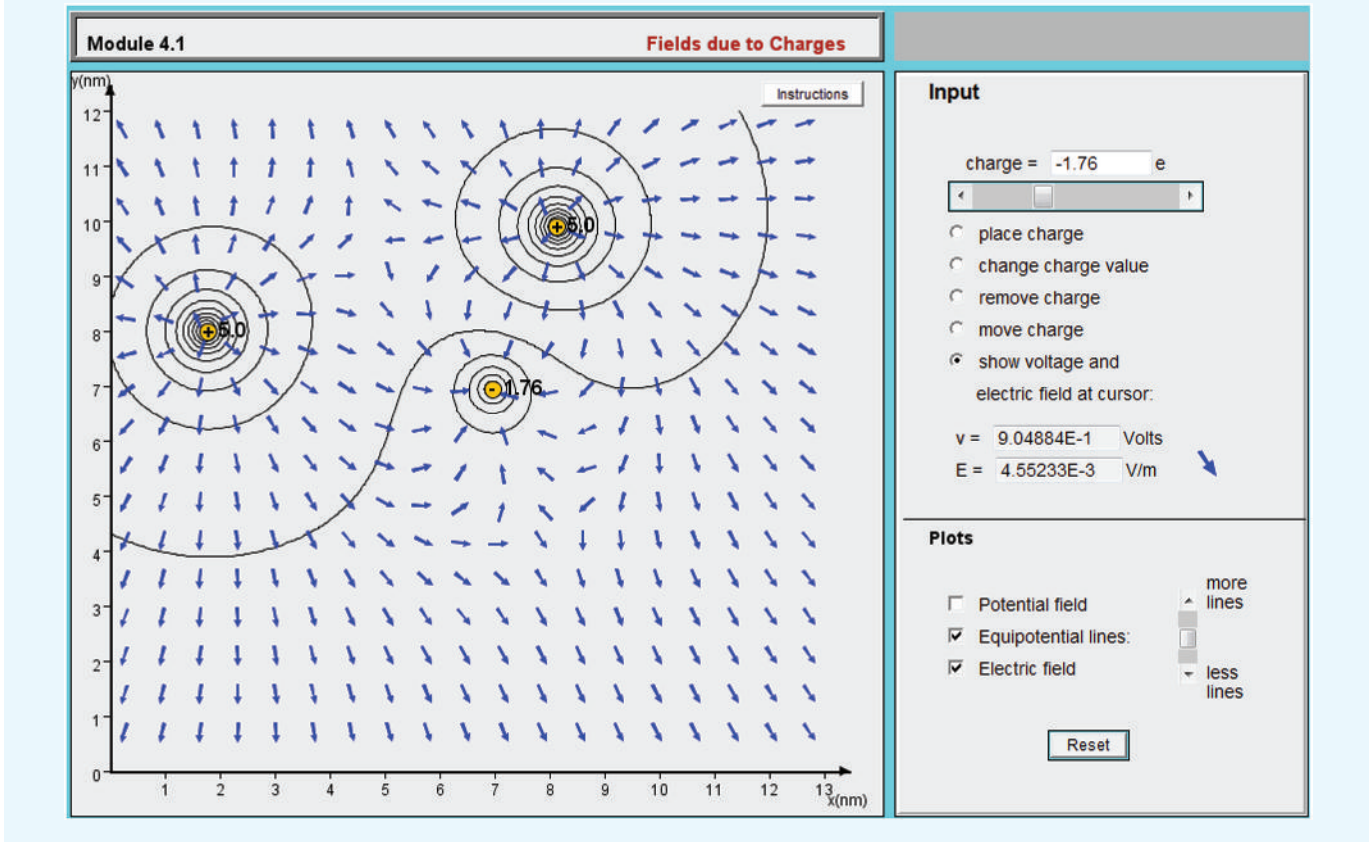
那是，

Perfect dielectric:	$\mathbf{J} = 0,$
Perfect conductor:	$\mathbf{E} = 0.$

由于大多数金属（例如银、铜、金和铝）的 σ 约为 10⁶ S /m（表 4-1），因此通常的做法是将它们视为理想导体并在其中设置 $\mathbf{E} = 0$ 。

理想导体是等电位介质，这意味着导体中每一点的电位都相同。此属性源自以下事实： V_{21} ，即导体中两点之间的电压差，

Module 4.1 Fields Due to Charges For any group of point charges, this module calculates and displays the electric field $\mathbf{E}$ and potential V across a 2-D grid. The user can specify the locations, magnitudes and polarities of the charges.



equals the line integral of $\mathbf{E}$ between them, as indicated by Eq. (4.39). Since $\mathbf{E} = 0$ everywhere in the perfect conductor, the voltage difference $V_{21} = 0$. The fact that the conductor is an equipotential medium, however, does not necessarily imply that the potential difference between the conductor and some other conductor is zero. Each conductor is an equipotential medium, but the presence of different distributions of charges on their two surfaces can generate a potential difference between them.

Example 4-10: Conduction Current in a Copper Wire

A 2-mm diameter copper wire with conductivity of 5.8×10^7 S/m and electron mobility of 0.0032 ($\text{m}^2/\text{V}\cdot\text{s}$) is subjected to an electric field of 20 (mV/m). Find (a) the volume charge density of the free electrons, (b) the current

density, (c) the current flowing in the wire, (d) the electron drift velocity, and (e) the volume density of the free electrons.

Solution: (a)

$$\rho_{ve} = -\frac{\sigma}{\mu_e} = -\frac{5.8 \times 10^7}{0.0032} = -1.81 \times 10^{10} \text{ (C/m}^3\text{)}.$$

(b)

$$J = \sigma E = 5.8 \times 10^7 \times 20 \times 10^{-3} = 1.16 \times 10^6 \text{ (A/m}^2\text{)}.$$

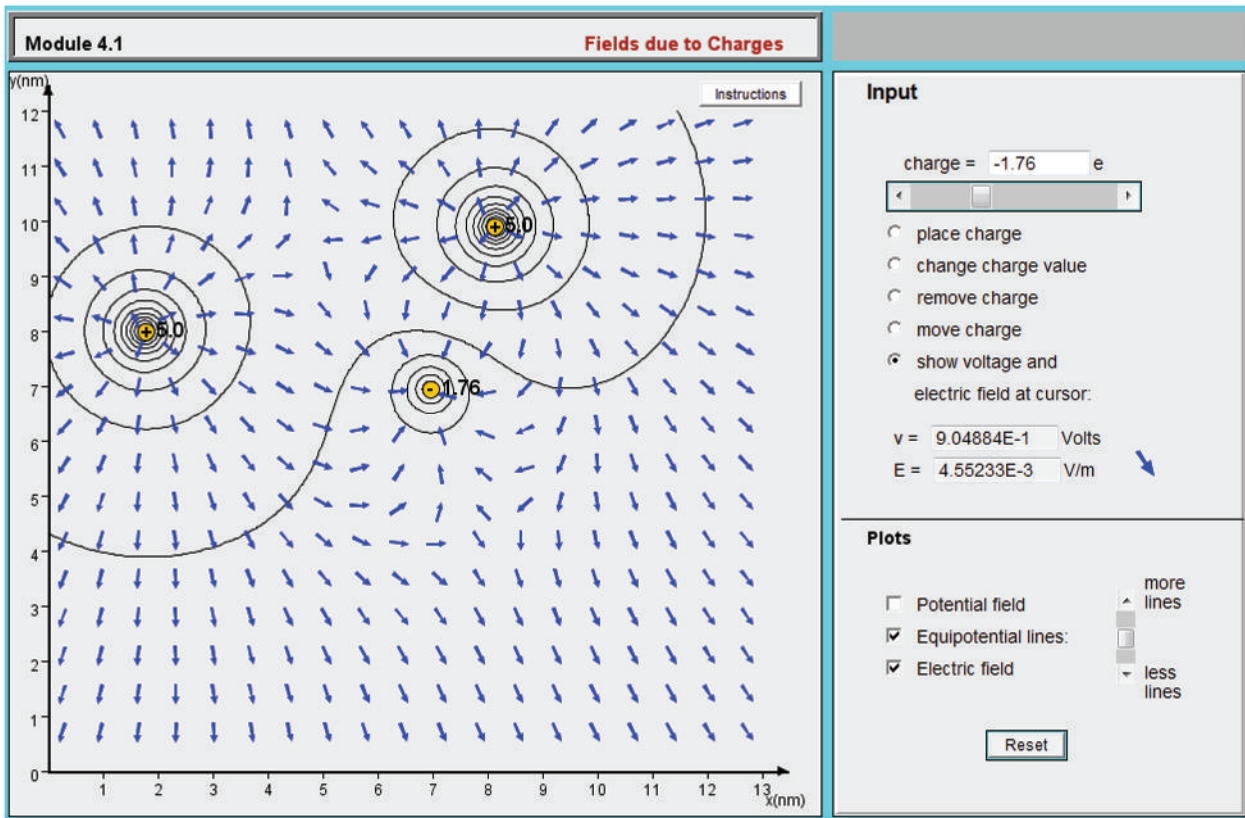
(c)

$$\begin{aligned} I &= JA \\ &= J \left(\frac{\pi d^2}{4} \right) = 1.16 \times 10^6 \left(\frac{\pi \times 4 \times 10^{-6}}{4} \right) = 3.64 \text{ A.} \end{aligned}$$

(d)

$$u_e = -\mu_e E = -0.0032 \times 20 \times 10^{-3} = -6.4 \times 10^{-5} \text{ m/s.}$$

模块 4.1 电荷产生的场 对于任何一组点电荷，该模块计算并显示二维网格上的电场 $\mathbf{E}$ 和电势 V 。用户可以指定电荷的位置、大小和极性。



等于它们之间 $\mathbf{E}$ 的线积分，如式 (1) 所示。 (4.39)。由于理想导体中到处 $\mathbf{E} = 0$ ，因此电压差 $V_{21} = 0$ 。然而，导体是等电位介质的事实并不意味着该导体与其他导体之间的电位差为零。每个导体都是等电位介质，但其两个表面上不同电荷分布的存在会在它们之间产生电位差。

例 4-10：铜线中的传导电流

一根直径为 2 毫米的铜线，其电导率为 5.8×10^7 S/m，电子迁移率为 0.0032 ($\text{m}^2/\text{V}\cdot\text{s}$)，该铜线承受 20 (mV/m) 的电场。求 (a) 自由电子的体积电荷密度，(b) 电流

密度，(c) 导线中流动的电流，(d) 电子漂移速度，以及 (e) 自由电子的体积密度。

解： (一)

$$\rho_{ve} = -\frac{\sigma}{\mu_e} = -\frac{5.8 \times 10^7}{0.0032} = -1.81 \times 10^{10} \text{ (C/m}^3\text{)}.$$

(b)

$$J = \sigma E = 5.8 \times 10^7 \times 20 \times 10^{-3} = 1.16 \times 10^6 \text{ (A/m}^2\text{)}.$$

(c)

$$\begin{aligned} I &= JA \\ &= J \left(\frac{\pi d^2}{4} \right) = 1.16 \times 10^6 \left(\frac{\pi \times 4 \times 10^{-6}}{4} \right) = 3.64 \text{ A.} \end{aligned}$$

(d)

$$u_e = -\mu_e E = -0.0032 \times 20 \times 10^{-3} = -6.4 \times 10^{-5} \text{ m/s.}$$

The minus sign indicates that $\mathbf{u}_e$ is in the opposite direction of $\mathbf{E}$.

(e)

$$N_e = -\frac{\rho_{ve}}{e} = \frac{1.81 \times 10^{10}}{1.6 \times 10^{-19}} = 1.13 \times 10^{29} \text{ electrons/m}^3.$$

Exercise 4-12: Determine the density of free electrons in aluminum given that its conductivity is 3.5×10^7 (S/m) and its electron mobility is 0.0015 ($\text{m}^2/\text{V} \cdot \text{s}$).

Answer: $N_e = 1.46 \times 10^{29}$ electrons/ m^3 . (See [EM](#).)

Exercise 4-13: The current flowing through a 100-m long conducting wire of uniform cross section has a density of 3×10^5 (A/ m^2). Find the voltage drop along the length of the wire if the wire material has a conductivity of 2×10^7 (S/m).

Answer: $V = 1.5$ V. (See [EM](#).)

4-6.2 Resistance

To demonstrate the utility of the point form of Ohm’s law, we apply it to derive an expression for the resistance R of a conductor of length l and uniform cross section A , as shown in [Fig. 4-16](#). The conductor axis is along the x direction and extends between points x_1 and x_2 , with $l = x_2 - x_1$. A voltage V applied across the conductor terminals establishes an electric field $\mathbf{E} = \hat{\mathbf{x}}E_x$; the direction of $\mathbf{E}$ is from the point with higher potential (point 1 in [Fig. 4-16](#)) to the point with lower

potential (point 2). The relation between V and E_x is obtained by applying Eq. (4.39):

$$V = V_1 - V_2 = -\int_{x_2}^{x_1} \mathbf{E} \cdot d\mathbf{l} = -\int_{x_2}^{x_1} \hat{\mathbf{x}}E_x \cdot \hat{\mathbf{x}} dl = E_x l \quad (\text{V}). \tag{4.78}$$

Using Eq. (4.73), the current flowing through the cross section A at x_2 is

$$I = \int_A \mathbf{J} \cdot d\mathbf{s} = \int_A \sigma \mathbf{E} \cdot d\mathbf{s} = \sigma E_x A \quad (\text{A}). \tag{4.79}$$

From $R = V/I$, the ratio of Eq. (4.78) to Eq. (4.79) gives

$$R = \frac{l}{\sigma A} \quad (\Omega). \tag{4.80}$$

We now generalize our result for R to any resistor of arbitrary shape by noting that the voltage V across the resistor is equal to the line integral of $\mathbf{E}$ over a path l between two specified points and the current I is equal to the flux of $\mathbf{J}$ through the surface S of the resistor. Thus,

$$R = \frac{V}{I} = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \mathbf{J} \cdot d\mathbf{s}} = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}. \tag{4.81}$$

The reciprocal of R is called the **conductance** G , and the unit of G is (Ω^{-1}) or siemens (S). For the linear resistor,

$$G = \frac{1}{R} = \frac{\sigma A}{l} \quad (\text{S}). \tag{4.82}$$

Example 4-11: Conductance of Coaxial Cable

The radii of the inner and outer conductors of a coaxial cable of length l are a and b , respectively ([Fig. 4-17](#)). The insulation material has conductivity σ . Obtain an expression for G' , which is the conductance per unit length of the insulation layer.

Solution: Let I be the total current flowing radially (along $\hat{\mathbf{r}}$) from the inner conductor to the outer conductor through the insulation material. At any radial distance r from the axis of the center conductor, the area through which the current flows is $A = 2\pi r l$. Hence,

$$\mathbf{J} = \hat{\mathbf{r}} \frac{I}{A} = \hat{\mathbf{r}} \frac{I}{2\pi r l}, \tag{4.83}$$

and from $\mathbf{J} = \sigma \mathbf{E}$,

$$\mathbf{E} = \hat{\mathbf{r}} \frac{I}{2\pi \sigma r l}. \tag{4.84}$$

负号表示 $\mathbf{u}_e$ 与 $\mathbf{E}$ 的方向相反。

(e)

$$N_e = -\frac{\rho_{ve}}{e} = \frac{1.81 \times 10^{10}}{1.6 \times 10^{-19}} = 1.13 \times 10^{29} \text{ electrons/m}^3.$$

练习 4-12：假设铝的电导率为 3.5×10^7 (S/m)，电子迁移率为 0.0015 ($\text{m}^2/\text{V} \cdot \text{s}$)，确定铝中自由电子的密度。

Answer: $N_e = 1.46 \times 10^{29}$ electrons/ m^3 . (See [EM](#).)

练习4-13：流过100米长、截面均匀的导线的电流密度为 3×10^5 (A/ m^2)。如果导线材料的电导率为 2×10^7 (S/m)，则求沿导线长度的电压降。

Answer: $V = 1.5$ V. (See [EM](#).)

4-6.2 电阻

为了证明欧姆定律点形式的实用性，我们应用它推导了长度为 l 、均匀横截面为 A 的导体的电阻 R 的表达式，如图 4-16 所示。导体轴沿 x 方向并在点 x_1 和 x_2 之间延伸，其中 $l = x_2 - x_1$ 。施加在导体端子上的电压 V 建立电场 $\mathbf{E} = \hat{\mathbf{x}}E_x$ ； $\mathbf{E}$ 的方向是从电势较高的点（图4-16中的点1）到电势较低的点

潜力（第 2 点）。 V 和 E_x 之间的关系通过应用式（1）获得。（4.39）：

$$V = V_1 - V_2 = -\int_{x_2}^{x_1} \mathbf{E} \cdot d\mathbf{l} = -\int_{x_2}^{x_1} \hat{\mathbf{x}}E_x \cdot \hat{\mathbf{x}} dl = E_x l \quad (\text{V}). \tag{4.78}$$

使用方程式（4.73），流过 x_2 处横截面 A 的电流为

$$I = \int_A \mathbf{J} \cdot d\mathbf{s} = \int_A \sigma \mathbf{E} \cdot d\mathbf{s} = \sigma E_x A \quad (\text{A}). \tag{4.79}$$

从 $R = V/I$ 可以得出方程式的比率：（4.78）至方程式(4.79) 给出

$$R = \frac{l}{\sigma A} \quad (\Omega). \tag{4.80}$$

现在，我们将 R 的结果推广到任意形状的电容器，注意电阻器两端的电压 V 等于 $\mathbf{E}$ 在两个指定点之间的路径 l 上的线积分，电流 I 等于通过电阻器表面 S 的通量 $\mathbf{J}$ 。因此，

$$R = \frac{V}{I} = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \mathbf{J} \cdot d\mathbf{s}} = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}. \tag{4.81}$$

R 的倒数称为电导 G ， G 的单位为(Ω^{-1})或西门子(S)。对于线性电阻，

$$G = \frac{1}{R} = \frac{\sigma A}{l} \quad (\text{S}). \tag{4.82}$$

例 4-11：同轴电缆的电导

长度为 l 的同轴电缆的内导体和外导体的半径分别为 a 和 b （图4-17）。绝缘材料具有电导率 σ 。求出 G' 的表达式，即绝缘层单位长度的电导。

解：设 I 为从内导体通过绝缘材料径向（沿 $\hat{\mathbf{r}}$ ）流到外导体的总电流。在距中心导体轴任意径向距离 r 处，电流流过的面积为 $A = 2\pi r l$ 。因此，

$$\mathbf{J} = \hat{\mathbf{r}} \frac{I}{A} = \hat{\mathbf{r}} \frac{I}{2\pi r l}, \tag{4.83}$$

并从 $\mathbf{J} = \sigma \mathbf{E}$,

$$\mathbf{E} = \hat{\mathbf{r}} \frac{I}{2\pi \sigma r l}. \tag{4.84}$$

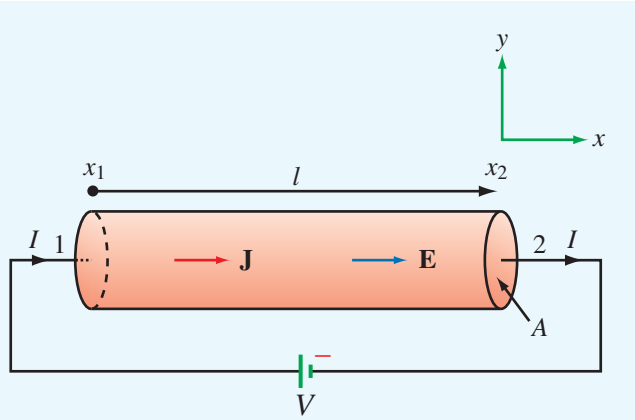


Figure 4-16 Linear resistor of cross section A and length l connected to a dc voltage source V .

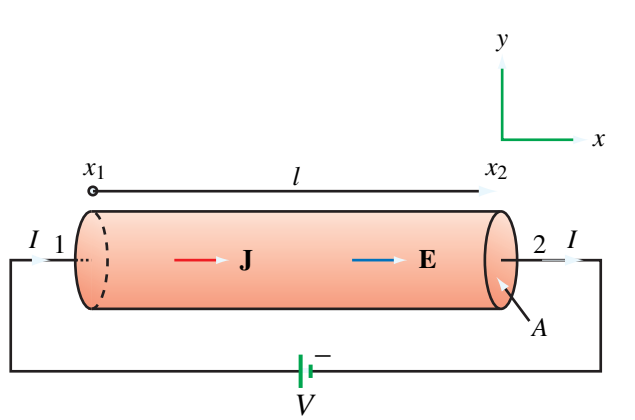
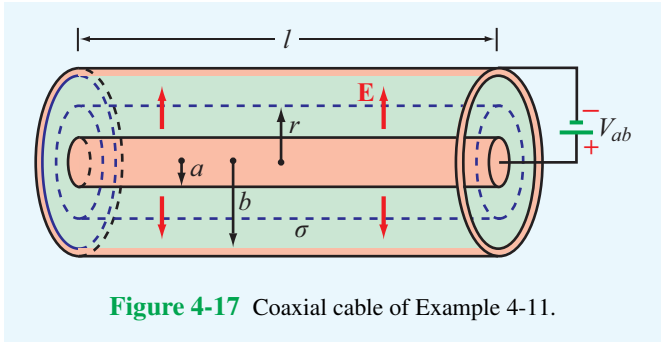


图 4-16 连接到直流电压源 V 的横截面为 A 、长度为 l 的线性电阻。



In a resistor, the current flows from higher electric potential to lower potential. Hence, if $\mathbf{J}$ is in the $\hat{\mathbf{r}}$ direction, the inner conductor must be at a potential higher than that at the outer conductor. Accordingly, the voltage difference between the conductors is

$$V_{ab} = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \int_b^a \frac{I}{2\pi\sigma l} \frac{\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} dr}{r} = \frac{I}{2\pi\sigma l} \ln\left(\frac{b}{a}\right). \quad (4.85)$$

The conductance per unit length is then

$$G' = \frac{G}{l} = \frac{1}{Rl} = \frac{I}{V_{abl}} = \frac{2\pi\sigma}{\ln(b/a)} \quad (\text{S/m}). \quad (4.86)$$

4-6.3 Joule’s Law

We now consider the power dissipated in a conducting medium in the presence of an electrostatic field $\mathbf{E}$. The medium contains free electrons and holes with volume charge densities ρ_{ve} and ρ_{vh} , respectively. The electron and hole charge contained in an elemental volume $\Delta\mathcal{V}$ is $q_e = \rho_{ve} \Delta\mathcal{V}$ and $q_h = \rho_{vh} \Delta\mathcal{V}$, respectively. The electric forces acting on q_e and q_h are $\mathbf{F}_e = q_e \mathbf{E} = \rho_{ve} \mathbf{E} \Delta\mathcal{V}$ and $\mathbf{F}_h = q_h \mathbf{E} = \rho_{vh} \mathbf{E} \Delta\mathcal{V}$. The work (energy) expended by the electric field in moving q_e a differential distance Δl_e and moving q_h a distance Δl_h is

$$\Delta W = \mathbf{F}_e \cdot \Delta \mathbf{l}_e + \mathbf{F}_h \cdot \Delta \mathbf{l}_h. \quad (4.87)$$

Power P is measured in watts (W) and is defined as the time rate of change of energy. The power corresponding to ΔW is

$$\begin{aligned} \Delta P &= \frac{\Delta W}{\Delta t} = \mathbf{F}_e \cdot \frac{\Delta \mathbf{l}_e}{\Delta t} + \mathbf{F}_h \cdot \frac{\Delta \mathbf{l}_h}{\Delta t} \\ &= \mathbf{F}_e \cdot \mathbf{u}_e + \mathbf{F}_h \cdot \mathbf{u}_h \\ &= (\rho_{ve} \mathbf{E} \cdot \mathbf{u}_e + \rho_{vh} \mathbf{E} \cdot \mathbf{u}_h) \Delta \mathcal{V} = \mathbf{E} \cdot \mathbf{J} \Delta \mathcal{V}, \end{aligned} \quad (4.88)$$

where $\mathbf{u}_e = \Delta \mathbf{l}_e / \Delta t$ and $\mathbf{u}_h = \Delta \mathbf{l}_h / \Delta t$ are the electron and hole drift velocities, respectively. Equation (4.75) was used

in the last step of the derivation leading to Eq. (4.88). For a volume $\mathcal{V}$, the total dissipated power is

$$P = \int_{\mathcal{V}} \mathbf{E} \cdot \mathbf{J} d\mathcal{V} \quad (\text{W}) \quad (\text{Joule’s law}), \quad (4.89)$$

and in view of Eq. (4.73),

$$P = \int_{\mathcal{V}} \sigma |\mathbf{E}|^2 d\mathcal{V} \quad (\text{W}). \quad (4.90)$$

Equation (4.89) is a mathematical statement of **Joule’s law**. For the resistor example considered earlier, $|\mathbf{E}| = E_x$ and its volume is $\mathcal{V} = lA$. Separating the volume integral in Eq. (4.90) into a product of a surface integral over A and a line integral over l , we have

$$\begin{aligned} P &= \int_{\mathcal{V}} \sigma |\mathbf{E}|^2 d\mathcal{V} = \int_A \sigma E_x ds \int_l E_x dl \\ &= (\sigma E_x A)(E_x l) = IV \quad (\text{W}), \end{aligned} \quad (4.91)$$

where use was made of Eq. (4.78) for the voltage V and Eq. (4.79) for the current I . With $V = IR$, we obtain the familiar expression

$$P = I^2 R \quad (\text{W}). \quad (4.92)$$

Concept Question 4-17: What is the fundamental difference between an insulator, a semiconductor, and a conductor?

Concept Question 4-18: Show that the power dissipated in the coaxial cable of **Fig. 4-17** is

$$P = \frac{I^2 \ln(b/a)}{2\pi\sigma l}.$$

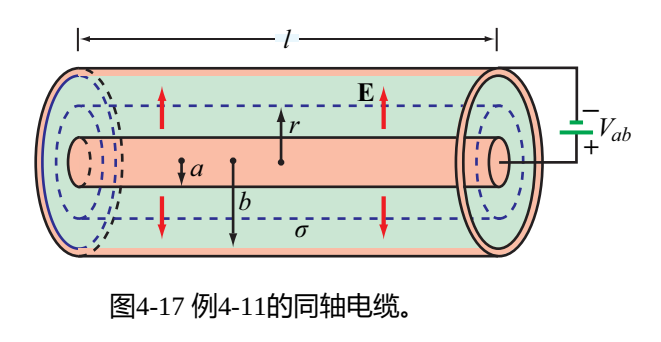
Exercise 4-14: A 50-m long copper wire has a circular cross section with radius $r = 2$ cm. Given that the conductivity of copper is 5.8×10^7 S/m, determine (a) the resistance R of the wire and (b) the power dissipated in the wire if the voltage across its length is 1.5 mV.

Answer: (a) $R = 6.9 \times 10^{-4} \Omega$, (b) $P = 3.3$ mW. (See )

Exercise 4-15: Repeat part (b) of Exercise 4-14 by applying Eq. (4.90). (See )

4-7 Dielectrics

The fundamental difference between a conductor and a di-electric is that electrons in the outermost atomic shells of



在电阻器中，电流从较高电势流向较低电势。因此，如果 $\mathbf{J}$ 沿 $\hat{\mathbf{r}}$ 方向，则内导体的电势必定高于外导体的电势。因此，导体之间的电压差为

$$V_{ab} = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \int_b^a \frac{I}{2\pi\sigma l} \frac{\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} dr}{r} = \frac{I}{2\pi\sigma l} \ln\left(\frac{b}{a}\right). \quad (4.85)$$

那么单位长度的电导为

$$G' = \frac{G}{l} = \frac{1}{Rl} = \frac{I}{V_{abl}} = \frac{2\pi\sigma}{\ln(b/a)} \quad (\text{S/m}). \quad (4.86)$$

4-6.3 焦耳定律

现在我们考虑存在静电场 $\mathbf{E}$ 时导电介质中耗散的功率。该介质包含体积电荷密度分别为 ρ_{ve} 和 ρ_{vh} 的自由电子和空穴。元素体积 $\Delta\mathcal{V}$ 中包含的电子和空穴电荷分别为 $q_e = \rho_{ve} \Delta\mathcal{V}$ 和 $q_h = \rho_{vh} \Delta\mathcal{V}$ 。作用在 q_e 和 q_h 上的电力为 $\mathbf{F}_e = q_e \mathbf{E} = \rho_{ve} \mathbf{E} \Delta\mathcal{V}$ 和 $\mathbf{F}_h = q_h \mathbf{E} = \rho_{vh} \mathbf{E} \Delta\mathcal{V}$ 。电场将 q_e 移动微分距离 Δl_e 和将 q_h 移动距离 Δl_h 所消耗的功（能量）为

$$\Delta W = \mathbf{F}_e \cdot \Delta \mathbf{l}_e + \mathbf{F}_h \cdot \Delta \mathbf{l}_h. \quad (4.87)$$

功率 P 以瓦 (W) 为单位测量，定义为能量的时间变化率。 ΔW 对应的功率为

$$\begin{aligned} \Delta P &= \frac{\Delta W}{\Delta t} = \mathbf{F}_e \cdot \frac{\Delta \mathbf{l}_e}{\Delta t} + \mathbf{F}_h \cdot \frac{\Delta \mathbf{l}_h}{\Delta t} \\ &= \mathbf{F}_e \cdot \mathbf{u}_e + \mathbf{F}_h \cdot \mathbf{u}_h \\ &= (\rho_{ve} \mathbf{E} \cdot \mathbf{u}_e + \rho_{vh} \mathbf{E} \cdot \mathbf{u}_h) \Delta \mathcal{V} = \mathbf{E} \cdot \mathbf{J} \Delta \mathcal{V}, \end{aligned} \quad (4.88)$$

其中 $\mathbf{u}_e = \Delta \mathbf{l}_e / \Delta t$ 和 $\mathbf{u}_h = \Delta \mathbf{l}_h / \Delta t$ 分别是电子和空穴漂移速度。使用方程 (4.75)

在导出方程的最后一步中 (4.88)。对于体积 $\mathcal{V}$ ，总耗散功率为

$$P = \int_{\mathcal{V}} \mathbf{E} \cdot \mathbf{J} d\mathcal{V} \quad (\text{W}) \quad (\text{Joule’s law}), \quad (4.89)$$

并考虑到等式。 (4.73),

$$P = \int_{\mathcal{V}} \sigma |\mathbf{E}|^2 d\mathcal{V} \quad (\text{W}). \quad (4.90)$$

方程 (4.89) 是焦耳定律的数学表述。对于前面考虑的电阻器示例，为 $|\mathbf{E}| = E_x$ ，其体积为 $\mathcal{V} = lA$ 。分离方程中的体积积分。将 (4.90) 代入 A 上的面积分和 l 上的线积分的乘积，我们有

$$\begin{aligned} P &= \int_{\mathcal{V}} \sigma |\mathbf{E}|^2 d\mathcal{V} = \int_A \sigma E_x ds \int_l E_x dl \\ &= (\sigma E_x A)(E_x l) = IV \quad (\text{W}), \end{aligned} \quad (4.91)$$

其中使用了方程式。 (4.78) 为电压 V 和方程。 (4.79) 为电流 I 。通过 $V = IR$ ，我们得到熟悉的表达式

$$P = I^2 R \quad (\text{W}). \quad (4.92)$$

概念问题 4-17: 绝缘体、半导体和导体之间的根本区别是什么?

概念问题 4-18: 证明图 4-17 的同轴电缆中消耗的功率为

$$P = \frac{I^2 \ln(b/a)}{2\pi\sigma l}.$$

练习 4-14: 一根 50 米长的铜线，其横截面为圆形，半径 $r = 2$ 厘米。假定铜的电导率为 5.8×10^7 S/m，请确定 (a) 导线的电阻 R 和 (b) 如果导线长度上的电压为 1.5 mV，则导线中消耗的功率。

Answer: (a) $R = 6.9 \times 10^{-4} \Omega$, (b) $P = 3.3$ mW. (See )

练习 4-15: 通过应用方程 4 重复练习 4-14 的 (b) 部分。(4.90)。(参见 )

4-7 电介质

导体和电介质之间的根本区别在于，电子位于导体的最外层原子壳层中。

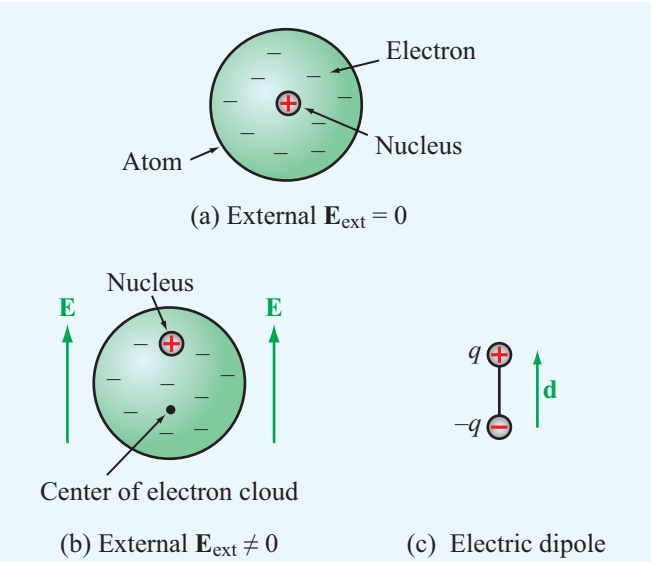


Figure 4-18 In the absence of an external electric field E , the center of the electron cloud is co-located with the center of the nucleus, but when a field is applied, the two centers are separated by a distance d .

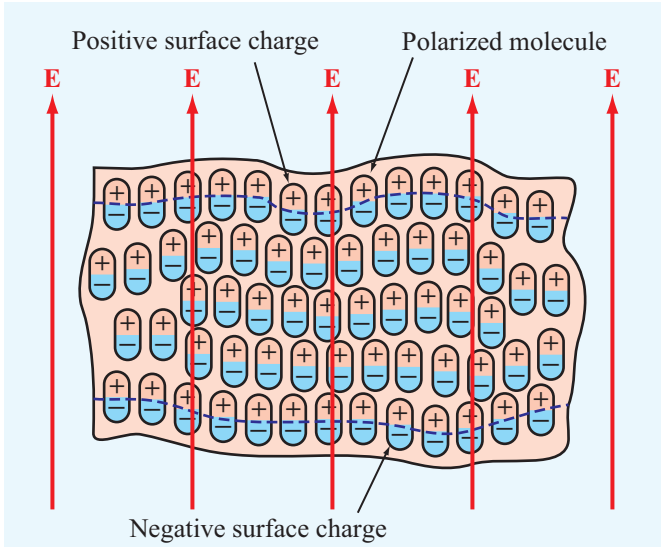


Figure 4-19 A dielectric medium polarized by an external electric field E .

dipole exhibits a dipole moment similar to that described in Example 4-9. Within a block of dielectric material subject to a uniform external field, the dipoles align themselves linearly, as shown in **Fig. 4-19**. Along the upper and lower edges of the material, the dipole arrangement exhibits positive and negative surface charge densities, respectively.

It is important to stress that this description applies to only nonpolar molecules that do not have permanent dipole moments. Nonpolar molecules become polarized only when an external electric field is applied; when the field is removed, the molecules return to their original unpolarized state.

In polar materials such as water, the molecules possess built-in **permanent dipole moments** that are randomly oriented in the absence of an applied electric field, and owing to their random orientations, the dipoles of polar materials produce no net macroscopic dipole moment (at the macroscopic scale, each point in the material represents a small volume containing thousands of molecules). Under the influence of an applied field, the permanent dipoles tend to align themselves along the direction of the electric field in a manner similar to that shown in **Fig. 4-19** for nonpolar materials.

4-7.1 Polarization Field

In free space $\mathbf{D} = \epsilon_0 \mathbf{E}$, the presence of microscopic dipoles in a dielectric material alters that relationship to

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (4.93)$$

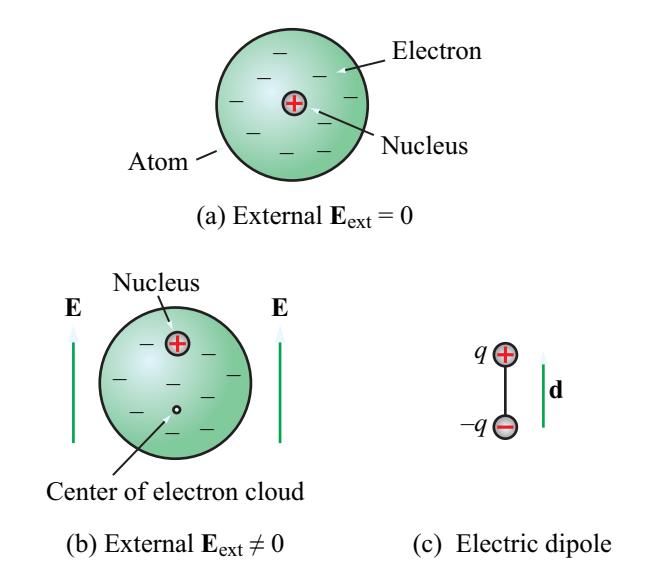


图4-18 在没有外部电场 E 的情况下，电子云的中心与原子核的中心位于同一位置，但是当施加电场时，两个中心相距距离 d 。

导体仅与原子弱结合，因此可以在材料中自由迁移，而在电介质中，它们与原子强结合。在没有电场的情况下，非极性分子中的电子在原子核周围形成对称的云，云的中心与原子核重合（图4-18（a））。带正电的原子核产生的电场吸引并保持其周围的电子云，相邻原子的电子云的相互排斥决定了其形状。当导体受到外部施加的电场时，每个原子中束缚最松的电子可以从一个原子跳到下一个原子，从而产生电流。然而，在电介质中，外部施加的电场 E 不能影响电荷的质量迁移，因为没有电荷能够自由移动。相反， E 会通过将电子云的中心移离原子核来极化材料中的原子或分子（图4-18(b)）。极化的原子或分子可以用电偶极子来表示，该电偶极子由原子核中的电荷 $+q$ 和电子云中心的 $-q$ 组成（图4-18（c））。每个这样的偶极子都会建立一个小电场，从带正电的原子核指向等量但带负电的电子云的中心。这种感应电场称为极化场，通常比 E 弱且方向相反。因此，介电材料中存在的净电场小于 E 。在微观层面上，每个

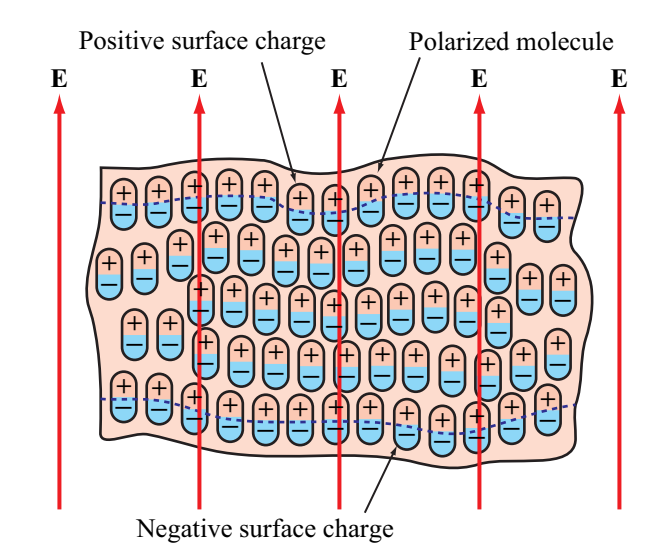


图 4-19 被外部电场 E 极化的介电介质

偶极子表现出类似于实施例4-9中所述的偶极矩。在一块受到均匀外场作用的介电材料内，偶极子会自行线性排列，如图4-19所示。沿着材料的上边缘和下边缘，偶极子排列分别表现出正表面电荷密度和负表面电荷密度。

需要强调的是，这种描述仅适用于不具有永久偶极矩的非极性分子。非极性分子只有在施加外部电场时才会极化；当电场移除时，分子恢复到原来的非极化状态。

在水等极性材料中，分子具有内置的永久偶极矩，在没有施加电场的情况下，这些偶极矩会随机定向，并且由于其随机定向，极性材料的偶极子不会产生净宏观偶极矩（在宏观尺度上，材料中的每个点代表包含数千个分子的小体积）。在外加电场的影响下，永久偶极子倾向于沿着电场方向排列，类似于图4-19所示的非极性材料。

4-7.1 偏振场

在自由空间 $\mathbf{D} = \epsilon_0 \mathbf{E}$ 中，介电材料中微观偶极子的存在改变了这种关系：

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (4.93)$$

where $\mathbf{P}$, called the *electric polarization field*, accounts for the polarization properties of the material. The polarization field is produced by the electric field $\mathbf{E}$ and depends on the material properties. A dielectric medium is said to be *linear* if the magnitude of the induced polarization field $\mathbf{P}$ is directly proportional to the magnitude of $\mathbf{E}$, and *isotropic* if $\mathbf{P}$ and $\mathbf{E}$ are in the same direction. Some crystals allow more polarization to take place along certain directions, such as the crystal axes, than along others. In such *anisotropic* dielectrics, $\mathbf{E}$ and $\mathbf{P}$ may have different directions. A medium is said to be *homogeneous* if its constitutive parameters (ϵ , μ , and σ) are constant throughout the medium. Our present treatment will be limited to media that are linear, isotropic, and homogeneous. For such media, $\mathbf{P}$ is directly proportional to $\mathbf{E}$ and is expressed as

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad (4.94)$$

where χ_e is called the *electric susceptibility* of the material. Inserting Eq. (4.94) into Eq. (4.93), we have

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}, \quad (4.95)$$

which defines the permittivity ϵ of the material as

$$\epsilon = \epsilon_0 (1 + \chi_e). \quad (4.96)$$

It is often convenient to characterize the permittivity of a material relative to that of free space, ϵ_0 ; this is accommodated by the relative permittivity $\epsilon_r = \epsilon / \epsilon_0$. Values of ϵ_r are listed in **Table 4-2** for a few common materials, and a longer list is given in Appendix B. In free space $\epsilon_r = 1$, and *for most conductors*, $\epsilon_r \approx 1$. The dielectric constant of air is approximately 1.0006 at sea level and decreases toward unity with increasing altitude. Except in some special circumstances, such as when calculating electromagnetic wave refraction (bending) through the atmosphere over long distances, *air can be treated as if it were free space*.

4-7.2 Dielectric Breakdown

The preceding dielectric–polarization model presumes that the magnitude of $\mathbf{E}$ does not exceed a certain critical value, which is known as the *dielectric strength* E_{ds} of the material. Beyond this, electrons will detach from the molecules and accelerate through the material in the form of a conduction current. When this happens, sparking can occur, and the dielectric material can sustain permanent damage due to electron collisions with the molecular structure. This abrupt change in behavior is called *dielectric breakdown*.

► The dielectric strength E_{ds} is the largest magnitude of $\mathbf{E}$ that the material can sustain without breakdown. ◀

Dielectric breakdown can occur in gases, liquids, and solids. The dielectric strength E_{ds} depends on the material composition, as well as other factors such as temperature and humidity. For air, E_{ds} is roughly 3 (MV/m); for glass, 25 to 40 (MV/m); and for mica, 200 (MV/m) (see **Table 4-2**).

A charged thundercloud at electric potential V relative to the ground induces an electric field $E = V/d$ in the air beneath it, where d is the height of the cloud base above the ground. If V is sufficiently large so that E exceeds the dielectric strength of air, ionization occurs and a lightning discharge follows. The *breakdown voltage* V_{br} of a parallel-plate capacitor is discussed in Example 4-12.

Example 4-12: Dielectric Breakdown

In a parallel-plate capacitor with a separation d between the conducting plates, the electric field E in the dielectric material

Table 4-2 Relative permittivity (dielectric constant) and dielectric strength of common materials.

Material	Relative Permittivity, ϵ_r	Dielectric Strength, E_{ds} (MV/m)
Air (at sea level)	1.0006	3
Petroleum oil	2.1	12
Polystyrene	2.6	20
Glass	4.5–10	25–40
Quartz	3.8–5	30
Bakelite	5	20
Mica	5.4–6	200

Note: $\epsilon = \epsilon_r \epsilon_0$ and $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

其中 $\mathbf{P}$, 称为电极化场, 说明了材料的极化特性。极化场由电场 $\mathbf{E}$ 产生, 取决于材料特性。如果感应极化场 $\mathbf{P}$ 的大小与 $\mathbf{E}$ 的大小成正比, 则称介电介质是线性的; 如果 $\mathbf{P}$ 和 $\mathbf{E}$ 方向相同, 则称介电介质是各向同性的。有些晶体允许沿着某些方向 (例如晶轴) 比沿着其他方向发生更多的偏振。在这种各向异性电介质中, $\mathbf{E}$ 和 $\mathbf{P}$ 可以具有不同的方向。如果介质的本构参数 (ϵ 、 μ 和 σ) 在整个介质中恒定, 则称该介质是均匀的。我们目前的处理仅限于线性、各向同性和均匀的介质。对于此类介质, $\mathbf{P}$ 与 $\mathbf{E}$ 成正比, 表示为

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad (4.94)$$

其中 χ_e 称为材料的电化率。插入方程式 (4.94) 代入等式 (4.93), 我们有

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}, \quad (4.95)$$

将材料的介电常数 ϵ 定义为

$$\epsilon = \epsilon_0 (1 + \chi_e). \quad (4.96)$$

通常可以方便地表征材料相对于自由空间的介电常数 ϵ_0 ; 这是通过相对介电常数 $\epsilon_r = \epsilon / \epsilon_0$ 来实现的。表 4-2 中列出了一些常见材料的 ϵ_r 值, 附录 B 中给出了更长的列表。在自由空间中 $\epsilon_r = 1$, 对于大多数导体, $\epsilon_r \approx 1$ 。空气的介电常数在海平面处约为 1.0006, 并随着高度的增加而减小到 1。除某些特殊情况外, 例如计算长距离穿过大气层的电磁波折射 (弯曲) 时, 空气可以视为自由空间。

4-7.2 介质击穿

前面的介电极化模型假设 $\mathbf{E}$ 的大小不超过某个临界值, 该临界值称为材料的介电强度 E_{ds} 。除此之外, 电子将从分子中分离出来, 并以传导电流的形式加速穿过材料。发生这种情况时, 可能会产生火花, 并且电介质材料可能会因电子与分子结构的碰撞而遭受永久性损坏。这种行为的突然变化称为介电击穿。

► 介电强度 E_{ds} 是材料在不击穿的情况下可以承受的最大 $\mathbf{E}$ 值。 ◀

气体、液体和固体中都可能发生电介质击穿。介电强度 E_{ds} 取决于材料成分以及温度和湿度等其他因素。对于空气, E_{ds} 大约为 3 (MV/m); 对于玻璃, 25至40 (MV/m); 云母为200 (MV/m) (见表4-2)。

相对于地面的电势为 V 的带电雷云会在其下方的空气中感应出电场 $E = V/d$, 其中 d 是云底距地面的高度。如果 V 足够大, 使得 E 超过空气的介电强度, 就会发生电离并随后发生闪电放电。示例 4-12 讨论了平行板电容器的击穿电压 V_{br} 。

例 4-12: 介电击穿

在导电板之间间距为 d 的平行板电容器中, 介电材料中的电场 $\mathbf{E}$

表4-2 常见材料的相对介电常数 (介电常数) 和介电强度。

Material	Relative Permittivity, ϵ_r	Dielectric Strength, E_{ds} (MV/m)
Air (at sea level)	1.0006	3
Petroleum oil	2.1	12
Polystyrene	2.6	20
Glass	4.5–10	25–40
Quartz	3.8–5	30
Bakelite	5	20
Mica	5.4–6	200

注: $\epsilon = \epsilon_r \epsilon_0$ 和 $\epsilon_0 = 8.854 \times 10^{-12}$ F/m。

separating the two plates is related to the voltage V between the two plates by

$$E = \frac{V}{d}.$$

The breakdown voltage V_{br} corresponds to the value of V at which $E = E_{\text{ds}}$, where E_{ds} is the dielectric strength of the material contained between the plates. That is,

$$V_{\text{br}} = E_{\text{ds}}d.$$

If V exceeds V_{br} , the electric charges will “spark” their way between the two plates.

A thin capacitor filled with quartz operates at 60 V. If $d = 0.01$ mm, what is the breakdown voltage, and how does it compare with the operating voltage?

Solution: From Table 4-2, $E_{\text{ds}} = 30 \times 10^6$ V/m for quartz. Hence, the breakdown voltage is

$$V_{\text{br}} = E_{\text{ds}}d = 30 \times 10^6 \times 10^{-5} = 300 \text{ V},$$

which is much higher than the operating voltage of 60 V. Therefore, the capacitor should experience no issues with dielectric breakdown.

Concept Question 4-19: What is a polar material? A nonpolar material?

Concept Question 4-20: Do $\mathbf{D}$ and $\mathbf{E}$ always point in the same direction? If not, when do they not?

Concept Question 4-21: What happens when dielectric breakdown occurs?

4-8 Electric Boundary Conditions

► A vector field is said to be *spatially continuous* if it does not exhibit abrupt changes in either magnitude or direction as a function of position. ◀

Even though the electric field may be continuous in adjoining dissimilar media, it may well be discontinuous at the boundary between them. Boundary conditions specify how the components of fields tangential and normal to an interface between two media relate across the interface Here we derive a general set of boundary conditions for $\mathbf{E}$, $\mathbf{D}$, and $\mathbf{J}$ that is applicable at the interface between *any* two dissimilar media—be they two dielectrics or a conductor and a dielectric. Of course, any of the dielectrics may be free space. Even though these boundary conditions are derived assuming electrostatic conditions, they remain valid for time-varying electric fields as well. Figure 4-20 shows an interface between medium 1 with permittivity ϵ_1 and medium 2 with permittivity ϵ_2 . In the general case, the interface may contain a surface charge density ρ_s (unrelated to the dielectric polarization charge density).

To derive the boundary conditions for the tangential components of $\mathbf{E}$ and $\mathbf{D}$, we consider the closed rectangular loop $abcd$ shown in Fig. 4-20 and apply the conservative property of the electric field expressed by Eq. (4.40), which states that the line integral of the electrostatic field around a closed path is always zero. By letting $\Delta h \rightarrow 0$, the contributions to the line integral by segments bc and da vanish. Hence,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{E}_1 \cdot \hat{\ell}_1 dl + \int_c^d \mathbf{E}_2 \cdot \hat{\ell}_2 dl = 0, \quad (4.97)$$

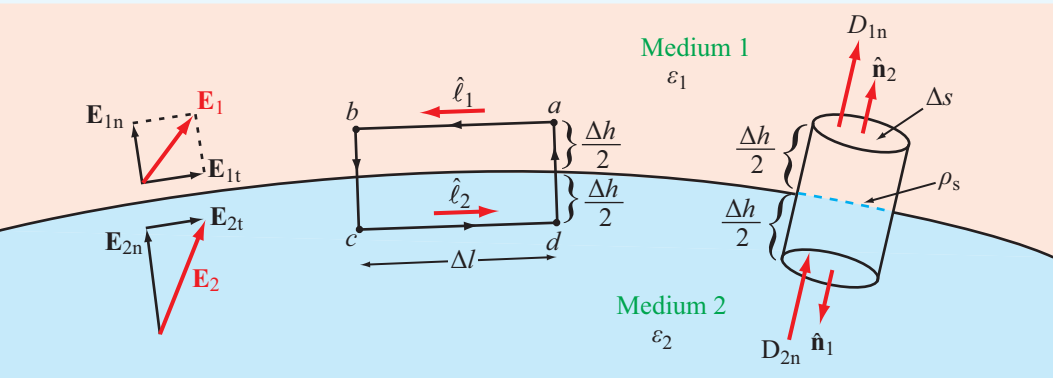


Figure 4-20 Interface between two dielectric media.

两板之间的距离与两板之间的电压 V 相关：

$$E = \frac{V}{d}.$$

击穿电压 V_{br} 对应于 $E = E_{\text{ds}}$ 时的 V 值，其中 E_{ds} 是板之间所含材料的介电强度。那是，

$$V_{\text{br}} = E_{\text{ds}}d.$$

如果 V 超过 V_{br} ，电荷将在两个板之间 “激发”。

填充石英的薄电容器在 60 V 下工作。如果 $d = 0.01$ m，击穿电压是多少？它与工作电压相比如何？

解：根据表 4-2，石英的 $E_{\text{ds}} = 30 \times 10^6$ V/m。因此，击穿电压为

$$V_{\text{br}} = E_{\text{ds}}d = 30 \times 10^6 \times 10^{-5} = 300 \text{ V},$$

该电压远高于 60 V 的工作电压。因此，电容器不应出现介电击穿问题。

概念问题 4-19：什么是极性材料？非极性材料？

概念问题 4-20：D 和 E 总是指向同一个方向吗？如果没有，什么时候没有？

概念问题 4-21：发生电介质击穿时会发生什么？

4-8 电边界条件

► 如果矢量场在大小或方向上不随位置发生突变，则称该矢量场是空间连续的。 ◀

尽管电场在相邻的不同介质中可能是连续的，但在它们之间的边界处很可能是不连续的。边界条件指定与两种介质之间的界面相切和法向的场分量如何在界面上相关。这里我们推导出 $\mathbf{E}$ 、 $\mathbf{D}$ 和 $\mathbf{J}$ 的一组通用边界条件，适用于任何两种不同介质（无论是两种电介质还是导体和电介质）之间的界面。当然，任何电介质都可以是自由空间。尽管这些边界条件是在假设静电条件的情况下得出的，但它们对于时变电场仍然有效。图 4-20 显示了介电常数为 ϵ_1 的介质 1 和介电常数为 ϵ_2 的介质 2 之间的界面。在一般情况下，界面可以包含与介电极化电荷密度无关的表面电荷密度 ρ_s 。

为了导出 $\mathbf{E}$ 和 $\mathbf{D}$ 切向分量的边界条件，我们考虑图 4-20 中所示的闭矩形环 $abcd$ ，并应用方程 1 表示的电场的保守性质。(4.40)，它指出闭合路径周围静电场的线积分始终为零。通过让 $\Delta h \rightarrow 0$ ，线段 bc 和 da 对线积分的贡献消失。因此，

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{E}_1 \cdot \hat{\ell}_1 dl + \int_c^d \mathbf{E}_2 \cdot \hat{\ell}_2 dl = 0, \quad (4.97)$$

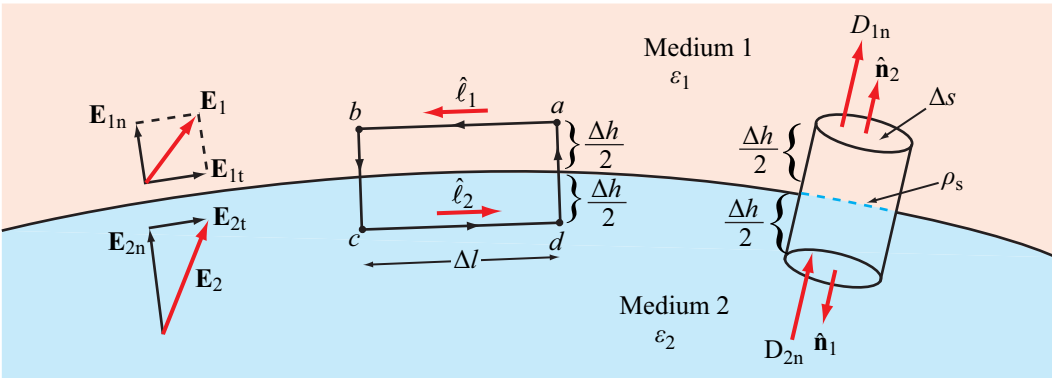


图 4-20 两种电介质之间的界面。

where $\hat{\ell}_1$ and $\hat{\ell}_2$ are unit vectors along segments ab and cd and $\mathbf{E}_1$ and $\mathbf{E}_2$ are the electric fields in media 1 and 2. Next, we decompose $\mathbf{E}_1$ and $\mathbf{E}_2$ into components tangential and normal to the boundary (**Fig. 4-20**),

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n}, \tag{4.98a}$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}. \tag{4.98b}$$

Noting that $\hat{\ell}_1 = -\hat{\ell}_2$, it follows that

$$(\mathbf{E}_1 - \mathbf{E}_2) \cdot \hat{\ell}_1 = 0. \tag{4.99}$$

In other words, the component of $\mathbf{E}_1$ along $\hat{\ell}_1$ equals that of $\mathbf{E}_2$ along $\hat{\ell}_1$, for all $\hat{\ell}_1$ tangential to the boundary. Hence,

$$\mathbf{E}_{1t} = \mathbf{E}_{2t} \quad (\text{V/m}). \tag{4.100}$$

► Thus, the tangential component of the electric field is **continuous** across the boundary between any two media. ◀

Upon decomposing $\mathbf{D}_1$ and $\mathbf{D}_2$ into tangential and normal components (in the manner of Eq. (4.98)) and noting that $\mathbf{D}_{1t} = \epsilon_1 \mathbf{E}_{1t}$ and $\mathbf{D}_{2t} = \epsilon_2 \mathbf{E}_{2t}$, the boundary condition on the tangential component of the electric flux density is

$$\frac{\mathbf{D}_{1t}}{\epsilon_1} = \frac{\mathbf{D}_{2t}}{\epsilon_2}. \tag{4.101}$$

Next, we apply Gauss’s law, as expressed by Eq. (4.29), to determine boundary conditions on the normal components of $\mathbf{E}$ and $\mathbf{D}$. According to Gauss’s law, the total outward flux of $\mathbf{D}$ through the three surfaces of the small cylinder shown in **Fig. 4-20** must equal the total charge enclosed in the cylinder. By letting the cylinder’s height $\Delta h \rightarrow 0$, the contribution to the total flux through the side surface goes to zero. Also, even if each of the two media happens to contain free charge densities, the only charge remaining in the collapsed cylinder is that distributed on the boundary. Thus, $Q = \rho_s \Delta s$, and

$$\begin{aligned} \oint_S \mathbf{D} \cdot d\mathbf{s} &= \int_{\text{top}} \mathbf{D}_1 \cdot \hat{\mathbf{n}}_2 \, ds + \int_{\text{bottom}} \mathbf{D}_2 \cdot \hat{\mathbf{n}}_1 \, ds \\ &= \rho_s \Delta s, \end{aligned} \tag{4.102}$$

where $\hat{\mathbf{n}}_1$ and $\hat{\mathbf{n}}_2$ are the outward normal unit vectors of the bottom and top surfaces, respectively. It is important to remember that *the normal unit vector at the surface of any*

medium is always defined to be in the outward direction away from that medium. Since $\hat{\mathbf{n}}_1 = -\hat{\mathbf{n}}_2$, Eq. (4.102) simplifies to

$$\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \quad (\text{C/m}^2). \tag{4.103}$$

If D_{1n} and D_{2n} denote as the normal components of $\mathbf{D}_1$ and $\mathbf{D}_2$ along $\hat{\mathbf{n}}_2$, we have

$$D_{1n} - D_{2n} = \rho_s \quad (\text{C/m}^2). \tag{4.104}$$

► The normal component of $\mathbf{D}$ changes abruptly at a charged boundary between two different media in an amount equal to the surface charge density. If no charge exists at the boundary, then D_n is continuous across the boundary. ◀

The corresponding boundary condition for $\mathbf{E}$ is

$$\hat{\mathbf{n}}_2 \cdot (\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2) = \rho_s, \tag{4.105a}$$

or equivalently

$$\epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s. \tag{4.105b}$$

In summary, (1) the conservative property of $\mathbf{E}$,

$$\nabla \times \mathbf{E} = 0 \quad \longleftrightarrow \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \tag{4.106}$$

led to the result that $\mathbf{E}$ has a continuous tangential component across a boundary, and (2) the divergence property of $\mathbf{D}$,

$$\nabla \cdot \mathbf{D} = \rho_v \quad \longleftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q, \tag{4.107}$$

led to the result that the normal component of $\mathbf{D}$ changes by ρ_s across the boundary. A summary of the conditions that apply at the boundary between different types of media is given in **Table 4-3**.

其中 $\hat{\ell}_1$ 和 $\hat{\ell}_2$ 是沿线段ab和cd的单位向量， $\mathbf{E}_1$ 和 $\mathbf{E}_2$ 是介质1和2中的电场。接下来，我们将 $\mathbf{E}_1$ 和 $\mathbf{E}_2$ 分解为与边界相切和法向的分量（图4-20），

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n}, \tag{4.98a}$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}. \tag{4.98b}$$

注意到 $\hat{\ell}_1 = -\hat{\ell}_2$ ，因此

$$(\mathbf{E}_1 - \mathbf{E}_2) \cdot \hat{\ell}_1 = 0. \tag{4.99}$$

换句话说，对于所有与边界相切的 $\hat{\ell}_1$ ， $\mathbf{E}_1$ 沿 $\hat{\ell}_1$ 的分量等于 $\mathbf{E}_2$ 沿 $\hat{\ell}_1$ 的分量。因此，

$$\mathbf{E}_{1t} = \mathbf{E}_{2t} \quad (\text{V/m}). \tag{4.100}$$

► 因此，电场的切向分量在任意两种介质之间的边界上是连续的。 ◀

将 $\mathbf{D}_1$ 和 $\mathbf{D}_2$ 分解为切向分量和法向分量（按照方程（4.98））并注意到 $\mathbf{D}_{1t} = \epsilon_1 \mathbf{E}_{1t}$ 和 $\mathbf{D}_{2t} = \epsilon_2 \mathbf{E}_{2t}$ ，电通量密度切向分量的边界条件为

$$\frac{\mathbf{D}_{1t}}{\epsilon_1} = \frac{\mathbf{D}_{2t}}{\epsilon_2}. \tag{4.101}$$

接下来，我们应用高斯定律，如方程所示。（4.29），以确定 $\mathbf{E}$ 和 $\mathbf{D}$ 法向分量的边界条件。根据高斯定律， $\mathbf{D}$ 通过图 4-20 所示小圆柱体三个表面的总向外通量必须等于圆柱体中封闭的总电荷。通过让圆柱体的高度 $\Delta h \rightarrow 0$ ，对通过侧面的总通量的贡献变为零。此外，即使两种介质中的每一种碰巧都含有自由电荷密度，塌缩圆柱体中剩余的唯一电荷也是分布在边界上的电荷。因此， $Q = \rho_s \Delta s$ ，并且

$$\begin{aligned} \oint_S \mathbf{D} \cdot d\mathbf{s} &= \int_{\text{top}} \mathbf{D}_1 \cdot \hat{\mathbf{n}}_2 \, ds + \int_{\text{bottom}} \mathbf{D}_2 \cdot \hat{\mathbf{n}}_1 \, ds \\ &= \rho_s \Delta s, \end{aligned} \tag{4.102}$$

其中 $\hat{\mathbf{n}}_1$ 和 $\hat{\mathbf{n}}_2$ 分别是底面和顶面的向外法线单位向量。重要的是要记住，任何物体表面的法向单位向量

介质始终被定义为远离该介质的向外方向。由于 $\hat{\mathbf{n}}_1 = -\hat{\mathbf{n}}_2$ ，等式：(4.102) 简化为

$$\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \quad (\text{C/m}^2). \tag{4.103}$$

如果 D_{1n} 和 D_{2n} 表示为 $\mathbf{D}_1$ 和 $\mathbf{D}_2$ 沿 $\hat{\mathbf{n}}_2$ 的法向分量，我们有

$$D_{1n} - D_{2n} = \rho_s \quad (\text{C/m}^2). \tag{4.104}$$

► $\mathbf{D}$ 的法向分量在两种不同介质之间的带电边界处突然变化，其变化量等于表面电荷密度。如果边界处不存在电荷，则 D_n 在边界上连续。 ◀

$\mathbf{E}$ 对应的边界条件为

$$\hat{\mathbf{n}}_2 \cdot (\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2) = \rho_s, \tag{4.105a}$$

或同等地

$$\epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s. \tag{4.105b}$$

综上所述，(1) $\mathbf{E}$ 的保守性质，

$$\nabla \times \mathbf{E} = 0 \quad \longleftrightarrow \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \tag{4.106}$$

导致 $\mathbf{E}$ 具有跨越边界的连续切向分量，以及 (2) $\mathbf{D}$ 的发散性质，

$$\nabla \cdot \mathbf{D} = \rho_v \quad \longleftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q, \tag{4.107}$$

导致 $\mathbf{D}$ 的法向分量越过边界改变了 ρ_s 。表 4-3 总结了适用于不同类型介质之间边界的条件。

Table 4-3 Boundary conditions for the electric fields.

Field Component	Any Two Media	Medium 1 Dielectric ϵ_1	Medium 2 Conductor
Tangential $\mathbf{E}$	$\mathbf{E}_{1t} = \mathbf{E}_{2t}$	$\mathbf{E}_{1t} = \mathbf{E}_{2t} = 0$	
Tangential $\mathbf{D}$	$\mathbf{D}_{1t}/\epsilon_1 = \mathbf{D}_{2t}/\epsilon_2$	$\mathbf{D}_{1t} = \mathbf{D}_{2t} = 0$	
Normal $\mathbf{E}$	$\epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s$	$E_{1n} = \rho_s/\epsilon_1$	$E_{2n} = 0$
Normal $\mathbf{D}$	$D_{1n} - D_{2n} = \rho_s$	$D_{1n} = \rho_s$	$D_{2n} = 0$
Notes: (1) ρ_s is the surface charge density at the boundary; (2) normal components of $\mathbf{E}_1$, $\mathbf{D}_1$, $\mathbf{E}_2$, and $\mathbf{D}_2$ are along $\hat{\mathbf{n}}_2$, which is the outward normal unit vector of medium 2.			

Example 4-13: Application of Boundary Conditions

The x - y plane is a charge-free boundary separating two dielectric media with permittivities ϵ_1 and ϵ_2 , as shown in Fig. 4-21. If the electric field in medium 1 is

$$E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}}E_{1z},$$

and $\mathbf{E}_1 = \hat{\mathbf{x}}$, find (a) the electric field $\mathbf{E}_2$ in medium 2 and (b) the angles θ_1 and θ_2 .

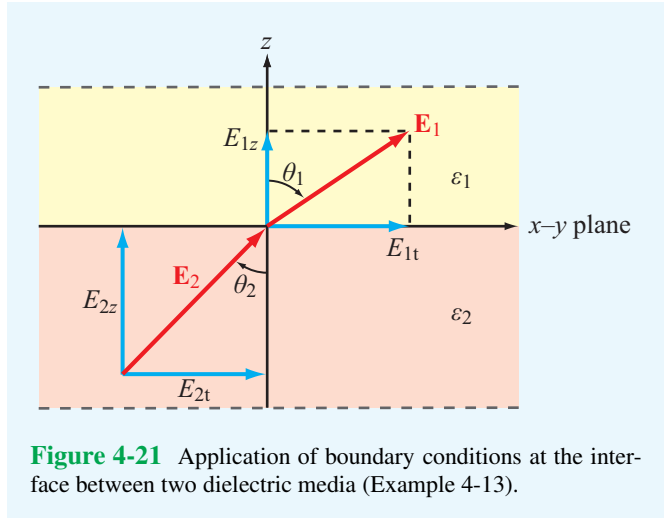


Figure 4-21 Application of boundary conditions at the interface between two dielectric media (Example 4-13).

Solution: (a) Let $\mathbf{E}_2 = \hat{\mathbf{x}}E_{2x} + \hat{\mathbf{y}}E_{2y} + \hat{\mathbf{z}}E_{2z}$. Our task is to find the components of $\mathbf{E}_2$ in terms of the given components

of $\mathbf{E}_1$. The normal to the boundary is $\hat{\mathbf{z}}$. Hence, the x and y components of the fields are tangential to the boundary and the z components are normal to the boundary. At a charge-free interface, the tangential components of $\mathbf{E}$ and the normal components of $\mathbf{D}$ are continuous. Consequently,

$$E_{2x} = E_{1x}, \quad E_{2y} = E_{1y},$$

and

$$D_{2z} = D_{1z} \quad \text{or} \quad \epsilon_2 E_{2z} = \epsilon_1 E_{1z}.$$

Hence,

$$\mathbf{E}_2 = \hat{\mathbf{x}}E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}} \frac{\epsilon_1}{\epsilon_2} E_{1z}. \quad (4.108)$$

(b) The tangential components of $\mathbf{E}_1$ and $\mathbf{E}_2$ are

$$E_{1t} = \sqrt{E_{1x}^2 + E_{1y}^2} \quad \text{and} \quad E_{2t} = \sqrt{E_{2x}^2 + E_{2y}^2}.$$

The angles θ_1 and θ_2 are then given by

$$\tan \theta_1 = \frac{E_{1t}}{E_{1z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{E_{1z}},$$

$$\tan \theta_2 = \frac{E_{2t}}{E_{2z}} = \frac{\sqrt{E_{2x}^2 + E_{2y}^2}}{E_{2z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{(\epsilon_1/\epsilon_2)E_{1z}},$$

and the two angles are related by

$$\frac{\tan \theta_2}{\tan \theta_1} = \frac{\epsilon_2}{\epsilon_1}. \quad (4.109)$$

表 4-3 电场的边界条件。

Field Component	Any Two Media	Medium 1 Dielectric ϵ_1	Medium 2 Conductor
Tangential $\mathbf{E}$	$\mathbf{E}_{1t} = \mathbf{E}_{2t}$	$\mathbf{E}_{1t} = \mathbf{E}_{2t} = 0$	
Tangential $\mathbf{D}$	$\mathbf{D}_{1t}/\epsilon_1 = \mathbf{D}_{2t}/\epsilon_2$	$\mathbf{D}_{1t} = \mathbf{D}_{2t} = 0$	
Normal $\mathbf{E}$	$\epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s$	$E_{1n} = \rho_s/\epsilon_1$	$E_{2n} = 0$
Normal $\mathbf{D}$	$D_{1n} - D_{2n} = \rho_s$	$D_{1n} = \rho_s$	$D_{2n} = 0$

Notes: (1) ρ_s is the surface charge density at the boundary; (2) normal components of $\mathbf{E}_1$, $\mathbf{D}_1$, $\mathbf{E}_2$, and $\mathbf{D}_2$ are along $\hat{\mathbf{n}}_2$, which is the outward normal unit vector of medium 2.

例 4-13：边界条件的应用

x - y 平面是一个无电荷边界，分隔了介电常数为 ϵ_1 和 ϵ_2 的两种介电介质，如图 4-21 所示。如果介质 1 中的电场为

$$E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}}E_{1z},$$

和 $\mathbf{E}_1 = \hat{\mathbf{x}}$ ，求 (a) 介质 2 中的电场 $\mathbf{E}_2$ 和 (b) 角度 θ_1 和 θ_2 。

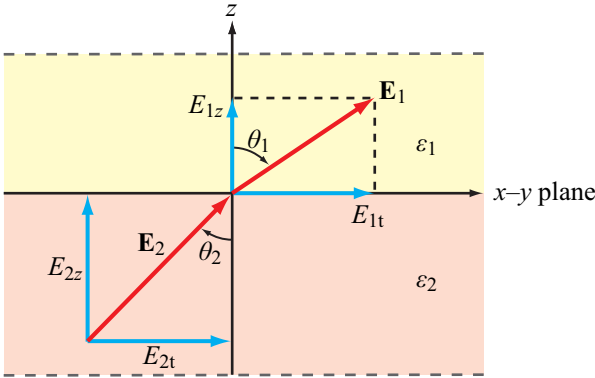


图 4-21 在两种介电介质之间的界面处应用边界条件 (示例 4-13)。

解：(a) 令 $\mathbf{E}_2 = \hat{\mathbf{x}}E_{2x} + \hat{\mathbf{y}}E_{2y} + \hat{\mathbf{z}}E_{2z}$ 。我们的任务是根据给定的分量找到 $\mathbf{E}_2$ 的分量

E_1 。边界法线为 $\hat{\mathbf{z}}$ 。因此，场的 x 和 y 分量与边界相切，而 z 分量与边界垂直。在无电荷界面处， $\mathbf{E}$ 的切向分量和 $\mathbf{D}$ 的法向分量是连续的。最后，

$$E_{2x} = E_{1x}, \quad E_{2y} = E_{1y},$$

和

$$D_{2z} = D_{1z} \quad \text{or} \quad \epsilon_2 E_{2z} = \epsilon_1 E_{1z}.$$

因此，

$$\mathbf{E}_2 = \hat{\mathbf{x}}E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}} \frac{\epsilon_1}{\epsilon_2} E_{1z}. \quad (4.108)$$

(b) $\mathbf{E}_1$ 和 $\mathbf{E}_2$ 的切向分量为

$$E_{1t} = \sqrt{E_{1x}^2 + E_{1y}^2} \quad \text{and} \quad E_{2t} = \sqrt{E_{2x}^2 + E_{2y}^2}.$$

角度 θ_1 和 θ_2 由下式给出

$$\tan \theta_1 = \frac{E_{1t}}{E_{1z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{E_{1z}},$$

$$\tan \theta_2 = \frac{E_{2t}}{E_{2z}} = \frac{\sqrt{E_{2x}^2 + E_{2y}^2}}{E_{2z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{(\epsilon_1/\epsilon_2)E_{1z}},$$

两个角度的关系为

$$\frac{\tan \theta_2}{\tan \theta_1} = \frac{\epsilon_2}{\epsilon_1}. \quad (4.109)$$

Technology Brief 9:
Capacitive Sensors

To sense is to respond to a stimulus. (See Technology Brief 7 on resistive sensors.) A capacitor can function as a sensor if the stimulus changes the capacitor's **geometry**—usually the spacing between its conductive elements—or the effective **dielectric properties** of the insulating material situated between them. Capacitive sensors are used in a multitude of applications. A few examples follow.

Fluid Gauge

The two metal electrodes in (Fig. TF9-1(a)), usually rods or plates, form a capacitor whose capacitance is directly proportional to the **permittivity** of the material between them. If the fluid section is of height h_f and the height of the empty space above it is $(h - h_f)$, then the overall capacitance is equivalent to two capacitors in parallel, or

$$C = C_f + C_a = \epsilon_f w \frac{h_f}{d} + \epsilon_a w \frac{(h - h_f)}{d},$$

where w is the electrode plate width, d is the spacing between electrodes, and ϵ_f and ϵ_a are the permittivities of the fluid and air, respectively. Rearranging the expression as a linear equation yields

$$C = kh_f + C_0,$$

where the constant coefficient is $k = (\epsilon_f - \epsilon_a)w/d$ and $C_0 = \epsilon_a wh/d$ is the capacitance of the tank when totally empty. Using the linear equation, the fluid height can be determined by measuring C with a bridge circuit (Fig. TF9-1(b)).

► The output voltage V_{out} assumes a functional form that depends on the source voltage v_g , the capacitance C_0 of the empty tank, and the unknown fluid height h_f . ◀

Humidity Sensor

Thin-film metal electrodes shaped in an **interdigitized pattern** (to enhance the ratio A/d) are fabricated on a silicon substrate (Fig. TF9-2). The spacing between digits is typically on the order of $0.2 \mu\text{m}$. The **effective permittivity of the material separating the electrodes varies with the relative humidity of the surrounding environment**. Hence, the capacitor becomes a humidity sensor.

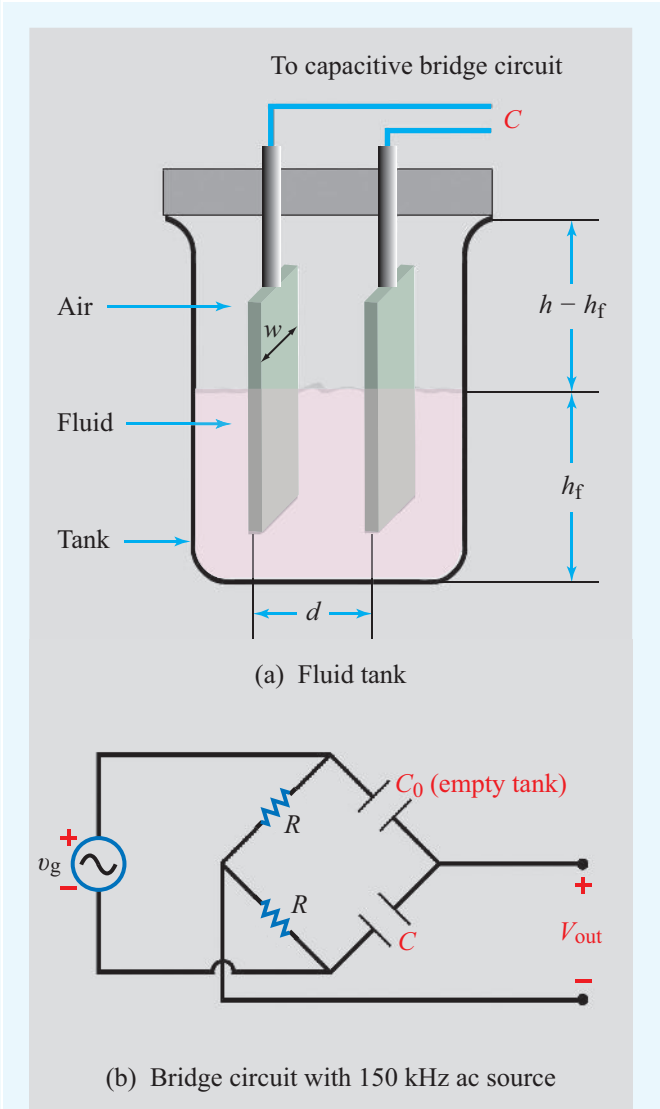


Figure TF9-1 Fluid gauge and associated bridge circuit with C_0 being the capacitance that an empty tank would have and C the capacitance of the tank under test.

Pressure Sensor

A flexible metal **diaphragm** separates an oil-filled chamber with reference pressure P_0 from a second chamber exposed to the gas or fluid whose pressure P is to be measured by the sensor (Fig. TF9-3(a)). The membrane is sandwiched—but electrically isolated—between two conductive parallel surfaces, forming two capacitors in series Fig. TF9-3(b). When $P > P_0$, the membrane bends

技术简介 9: 电容
式传感器

感知就是对刺激做出反应。(请参阅有关电阻传感器的技术简介 7。)如果刺激改变了电容器的几何形状(通常是其导电元件之间的间距)或位于它们之间的绝缘材料的有效介电特性,则电容器可以充当传感器。电容式传感器有多种应用。下面是一些例子。

液位计

(图TF9-1 (a)) 中的两个金属电极(通常是棒或板)形成一个电容器,其电容与它们之间材料的介电常数成正比。如果流体部分的高度为 h_f , 其上方空白空间的高度为 $(h - h_f)$, 则总电容相当于两个并联电容器, 或者

$$C = C_f + C_a = \epsilon_f w \frac{h_f}{d} + \epsilon_a w \frac{(h - h_f)}{d},$$

其中 w 是电极板宽度, d 是电极之间的间距, ϵ_f 和 ϵ_a 分别是流体和空气的介电常数。将表达式重新排列为线性方程得出

$$C = kh_f + C_0,$$

其中常数系数为 $k = (\epsilon_f - \epsilon_a)w/d$, $C_0 = \epsilon_a wh/d$ 为水箱完全排空时的电容。利用线性方程, 可以通过用电桥电路测量 C 来确定流体高度 (图TF9-1 (b))。

► 输出电压 V_{out} 呈现出取决于源电压 v_g 、空罐电容 C_0 和未知流体高度 h_f 的函数形式。 ◀

湿度传感器

在硅基板上制造了呈交叉指状图案形状的薄膜金属电极(以提高比率 A/d) (图 TF9-2)。数字之间的间距通常约为 $0.2 \mu\text{m}$ 。分隔电极的材料的有效介电常数随周围环境的相对湿度而变化。因此, 电容器成为湿度传感器。

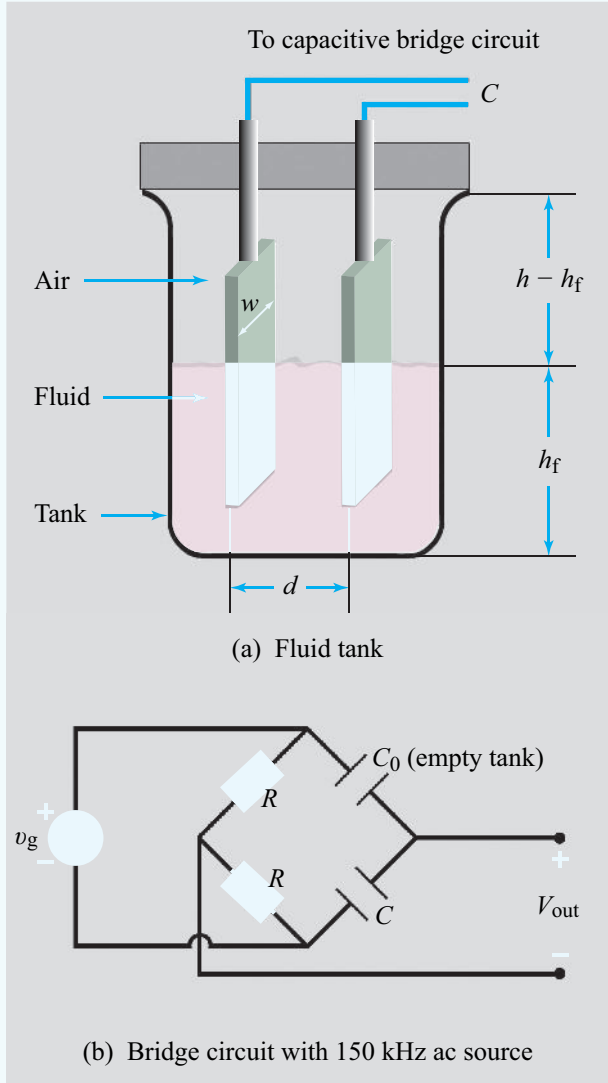
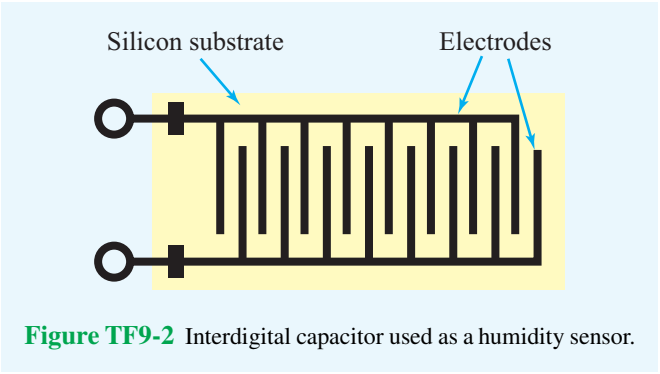


图 TF9-1 液位计和相关电桥电路, 其中 C_0 是空罐具有的电容, C 是被测罐的电容。

压力传感器

柔性金属隔膜将具有参考压力 P_0 的充油室与暴露于气体或流体的第二个室分开, 其压力 P 由传感器测量 (图 TF 9-3(a))。该膜夹在两个导电平行表面之间(但电隔离), 形成两个串联电容器 (图 TF9-3 (b))。当 $P > P_0$ 时, 膜弯曲



in the direction of the lower plate. Consequently, d_1 increases and d_2 decreases, and in turn, C_1 decreases and C_2 increases (Fig. TF9-3(c)). The converse happens when $P < P_0$. With the use of a capacitance bridge circuit, such as the one in Fig. TF9-1(b), the sensor can be calibrated to measure the pressure P with good precision.

Noncontact Sensors

Precision positioning is a critical ingredient in semiconductor device fabrication, as well as in the operation and control of many mechanical systems. **Noncontact capacitive sensors** are used to sense the position of silicon wafers during the deposition, etching, and cutting processes, without coming in direct contact with the wafers.

► Noncontact sensors are also used to sense and control robot arms in equipment manufacturing and to position hard disc drives, photocopier rollers, printing presses, and other similar systems. ◀

The concentric plate capacitor in Fig. TF9-4 consists of two metal plates sharing the same plane but electrically isolated from each other by an insulating material. When connected to a voltage source, charges of opposite polarity form on the two plates, resulting in the creation of electric field lines between them. The same principle applies to the adjacent plate's capacitor in Fig. TF9-5. In both cases, the capacitance is determined by the shapes and sizes of the conductive elements and by the effective permittivity of the dielectric medium containing the electric field lines between them. Often, the capacitor surface is covered by a thin film of nonconductive material, the

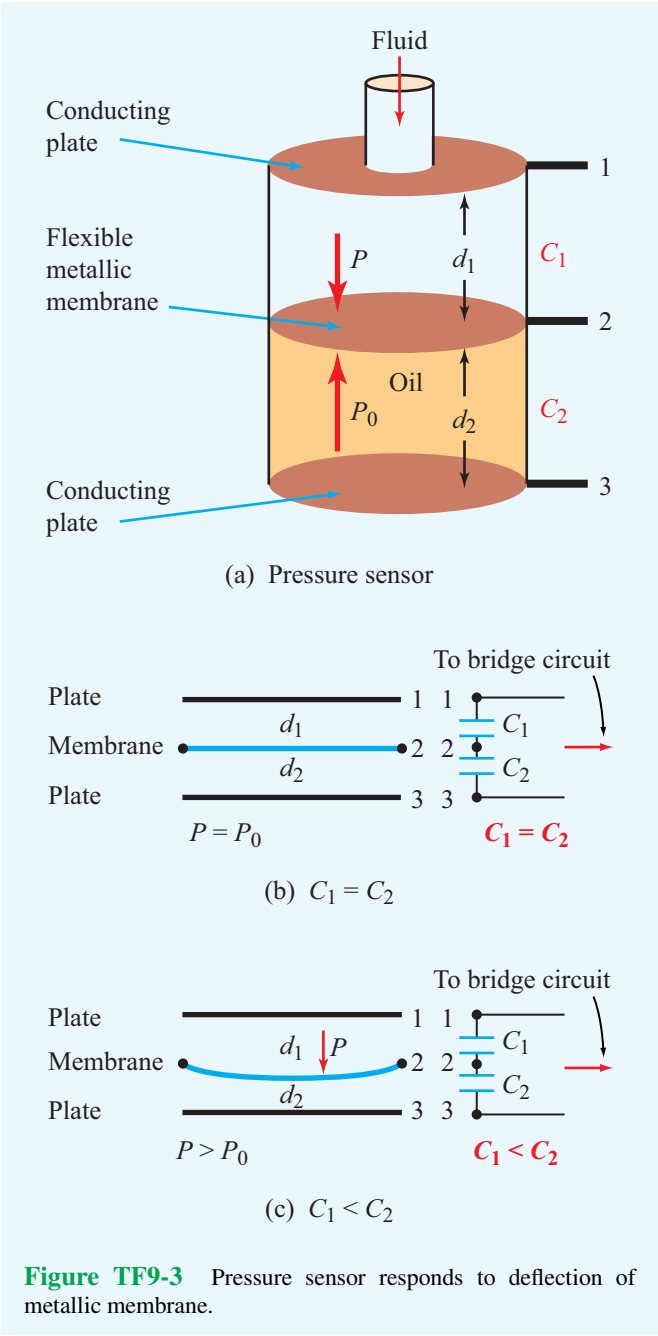
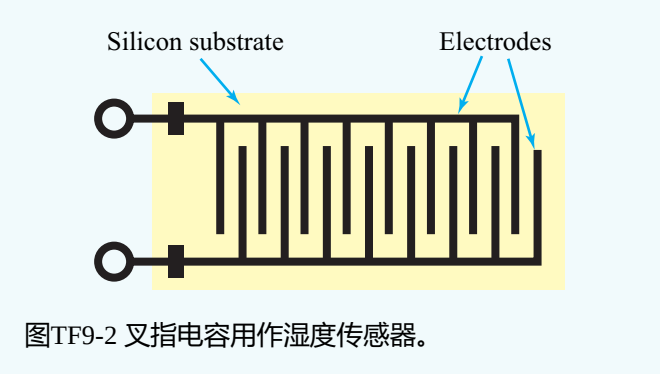


Figure TF9-3 Pressure sensor responds to deflection of metallic membrane.



沿下板的方向。因此， d_1 增加， d_2 减少，反过来， C_1 减少， C_2 增加（图 TF9-3(c)）。当 $P < P_0$ 时，情况相反。通过使用电容桥电路，如图TF9-1(b)所示，可以对传感器进行校准，以高精度测量压力 P 。

非接触式传感器

精确定位是半导体器件制造以及许多机械系统的操作和控制中的关键因素。非接触式电容传感器用于在沉积、蚀刻和切割过程中感测硅晶圆的位置，而无需与晶圆直接接触。

► 非接触式传感器还用于感测和控制设备制造中的机器人手臂，以及定位硬盘驱动器、复印机滚筒、印刷机和其他类似系统。 ◀

图TF9-4中的同心板电容器由共享同一平面但通过绝缘材料彼此电隔离的两个金属板组成。当连接到电压源时，两个极板上会形成相反极性的电荷，从而在它们之间产生电场线。同样的原理也适用于图 TF9-5 中相邻板的电容器。在这两种情况下，电容都由导电元件的形状和尺寸以及导电元件之间包含电场线的介电介质的有效介电常数决定。通常，电容器表面覆盖有一层非导电材料薄膜，

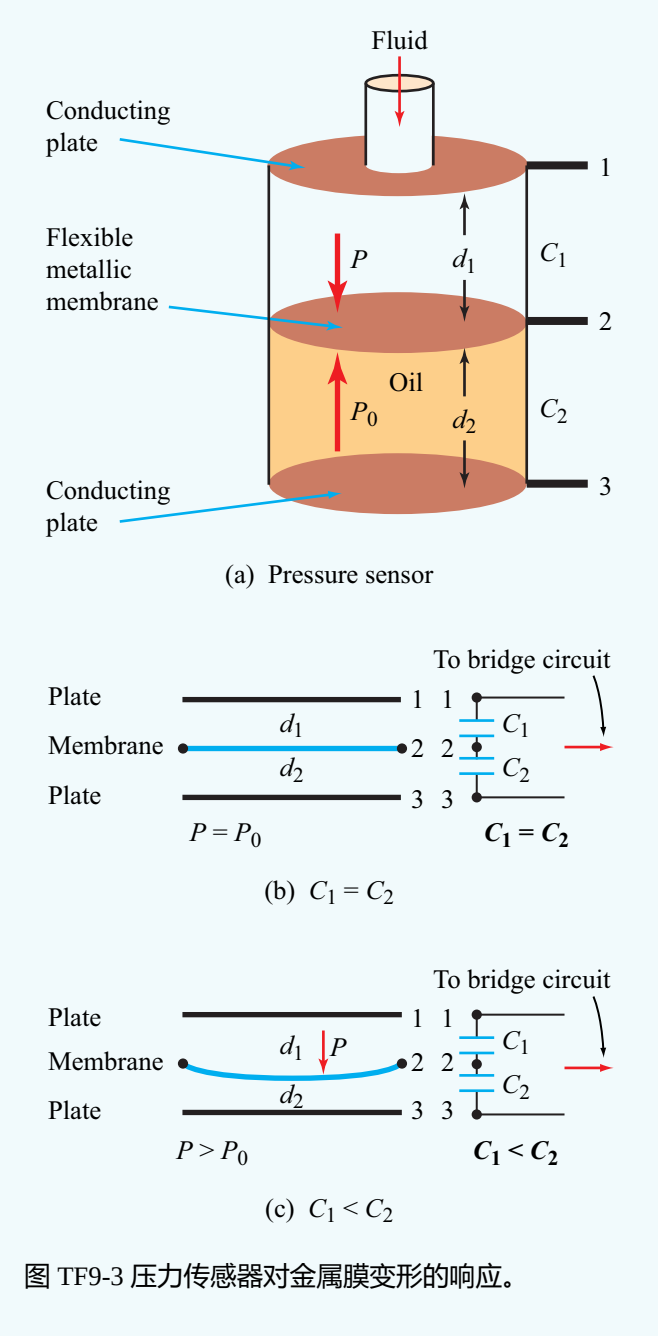


图 TF9-3 压力传感器对金属膜变形的响应。

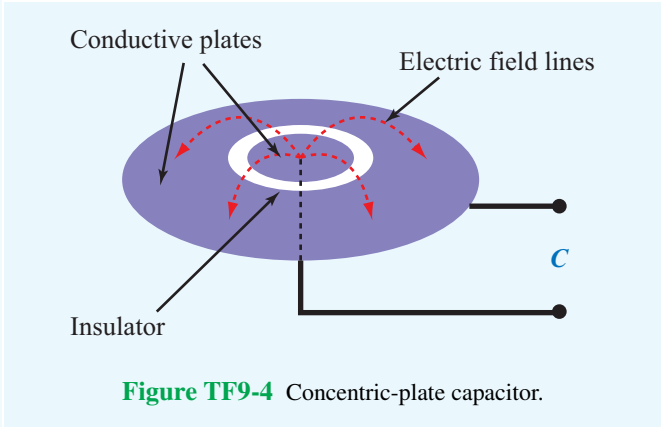


Figure TF9-4 Concentric-plate capacitor.

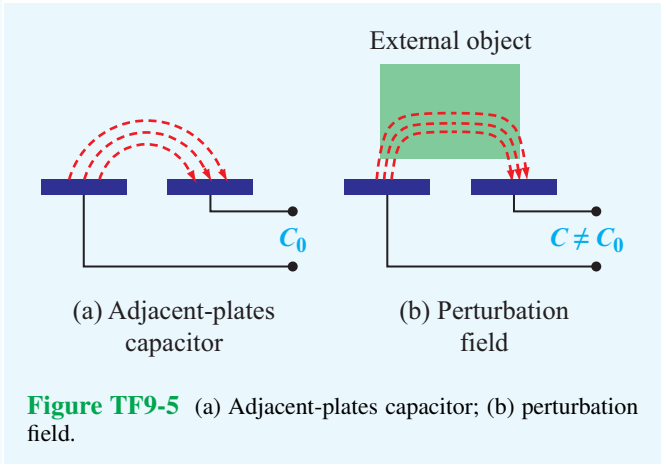


Figure TF9-5 (a) Adjacent-plates capacitor; (b) perturbation field.

purpose of which is to keep the plate surfaces clean and dust free.

► The introduction of an external object into the proximity of the capacitor (Fig. TF9-5(b)) changes the effective permittivity of the medium, perturbs the electric field lines, and modifies the charge distribution on the plates. ◀

This, in turn, changes the value of the capacitance as it would be measured by a capacitance meter or **bridge circuit**. Hence, the capacitor becomes a **proximity sensor**, and its sensitivity depends, in part, on how different the permittivity of the external object is from that of the unperturbed medium and on whether it is or is not made of a conductive material.

Fingerprint Imager

An interesting extension of noncontact capacitive sensors is the development of a fingerprint imager consisting of a two-dimensional array of capacitive sensor cells constructed to record an electrical representation of a fingerprint (Fig. TF9-6). Each sensor cell is composed of an adjacent-plates capacitor connected to a capacitance measurement circuit (Fig. TF9-7). The entire surface of the imager is covered by a thin layer of nonconductive oxide. When the finger is placed on the oxide surface, it perturbs the field lines of the individual sensor cells to varying degrees, depending on the distance between the ridges and valleys of the finger's surface from the sensor cells.

► Given that the dimensions of an individual sensor are on the order of 65 μm on the side, the imager is capable of recording a fingerprint image at a resolution corresponding to 400 dots per inch or better. ◀

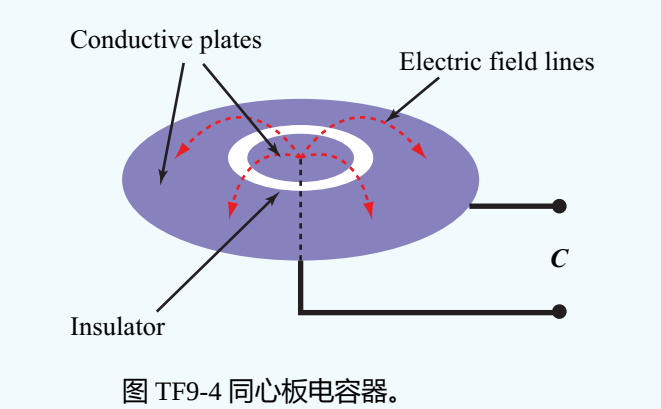
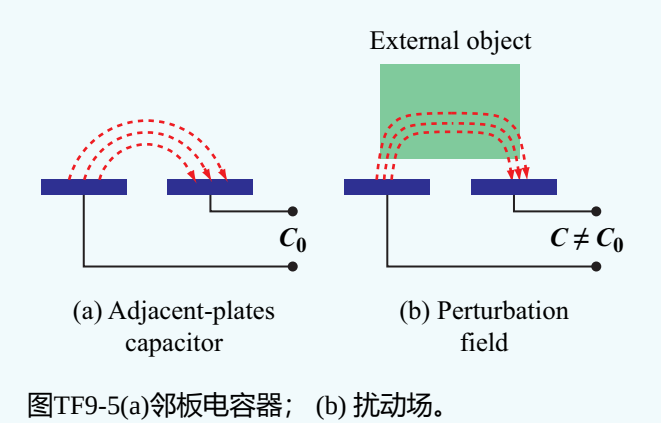


图 TF9-4 同心板电容器。



图TF9-5(a)邻板电容器；(b)扰动场。

其目的是保持板材表面清洁无灰尘。

► 将外部物体引入电容器附近（图TF9-5 (b) ）会改变介质的有效介电常数，扰乱电场线，并改变板上的电荷分布。 ◀

这反过来又改变了电容值，因为它是由电容表或电桥电路测量的。因此，电容器成为接近传感器，其灵敏度部分取决于外部物体的介电常数与未受干扰介质的介电常数的差异程度以及它是否由导电材料制成。

指纹成像仪

非接触式电容传感器的一个有趣的扩展是指纹成像仪的开发，该成像仪由二维电容传感器单元阵列组成，用于记录指纹的电气表示（图 TF9-6）。每个传感器单元由连接到电容测量电路的相邻板电容器组成（图 TF9-7）。成像仪的整个表面都覆盖有一层薄薄的非导电氧化物。当手指放在氧化物表面上时，它会不同程度地扰动各个传感器单元的磁力线，具体取决于手指表面的脊和谷与传感器单元之间的距离。

► 鉴于单个传感器的侧面尺寸约为 65 μm，成像器能够以相当于每英寸 400 点或更高的分辨率记录指纹图像。 ◀



Figure TF9-6 Elements of a fingerprint matching system. (Courtesy of IEEE Spectrum.)



图 TF9-6 指纹匹配系统的要素。（由 IEEE Spectrum 提供。）



Figure TF9-7 Fingerprint representation.



图 TF9-7 指纹表示。

Exercise 4-16: Find $\mathbf{E}_1$ in Fig. 4-21 if

$$\begin{aligned}\mathbf{E}_2 &= \hat{x}2 - \hat{y}3 + \hat{z}3 \text{ (V/m)}, \\ \epsilon_1 &= 2\epsilon_0, \\ \epsilon_2 &= 8\epsilon_0,\end{aligned}$$

and the boundary is charge-free.

Answer: $\mathbf{E}_1 = \hat{x}2 - \hat{y}3 + \hat{z}12 \text{ (V/m)}$. (See [EM](#).)

Exercise 4-17: Repeat Exercise 4.16 for a boundary with surface charge density $\rho_s = 3.54 \times 10^{-11} \text{ (C/m}^2\text{)}$.

Answer: $\mathbf{E}_1 = \hat{x}2 - \hat{y}3 + \hat{z}14 \text{ (V/m)}$. (See [EM](#).)

These two boundary conditions can be combined into

$$\mathbf{D}_1 = \epsilon_1 \mathbf{E}_1 = \hat{n} \rho_s, \quad (4.111)$$

(at conductor surface)

where $\hat{n}$ is a unit vector directed normally outward from the conducting surface.

► The electric field lines point directly away from the conductor surface when ρ_s is positive and directly toward the conductor surface when ρ_s is negative. ◀

Figure 4-22 shows an infinitely long conducting slab placed in a uniform electric field $\mathbf{E}_1$. The media above and below the slab have permittivity ϵ_1 . Because $\mathbf{E}_1$ points away from the upper surface, it induces a positive charge density $\rho_s = \epsilon_1 |\mathbf{E}_1|$ on the upper slab surface. On the bottom surface, $\mathbf{E}_1$ points toward the surface; therefore, the induced charge density is $-\rho_s$. The presence of these surface charges induces an electric field $\mathbf{E}_i$ in the conductor, resulting in a total field $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_i$. To satisfy the condition that $\mathbf{E}$ must be everywhere zero in the conductor, $\mathbf{E}_i$ must equal $-\mathbf{E}_1$.

If we place a metallic sphere in an electrostatic field (**Fig. 4-23**), positive and negative charges accumulate on the upper and lower hemispheres, respectively. The presence of the sphere causes the field lines to bend to satisfy the condition expressed by Eq. (4.111); that is, **$\mathbf{E}$ is always normal to a conductor boundary**.

$$E_{1t} = D_{1t} = 0, \quad (4.110a)$$

$$D_{1n} = \epsilon_1 E_{1n} = \rho_s. \quad (4.110b)$$

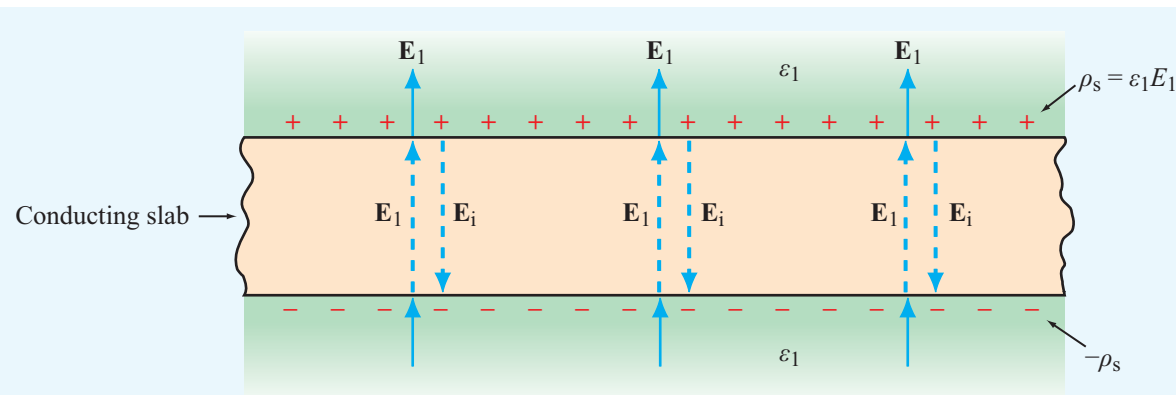


Figure 4-22 When a conducting slab is placed in an external electric field $\mathbf{E}_1$, charges that accumulate on the conductor surfaces induce an internal electric field $\mathbf{E}_i = -\mathbf{E}_1$. Consequently, the total field inside the conductor is zero.

练习 4-16: 在图 4-21 中求 $\mathbf{E}_1$, 如果

$$\begin{aligned}\mathbf{E}_2 &= \hat{x}2 - \hat{y}3 + \hat{z}3 \text{ (V/m)}, \\ \epsilon_1 &= 2\epsilon_0, \\ \epsilon_2 &= 8\epsilon_0,\end{aligned}$$

并且界面上无自由面电荷 ($\rho_s = 0$)。

Answer: $\mathbf{E}_1 = \hat{x}2 - \hat{y}3 + \hat{z}12 \text{ (V/m)}$. (See [EM](#).)

练习 4-17: 对表面电荷密度 $\rho_s = 3.54 \times 10^{-11} \text{ (C/m}^2\text{)}$ 的边界重复练习 4.16。

答案: $\mathbf{E}_1 = \hat{x}2 - \hat{y}3 + \hat{z}14 \text{ (V/m)}$ 。(参见 [EM](#)。)

这两个边界条件可以合并为

$$\mathbf{D}_1 = \epsilon_1 \mathbf{E}_1 = \hat{n} \rho_s, \quad (4.111)$$

(at conductor surface)

其中 $\hat{n}$ 是从导电表面法向向外的单位矢量。

► 当 ρ_s 为正时, 电场线直接远离导体表面; 当 ρ_s 为负时, 电场线直接指向导体表面。◀

4-8.1 电介质-导体边界

考虑介质 1 是电介质且介质 2 是理想导体的情况。在理想导体中, 由于电场和通量消失, 因此 $\mathbf{E}_2 = \mathbf{D}_2 = 0$, 这意味着 $\mathbf{E}_2$ 和 $\mathbf{D}_2$ 与界面相切和法向的分量为零。因此, 从方程 (4.100) 和等式 (4.104), 介电介质中与导体边界处的场满足

$$E_{1t} = D_{1t} = 0, \quad (4.110a)$$

$$D_{1n} = \epsilon_1 E_{1n} = \rho_s. \quad (4.110b)$$

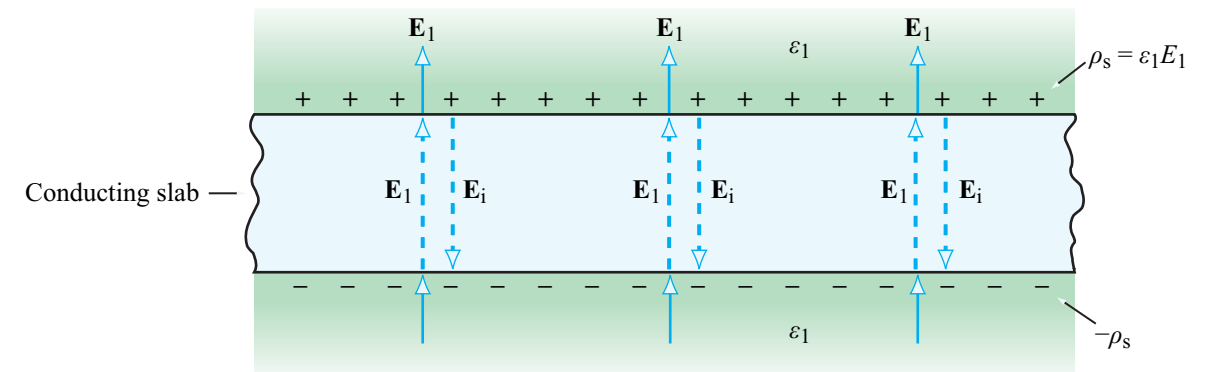
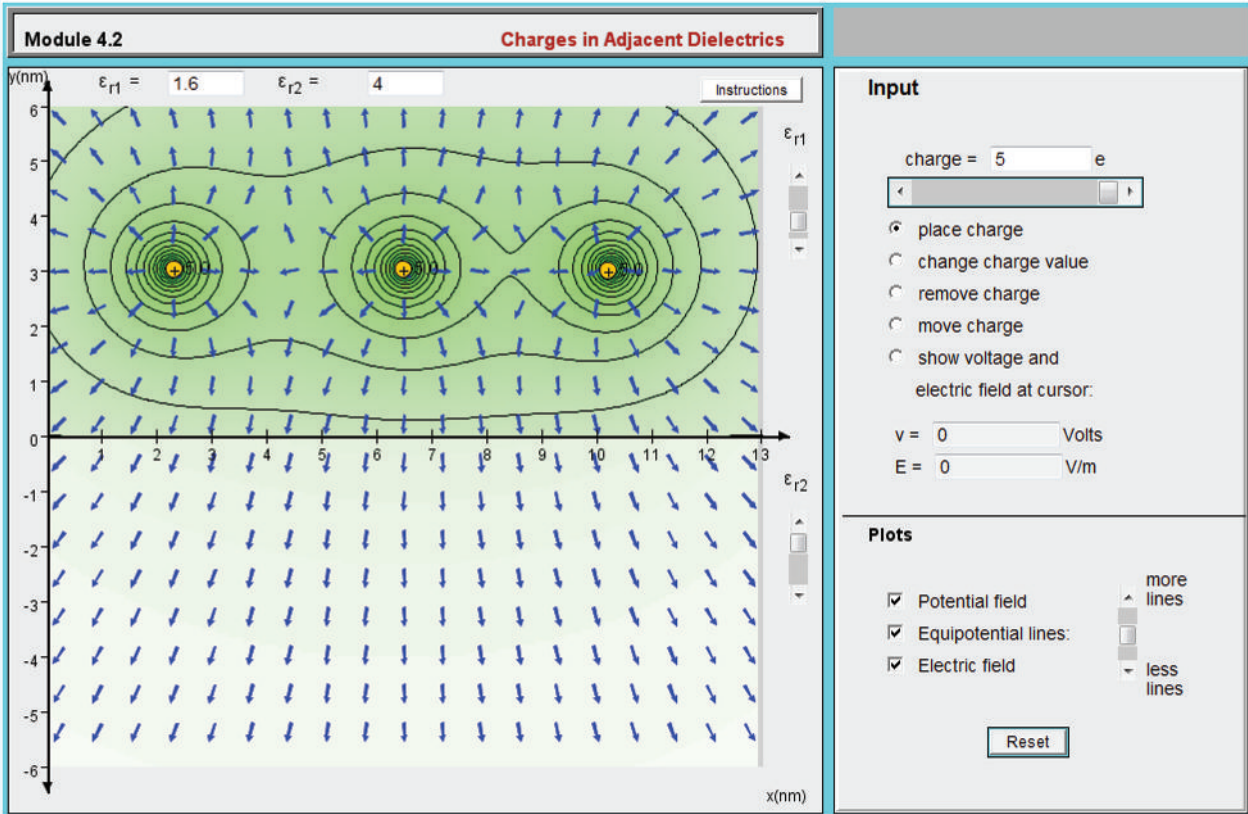


图 4-22 当导电板置于外部电场 $\mathbf{E}_1$ 中时, 积聚在导体表面的电荷会感应出内部电场 $\mathbf{E}_i = -\mathbf{E}_1$ 。因此, 导体内的总电场为零。

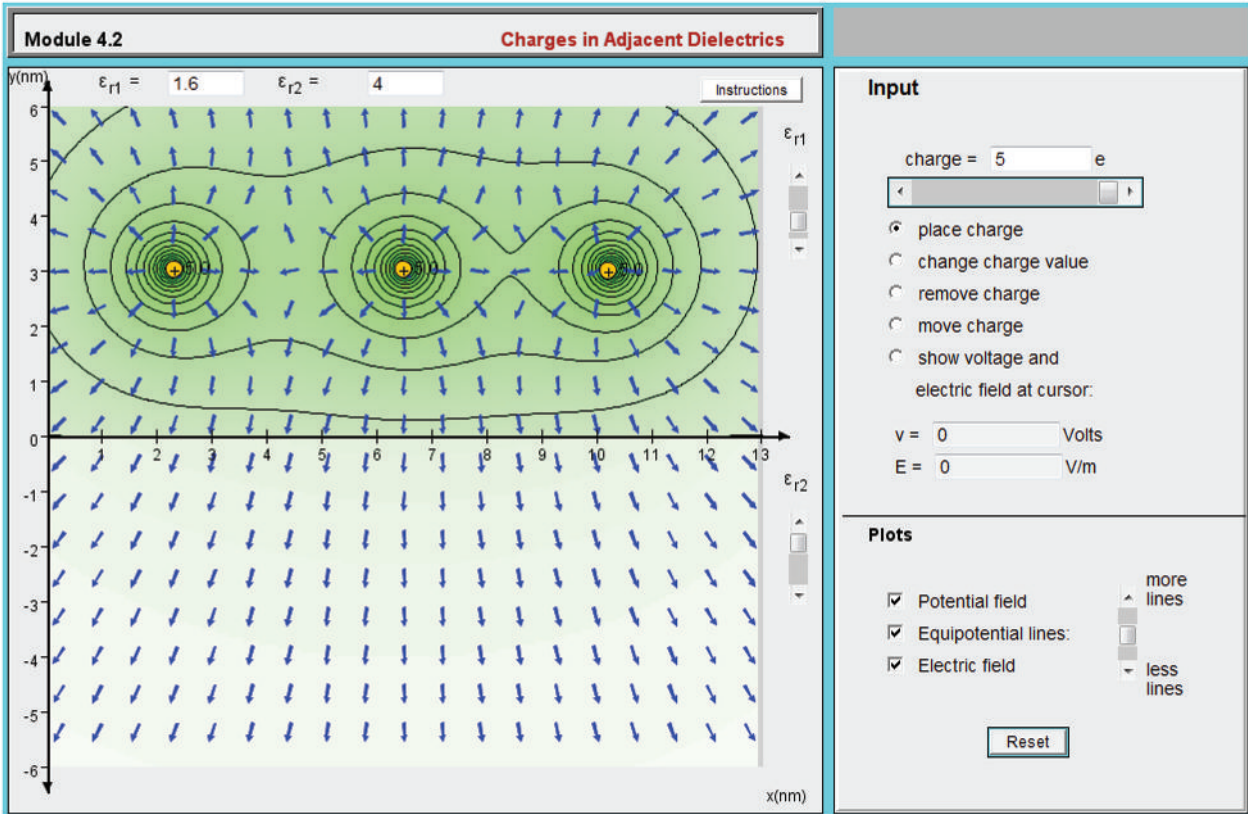
图 4-22 显示了放置在均匀电场 $\mathbf{E}_1$ 中的无限长导电板。板上方和下方的介质具有介电常数 ϵ_1 。由于 $\mathbf{E}_1$ 指向远离上表面的方向, 因此它会在上板表面上感应出正电荷密度 $\rho_s = \epsilon_1 |\mathbf{E}_1|$ 。在底面上, $\mathbf{E}_1$ 指向表面; 因此, 感应电荷密度为 $-\rho_s$ 。这些表面电荷的存在会在导体中感应出电场 $\mathbf{E}_i$, 从而产生总电场 $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_i$ 。为了满足导体中 $\mathbf{E}$ 必须处处为零的条件, $\mathbf{E}_i$ 必须等于 $-\mathbf{E}_1$ 。

如果我们将一个金属球放在静电场中 (图 4-23), 正电荷和负电荷分别积聚在上半球和下半球上。球体的存在导致场线弯曲以满足方程 (1) 表示的条件 (4.111); 也就是说, $\mathbf{E}$ 始终垂直于导体边界。

Module 4.2 Charges in Adjacent Dielectrics In two adjoining half-planes with selectable permittivities, the user can place point charges anywhere in space and select their magnitudes and polarities. The module then displays E , V , and the equipotential contours of V .



模块 4.2 相邻电介质中的电荷 在具有可选介电常数的两个相邻半平面中，用户可以将点电荷放置在空间中的任何位置，并选择它们的大小和极性。然后模块显示 E 、 V 和 V 的等势线。



4-8.2 Conductor–Conductor Boundary

We now examine the general case of the boundary between two media—neither of which is a perfect dielectric or a perfect conductor (**Fig. 4-24**). Medium 1 has permittivity ϵ_1 and conductivity σ_1 , medium 2 has ϵ_2 and σ_2 , and the interface between them holds a surface charge density ρ_s . For the electric fields, Eqs. (4.100) and (4.105b) give

$$\mathbf{E}_{1t} = \mathbf{E}_{2t}, \quad \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s. \quad (4.112)$$

Since we are dealing with conducting media, the electric fields give rise to current densities $\mathbf{J}_1 = \sigma_1 \mathbf{E}_1$ and $\mathbf{J}_2 = \sigma_2 \mathbf{E}_2$. Hence,

$$\frac{\mathbf{J}_{1t}}{\sigma_1} = \frac{\mathbf{J}_{2t}}{\sigma_2}, \quad \epsilon_1 \frac{J_{1n}}{\sigma_1} - \epsilon_2 \frac{J_{2n}}{\sigma_2} = \rho_s. \quad (4.113)$$

The tangential current components $\mathbf{J}_{1t}$ and $\mathbf{J}_{2t}$ represent currents flowing in the two media in a direction parallel to the

4-8.2 导体-导体边界

我们现在研究两种介质之间边界的一般情况——这两种介质都不是理想的电介质或理想的导体（图4-24）。介质1具有介电常数 ϵ_1 和电导率 σ_1 ，介质2具有 ϵ_2 和 σ_2 ，并且它们之间的界面具有表面电荷密度 ρ_s 。对于电场，方程（4.100）和（4.105b）给出

$$\mathbf{E}_{1t} = \mathbf{E}_{2t}, \quad \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s. \quad (4.112)$$

由于我们处理的是导电介质，电场会产生电流密度 $\mathbf{J}_1 = \sigma_1 \mathbf{E}_1$ 和 $\mathbf{J}_2 = \sigma_2 \mathbf{E}_2$ 。因此，

$$\frac{\mathbf{J}_{1t}}{\sigma_1} = \frac{\mathbf{J}_{2t}}{\sigma_2}, \quad \epsilon_1 \frac{J_{1n}}{\sigma_1} - \epsilon_2 \frac{J_{2n}}{\sigma_2} = \rho_s. \quad (4.113)$$

切向电流分量 $\mathbf{J}_{1t}$ 和 $\mathbf{J}_{2t}$ 表示在两种介质中沿平行于介质方向流动的电流。

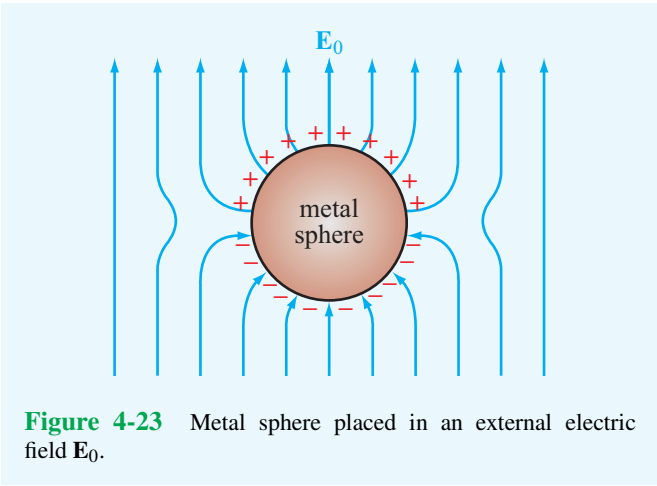


Figure 4-23 Metal sphere placed in an external electric field $\mathbf{E}_0$.

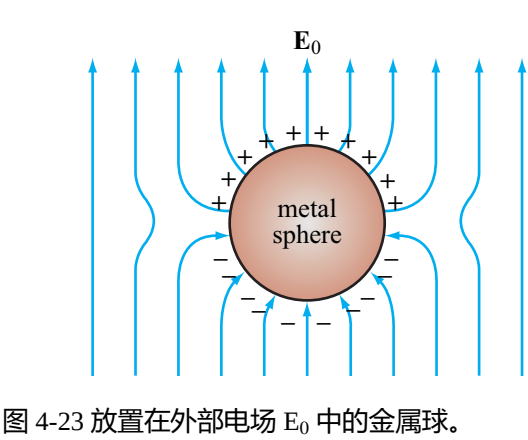
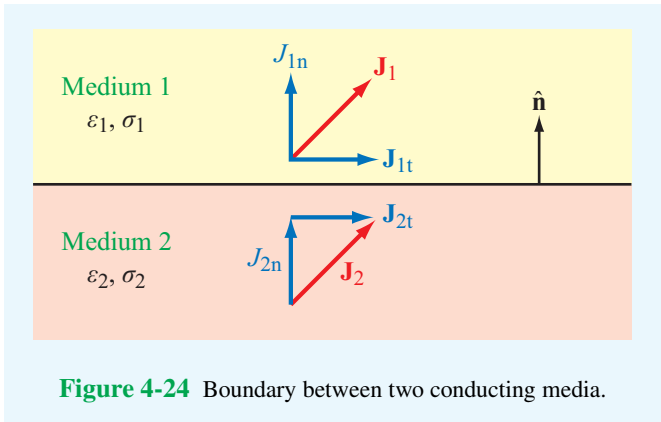


图 4-23 放置在外部电场 $\mathbf{E}_0$ 中的金属球。



boundary; hence, there is no transfer of charge between them. This is not the case for the normal components. If $J_{1n} \neq J_{2n}$, then a different amount of charge arrives at the boundary than leaves it. Hence, ρ_s cannot remain constant in time, which violates the condition of electrostatics requiring all fields and charges to remain constant. Consequently, *the normal component of $\mathbf{J}$ has to be continuous across the boundary between two different media under electrostatic conditions.* Upon setting $J_{1n} = J_{2n}$ in Eq. (4.113), we have

$$J_{1n} \left(\frac{\epsilon_1}{\sigma_1} - \frac{\epsilon_2}{\sigma_2} \right) = \rho_s \quad \text{(electrostatics).} \quad (4.114)$$

Concept Question 4-22: What are the boundary conditions for the electric field at a conductor–dielectric boundary?

Concept Question 4-23: Under electrostatic conditions, we require $J_{1n} = J_{2n}$ at the boundary between two conductors. Why?

4-9 Capacitance

When separated by an insulating (dielectric) medium, any two conducting bodies, regardless of their shapes and sizes, form a **capacitor**. If a dc voltage source is connected across them (Fig. 4-25) the surfaces of the conductors connected to the positive and negative source terminals accumulate charges $+Q$ and $-Q$, respectively.

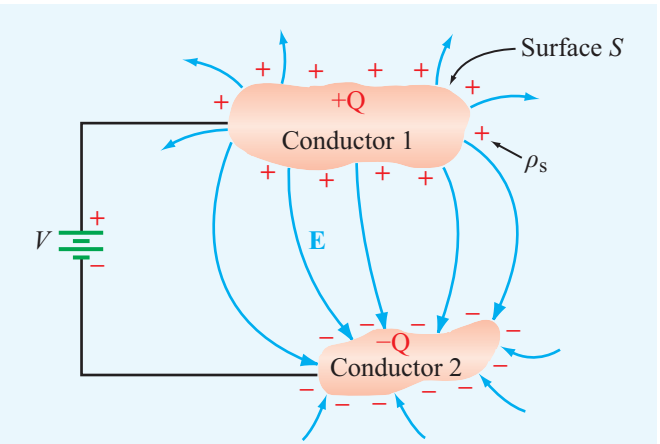


Figure 4-25 A dc voltage source connected to a capacitor composed of two conducting bodies.

► When a conductor has excess charge, it distributes the charge on its surface in such a manner as to maintain a zero electric field everywhere within the conductor, thereby ensuring that the electric potential is the same at every point in the conductor. ◀

The **capacitance** of a two-conductor configuration is defined as

$$C = \frac{Q}{V} \quad (\text{C/V or F}), \quad (4.115)$$

where V is the potential (voltage) difference between the conductors. Capacitance is measured in farads (F), which is equivalent to coulombs per volt (C/V).

The presence of free charges on the conductors' surfaces gives rise to an electric field $\mathbf{E}$ (Fig. 4-25) with field lines originating on the positive charges and terminating on the negative ones. Since the tangential component of $\mathbf{E}$ always vanishes at a conductor's surface, $\mathbf{E}$ is always perpendicular to the conducting surfaces. The normal component of $\mathbf{E}$ at any point on the surface of either conductor is given by

$$E_n = \hat{\mathbf{n}} \cdot \mathbf{E} = \frac{\rho_s}{\epsilon}, \quad (4.116)$$

(at conductor surface)

where ρ_s is the surface charge density at that point, $\hat{\mathbf{n}}$ is the outward normal unit vector at the same location, and ϵ is the permittivity of the dielectric medium separating the

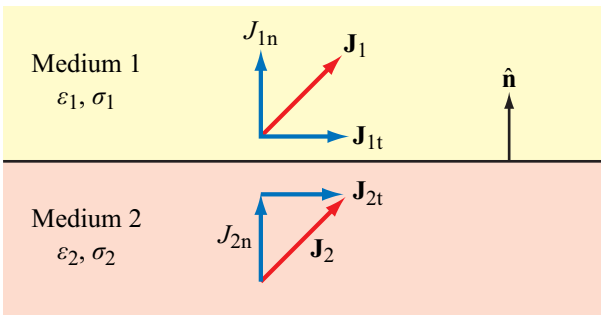


图 4-24 两种导电介质之间的边界。

边界;因此, 它们之间不存在电荷转移。对于普通组件来说, 情况并非如此。如果 $J_{1n} \neq J_{2n}$, 则到达边界的电荷量与离开边界的电荷量不同。因此, ρ_s 不能随时间保持恒定, 这违反了静电学要求所有场和电荷保持恒定的条件。因此, 在静电条件下, $\mathbf{J}$ 的法向分量必须连续穿过两种不同介质之间的边界。在等式中设置 $J_{1n} = J_{2n}$ 后 (4.113), 我们有

$$J_{1n} \left(\frac{\epsilon_1}{\sigma_1} - \frac{\epsilon_2}{\sigma_2} \right) = \rho_s \quad \text{(electrostatics).} \quad (4.114)$$

概念问题 4-22: 导体-电介质边界电场的边界条件是什么?

概念问题 4-23: 在静电条件下, 我们要求两个导体之间的边界处有 $J_{1n} = J_{2n}$ 。为什么?

4-9 电容

当被绝缘(介电)介质隔开时, 任何两个导体, 无论其形状和大小如何, 都会形成电容器。如果直流电压源跨接在它们之间(图 4-25), 则连接到正源极端子和负源极端子的导体表面将分别积聚电荷 $+Q$ 和 $-Q$ 。

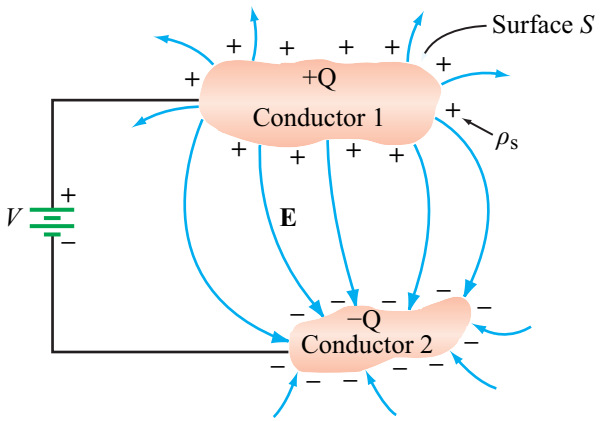


图 4-25 连接到由两个导体组成的电容器的直流电压源。

► 当导体有多余的电荷时, 它会将电荷分布在其表面, 以保持导体内各处的电场为零, 从而确保导体内每一点的电势相同。 ◀

双导体配置的电容定义为

$$C = \frac{Q}{V} \quad (\text{C/V or F}), \quad (4.115)$$

其中 V 是导体之间的电势(电压)差。电容以法拉(F)为单位, 相当于库仑每伏(C/V)。

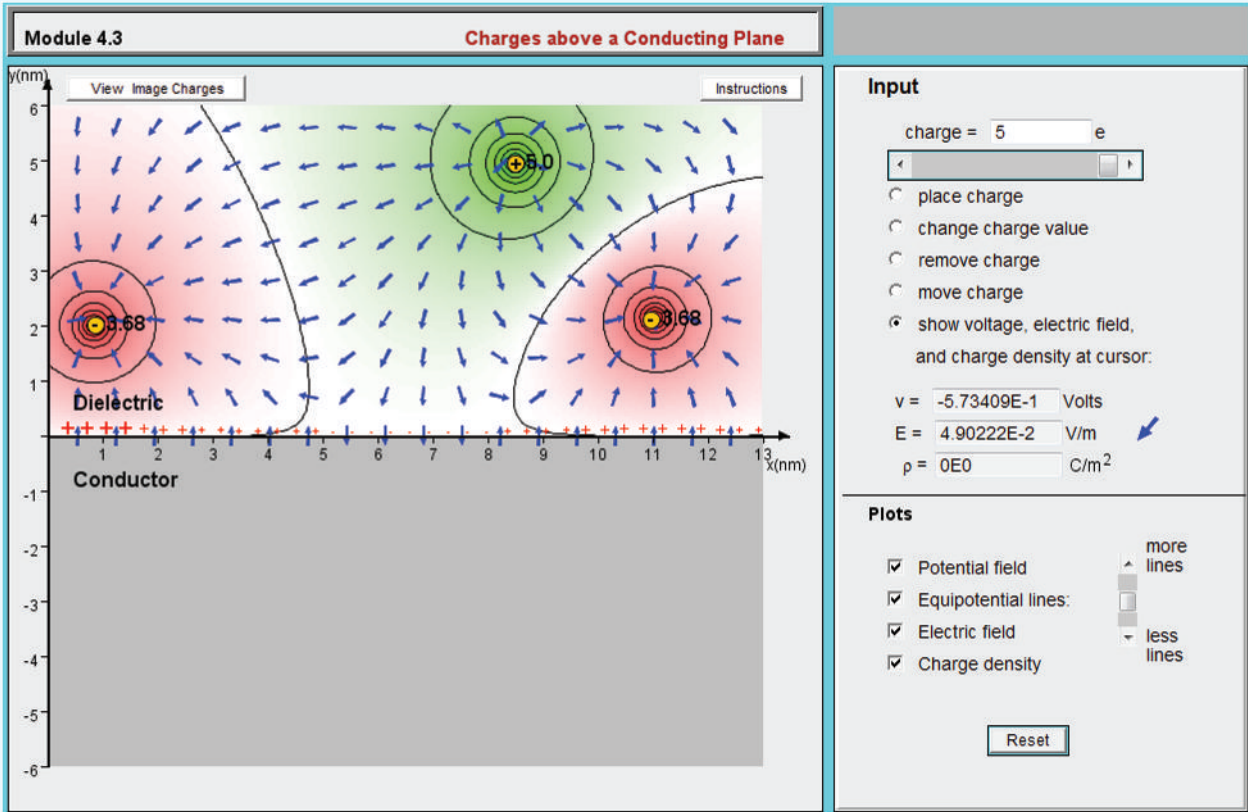
导体表面自由电荷的存在会产生电场 $\mathbf{E}$ (图 4-25), 其电场线始于正电荷, 终止于负电荷。由于 $\mathbf{E}$ 的切向分量在导体表面总是消失, 因此 $\mathbf{E}$ 始终垂直于导电表面。任一导体表面上任意点的 $\mathbf{E}$ 法向分量由下式给出

$$E_n = \hat{\mathbf{n}} \cdot \mathbf{E} = \frac{\rho_s}{\epsilon}, \quad (4.116)$$

(at conductor surface)

其中 ρ_s 是该点的表面电荷密度, $\hat{\mathbf{n}}$ 是同一位置的向外法向单位矢量, ϵ 是分隔该点的介电介质的介电常数。

Module 4.3 Charges above a Conducting Plane When electric charges are placed in a dielectric medium adjoining a conducting plane, some of the conductor’s electric charges move to its surface boundary, thereby satisfying the boundary conditions outlined in **Table 4-3**. This module displays **E** and **V** everywhere and ρ_s along the dielectric–conductor boundary.



conductors. The charge Q is equal to the integral of ρ_s over surface S (**Fig. 4-25**):

$$Q = \int_S \rho_s \, ds = \int_S \epsilon \hat{\mathbf{n}} \cdot \mathbf{E} \, ds = \int_S \epsilon \mathbf{E} \cdot d\mathbf{s}, \quad (4.117)$$

where use was made of Eq. (4.116). The voltage V is related to $\mathbf{E}$ by Eq. (4.39):

$$V = V_{12} = - \int_{P_2}^{P_1} \mathbf{E} \cdot d\mathbf{l}, \quad (4.118)$$

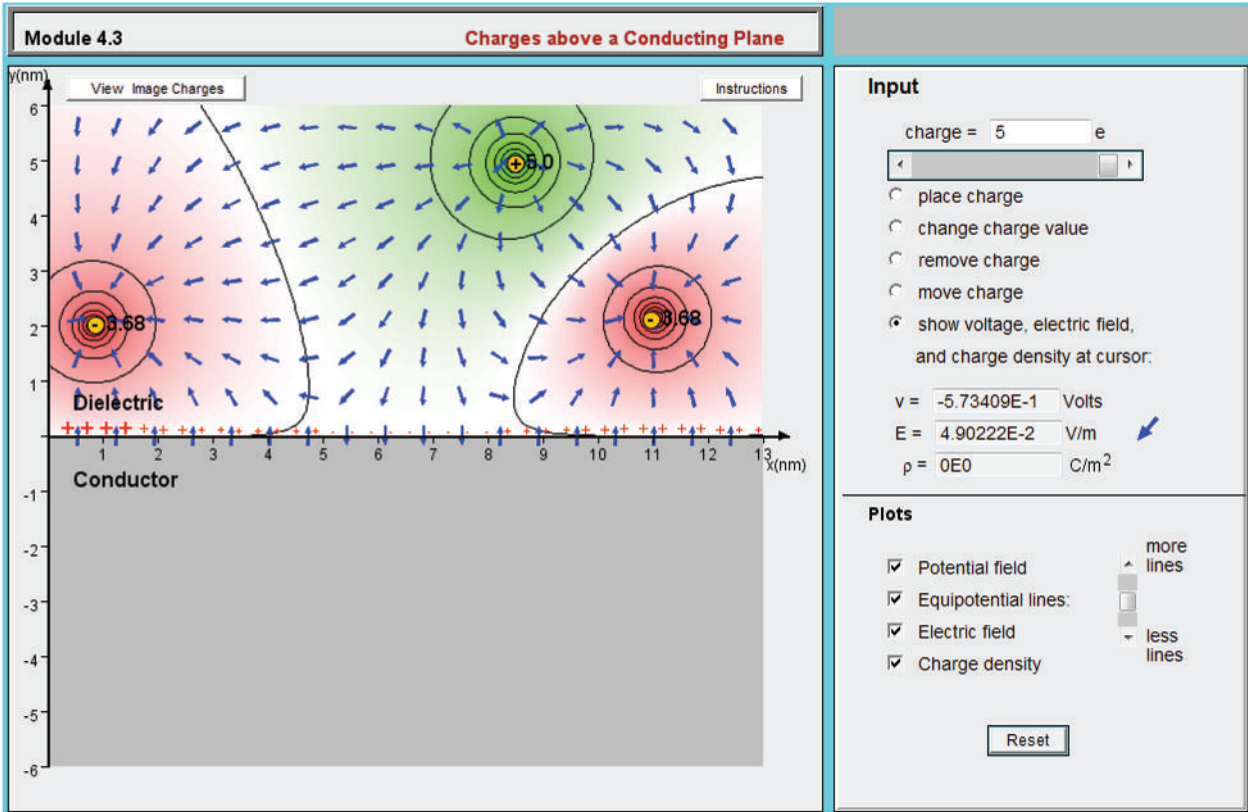
where points P_1 and P_2 are any two arbitrary points on conductors 1 and 2, respectively. Substituting Eqs. (4.117) and

(4.118) into Eq. (4.115) gives

$$C = \frac{\int_S \epsilon \mathbf{E} \cdot d\mathbf{s}}{- \int_l \mathbf{E} \cdot d\mathbf{l}} \quad (\text{F}), \quad (4.119)$$

where l is the integration path from conductor 2 to conductor 1. To avoid making sign errors when applying Eq. (4.119), it is important to remember that surface S is the $+Q$ surface and P_1 is on S . [Alternatively, if you compute C and it comes out negative, just change its sign.] Because $\mathbf{E}$ appears in both the numerator and denominator of Eq. (4.119), *the value of C obtained for any specific capacitor configuration is always independent of $\mathbf{E}$'s magnitude*. In fact, C depends only on the capacitor geometry (sizes, shapes and relative positions of the two conductors) and the permittivity of the insulating material.

模块 4.3 导电平面上方的电荷 当电荷放置在邻近导电平面的介电介质中时，导体的一些电荷会移动到其表面边界，从而满足表 4-3 中列出的边界条件。该模块随处显示 **E** 和 **V**，并沿电介质-导体边界显示 ρ_s 。



导体。电荷 Q 等于表面 S 上 ρ_s 的积分（图 4-25）：

$$Q = \int_S \rho_s \, ds = \int_S \epsilon \hat{\mathbf{n}} \cdot \mathbf{E} \, ds = \int_S \epsilon \mathbf{E} \cdot d\mathbf{s}, \quad (4.117)$$

其中使用了方程式。 (4.116) 。电压 V 与 $\mathbf{E}$ 的关系如下： (4.39)：

$$V = V_{12} = - \int_{P_2}^{P_1} \mathbf{E} \cdot d\mathbf{l}, \quad (4.118)$$

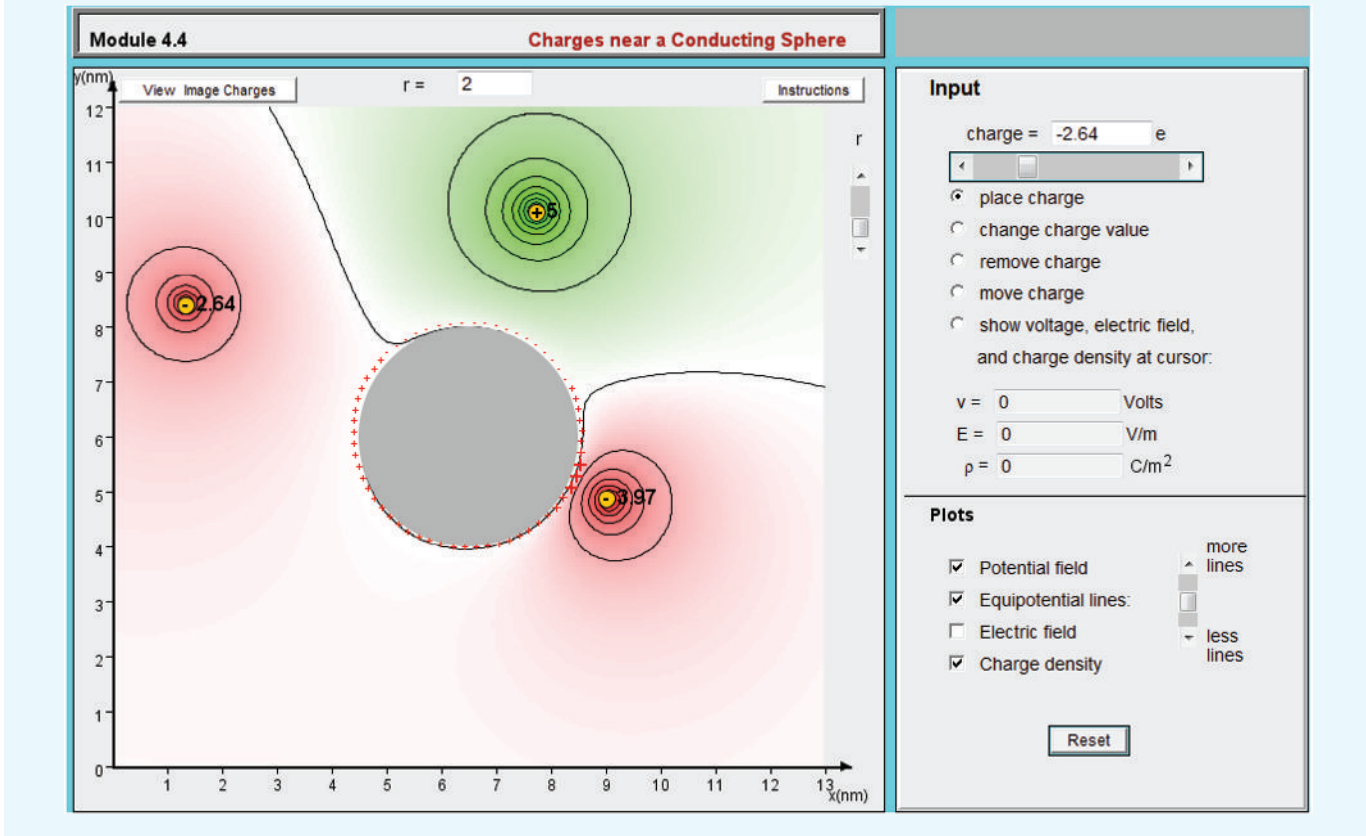
其中点 P_1 和 P_2 分别是导体 1 和 2 上的任意两个点。代入等式。 (4.117) 和

(4.118) 代入等式。 (4.115) 给出

$$C = \frac{\int_S \epsilon \mathbf{E} \cdot d\mathbf{s}}{- \int_l \mathbf{E} \cdot d\mathbf{l}} \quad (\text{F}), \quad (4.119)$$

其中 l 是从导体 2 到导体 1 的积分路径。为避免应用方程式时出现符号错误， (4.119)，重要的是要记住曲面 S 是 $+Q$ 曲面， P_1 在 S 上。[或者，如果计算 C 并且结果为负，只需更改其符号。]因为 $\mathbf{E}$ 出现在方程 1 的分子和分母中。 (4.119)，对于任何特定电容器配置获得的 C 值始终与 $\mathbf{E}$ 的大小无关。事实上， C 仅取决于电容器的几何形状（两个导体的尺寸、形状和相对位置）和绝缘材料的介电常数。

Module 4.4 Charges near a Conducting Sphere This module is similar to Module 4.3, except that now the conducting body is a sphere of selectable size.



If the material between the conductors is not a perfect dielectric (i.e., if it has a small conductivity σ), then current can flow through the material between the conductors, and the material exhibits a resistance R . The general expression for R for a resistor of arbitrary shape is given by Eq. (4.81):

$$R = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}} \quad (\Omega). \quad (4.120)$$

For a medium with uniform σ and ϵ , the product of Eqs. (4.119) and (4.120) gives

$$RC = \frac{\epsilon}{\sigma}. \quad (4.121)$$

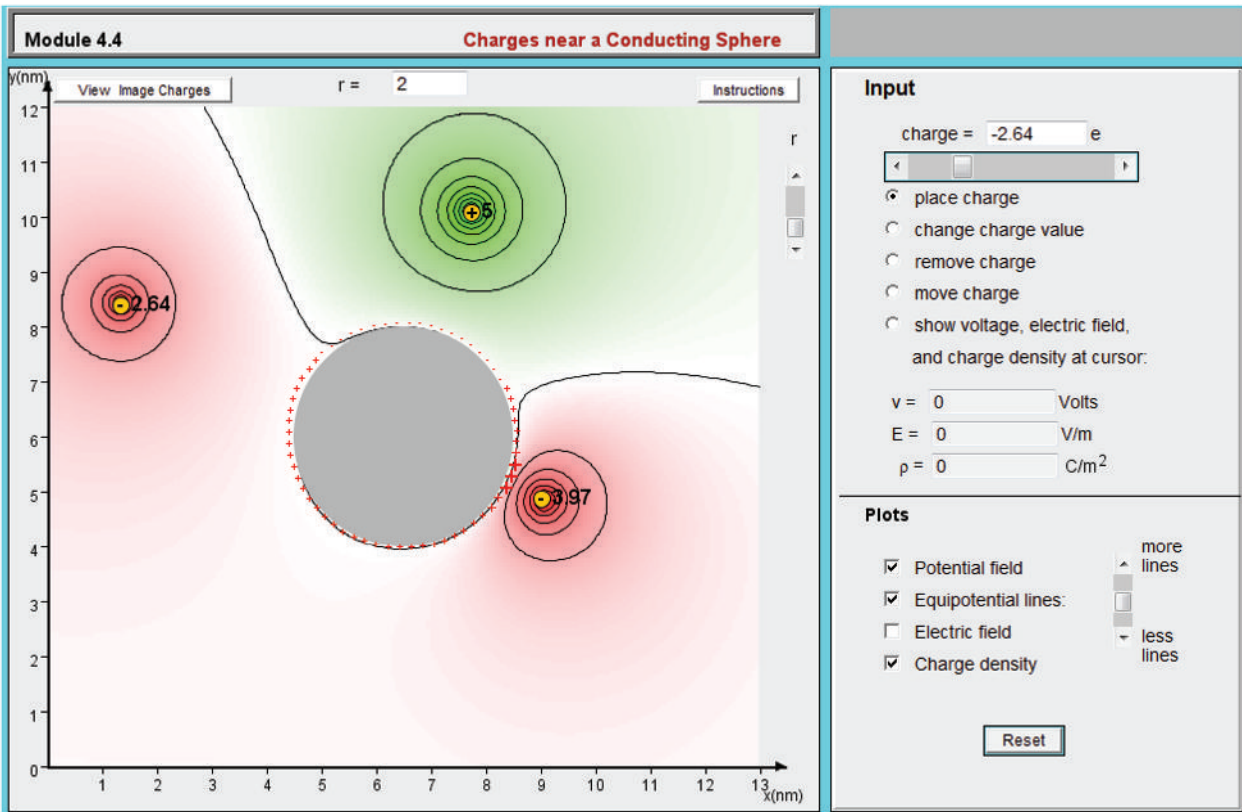
This simple relation allows us to find R if C is known, and vice versa.

Example 4-14: Capacitance of Parallel-Plate Capacitor

Obtain an expression for the capacitance C of a parallel-plate capacitor comprised of two parallel plates each of surface area A and separated by a distance d . The capacitor is filled with a dielectric material with permittivity ϵ .

Solution: In Fig. 4-26, we place the lower plate of the capacitor in the x - y plane and the upper plate in the plane $z = d$. Because of the applied voltage difference V , charges $+Q$ and $-Q$ accumulate on the top and bottom capacitor plates. If the plate dimensions are much larger than the separation d , then these charges distribute themselves quasi-uniformly across the plates, giving rise to a quasi-uniform field between them pointing in the $-\hat{z}$ direction. In addition, a **fringing field** will exist near the capacitor edges, but its effects may be ignored because the bulk of the electric field exists between the plates.

模块 4.4 导电球体附近的电荷 该模块与模块 4.3 类似，不同之处在于现在导体是可选择尺寸的球体。



如果导体之间的材料不是理想的电介质（即，如果它的电导率较小 σ ），则电流可以流过导体之间的材料，并且该材料表现出电阻 R 。任意形状的电阻器的 R 的一般表达式由式 (1) 给出。 (4.81) :

$$R = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}} \quad (\Omega). \quad (4.120)$$

对于具有均匀 σ 和 ϵ 的介质，方程的乘积： (4.119) 和 (4.120) 给出

$$RC = \frac{\epsilon}{\sigma}. \quad (4.121)$$

如果 C 已知，这个简单的关系允许我们找到 R ，反之亦然。

例 4-14：平行板电容器的电容

获得由两个平行板组成的平行板电容器的电容 C 的表达式，每个平行板的表面积为 A ，且间隔距离 d 。电容器充满介电常数为 ϵ 的介电材料。

解：在图4-26中，我们将电容器的下极板放置在 x - y 平面上，将上极板放置在平面 $z = d$ 上。由于施加的电压差 V ，电荷 $+Q$ 和 $-Q$ 积聚在顶部和底部电容器板上。如果板尺寸远大于间距 d ，则这些电荷会准均匀地分布在板上，从而在它们之间产生指向 $-\hat{z}$ 方向的准均匀场。此外，电容器边缘附近将存在边缘场，但其影响可以忽略不计，因为大部分电场存在于极板之间。

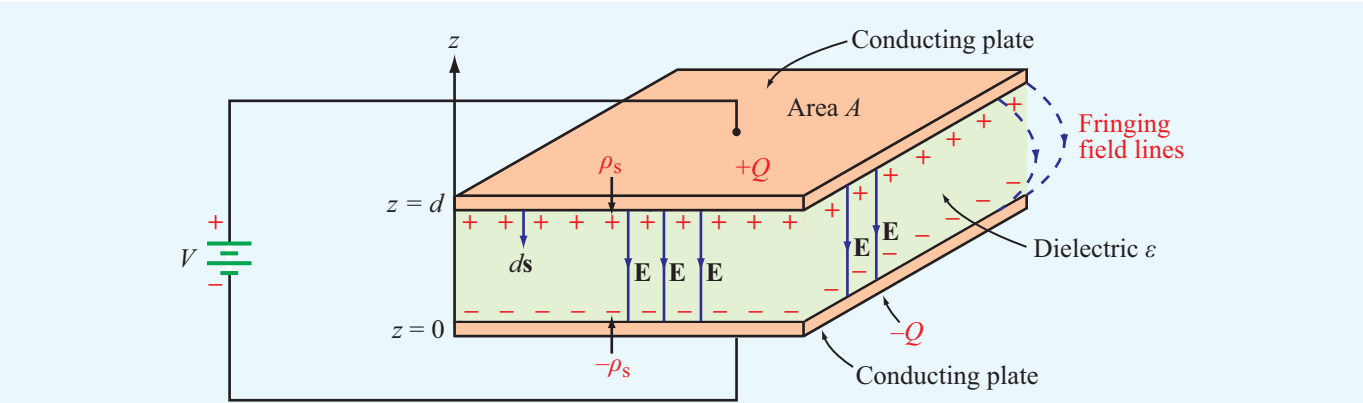


Figure 4-26 A dc voltage source connected to a parallel-plate capacitor (Example 4-14).

The charge density on the upper plate is $\rho_s = Q/A$. Hence, in the dielectric medium

$$\mathbf{E} = -\hat{\mathbf{z}}E,$$

and from Eq. (4.116), the magnitude of $\mathbf{E}$ at the conductor–dielectric boundary is $E = \rho_s/\epsilon = Q/\epsilon A$. From Eq. (4.118), the voltage difference is

$$V = -\int_0^d \mathbf{E} \cdot d\mathbf{l} = -\int_0^d (-\hat{\mathbf{z}}E) \cdot \hat{\mathbf{z}} dz = Ed, \quad (4.122)$$

and the capacitance is

$$C = \frac{Q}{V} = \frac{Q}{Ed} = \frac{\epsilon A}{d}, \quad (4.123)$$

where use was made of the relation $E = Q/\epsilon A$.

Example 4-15: Capacitance per Unit Length of Coaxial Line

Obtain an expression for the capacitance of the coaxial line shown in Fig. 4-27.

Solution: For a given voltage V across the capacitor, charges $+Q$ and $-Q$ accumulate on the surfaces of the outer and inner conductors, respectively. We assume that these charges are uniformly distributed along the length and circumference of the conductors with surface charge density $\rho'_s = Q/2\pi bl$ on the outer conductor and $\rho''_s = -Q/2\pi al$ on the inner one. Ignoring fringing fields near the ends of the coaxial line, we can construct a cylindrical Gaussian surface in the dielectric in between the conductors with the radius r such that $a < r < b$. Symmetry implies that the $\mathbf{E}$ -field is identical at all points on this surface, which is directed radially inward. From Gauss's law, it follows that the field magnitude equals the absolute

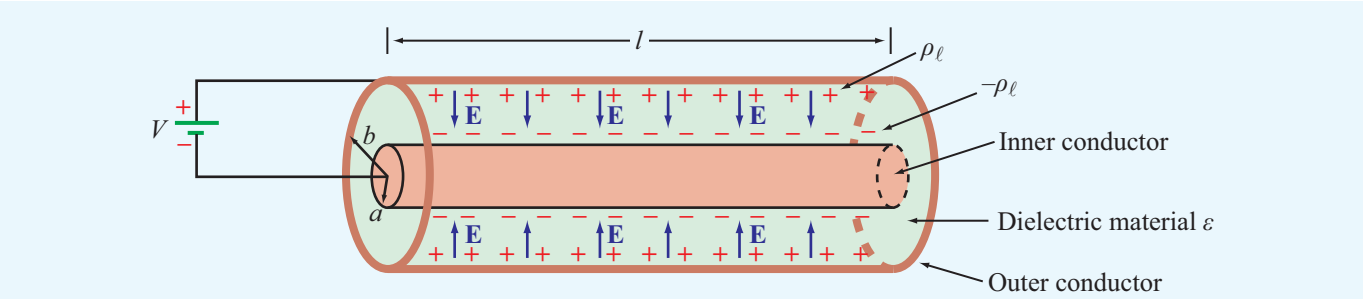


Figure 4-27 Coaxial capacitor filled with insulating material of permittivity ϵ (Example 4-15).

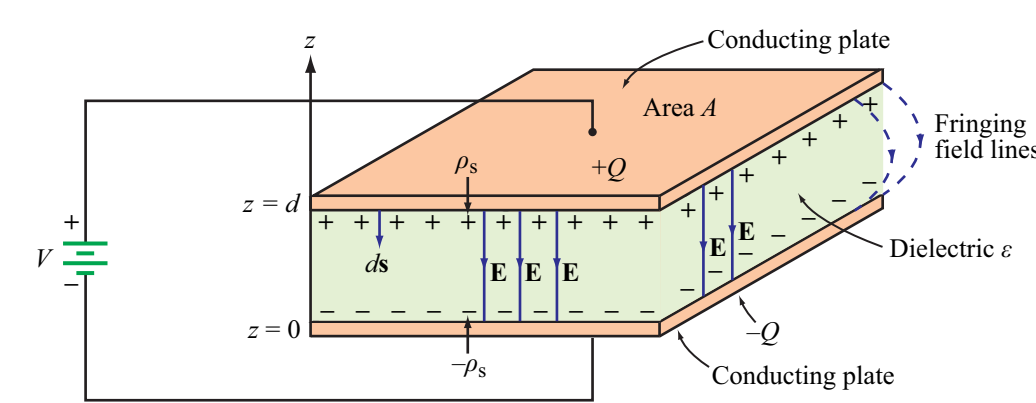


图 4-26 连接到平行板电容器的直流电压源（示例 4-14）。

上板上的电荷密度为 $\rho_s = Q/A$ 。因此，在介电介质中

$$\mathbf{E} = -\hat{\mathbf{z}}E,$$

并从方程式根据方程 (4.116) ，导体-电介质边界处的 E 大小为 $E = \rho_s/\epsilon = Q/\epsilon A$ 。从方程。 (4.118) ，电压差为

$$V = -\int_0^d \mathbf{E} \cdot d\mathbf{l} = -\int_0^d (-\hat{\mathbf{z}}E) \cdot \hat{\mathbf{z}} dz = Ed, \quad (4.122)$$

电容为

$$C = \frac{Q}{V} = \frac{Q}{Ed} = \frac{\epsilon A}{d}, \quad (4.123)$$

其中使用了关系 $E = Q/\epsilon A$ 。

例 4-15：同轴线单位长度的电容

求出图 4-27 所示同轴线电容的表达式。

解决方案：对于给定的电容器两端电压 V ，电荷 $+Q$ 和 $-Q$ 分别积聚在外导体和内导体的表面上。我们假设这些电荷沿着导体的长度和圆周均匀分布，外导体上的表面电荷密度为 $\rho'_s = Q/2\pi bl$ ，内导体上的表面电荷密度为 $\rho''_s = -Q/2\pi al$ 。忽略同轴线末端附近的边缘场，我们可以在导体之间的电介质中构造一个半径为 r 的圆柱形高斯表面，使得 $a < r < b$ 。对称性意味着该表面上所有点的电场都是相同的，并且径向向内指向。根据高斯定律，场强等于绝对场强

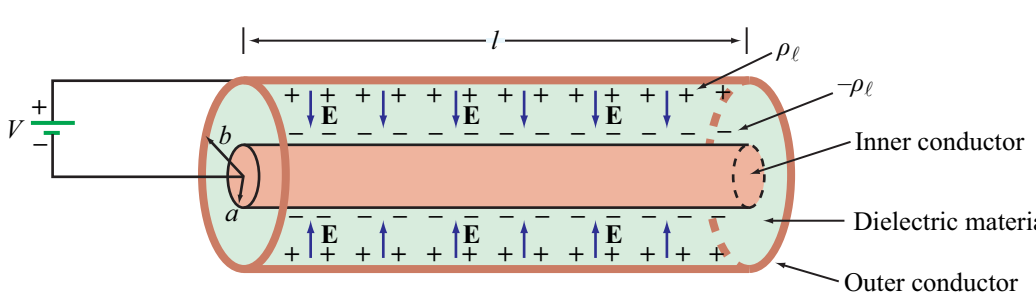


图4-27 填充介电常数为ε (的绝缘材料的同轴电容器示例4-15)。

value of the total charge enclosed, divided by the surface area. That is,

$$\mathbf{E} = -\hat{\mathbf{r}} \frac{Q}{2\pi\epsilon r l}.$$
(4.124)

The potential difference V between the outer and inner conductors is

$$\begin{aligned} V &= -\int_a^b \mathbf{E} \cdot d\mathbf{l} = -\int_a^b \left(-\hat{\mathbf{r}} \frac{Q}{2\pi\epsilon r l}\right) \cdot (\hat{\mathbf{r}} dr) \\ &= \frac{Q}{2\pi\epsilon l} \ln\left(\frac{b}{a}\right). \end{aligned}$$
(4.125)

The capacitance C is then given by

$$C = \frac{Q}{V} = \frac{2\pi\epsilon l}{\ln(b/a)},$$
(4.126)

and the capacitance per unit length of the coaxial line is

$$C' = \frac{C}{l} = \frac{2\pi\epsilon}{\ln(b/a)} \quad (\text{F/m}).$$
(4.127)

Concept Question 4-24: How is the capacitance of a two-conductor structure related to the resistance of the insulating material between the conductors?

Concept Question 4-25: What are fringing fields and when may they be ignored?

4-10 Electrostatic Potential Energy

A source connected to a capacitor expends energy in charging up the capacitor. If the capacitor plates are made of a good conductor with effectively zero resistance, and if the dielectric separating the two plates has negligible conductivity, then no real current can flow through the dielectric, and no ohmic losses occur anywhere in the capacitor. Where then does the energy expended in charging up the capacitor go? The energy ends up getting stored in the dielectric medium in the form of **electrostatic potential energy**. The amount of stored energy W_e is related to Q , C , and V .

Suppose we were to charge up a capacitor by ramping up the voltage across it from $v = 0$ to $v = V$. During the process, charge $+q$ accumulates on one conductor and $-q$ on the other. In effect, a charge q has been transferred from one of the conductors to the other. The voltage v across the capacitor is related to q by

$$v = \frac{q}{C}.$$
(4.128)

From the definition of v , the amount of work dW_e required to transfer an additional incremental charge dq from one conductor to the other is

$$dW_e = v dq = \frac{q}{C} dq.$$
(4.129)

If we transfer a total charge Q between the conductors of an initially uncharged capacitor, then the total amount of work performed is

$$W_e = \int_0^Q \frac{q}{C} dq = \frac{1}{2} \frac{Q^2}{C} \quad (\text{J}).$$
(4.130)

Using $C = Q/V$, where V is the final voltage, W_e also can be expressed as

$$W_e = \frac{1}{2} CV^2 \quad (\text{J}).$$
(4.131)

The capacitance of the parallel-plate capacitor discussed in Example 4-14 is given by Eq. (4.123) as $C = \epsilon A/d$, where A is the surface area of each of its plates and d is the separation between them. Also, the voltage V across the capacitor is related to the magnitude of the electric field E in the dielectric by $V = Ed$. Using these two expressions in Eq. (4.131) gives

$$W_e = \frac{1}{2} \frac{\epsilon A}{d} (Ed)^2 = \frac{1}{2} \epsilon E^2 (Ad) = \frac{1}{2} \epsilon E^2 \mathcal{V},$$
(4.132)

where $\mathcal{V} = Ad$ is the volume of the capacitor. This expression affirms the assertion made at the beginning of this section, namely that the energy expended in charging up the capacitor is being stored in the electric field present in the dielectric material in between the two conductors.

The **electrostatic energy density** w_e is defined as the electrostatic potential energy W_e per unit volume:

$$w_e = \frac{W_e}{\mathcal{V}} = \frac{1}{2} \epsilon E^2 \quad (\text{J/m}^3).$$
(4.133)

Even though this expression was derived for a parallel-plate capacitor, it is equally valid for any dielectric medium containing an electric field $\mathbf{E}$, including a vacuum. Furthermore, for any volume $\mathcal{V}$, the total electrostatic potential energy stored in it is

$$W_e = \frac{1}{2} \int_{\mathcal{V}} \epsilon E^2 d\mathcal{V} \quad (\text{J}).$$
(4.134)

Returning to the parallel-plate capacitor, the oppositely charged plates are attracted to each other by an **electrical force** $\mathbf{F}_e$. In terms of the coordinate system of **Fig. 4-28**, the

所包含的总电荷除以表面积的值。那是,

$$\mathbf{E} = -\hat{\mathbf{r}} \frac{Q}{2\pi\epsilon r l}.$$
(4.124)

外导体和内导体之间的电位差 V 为

$$\begin{aligned} V &= -\int_a^b \mathbf{E} \cdot d\mathbf{l} = -\int_a^b \left(-\hat{\mathbf{r}} \frac{Q}{2\pi\epsilon r l}\right) \cdot (\hat{\mathbf{r}} dr) \\ &= \frac{Q}{2\pi\epsilon l} \ln\left(\frac{b}{a}\right). \end{aligned}$$
(4.125)

电容 C 由下式给出

$$C = \frac{Q}{V} = \frac{2\pi\epsilon l}{\ln(b/a)},$$
(4.126)

同轴线单位长度的电容为

$$C' = \frac{C}{l} = \frac{2\pi\epsilon}{\ln(b/a)} \quad (\text{F/m}).$$
(4.127)

概念问题 4-24: 双导体结构的电容与导体之间绝缘材料的电阻有何关系?

概念问题 4-25: 什么是边缘场以及何时可以忽略它们?

4-10 静电势能

连接到电容器的电源在给电容器充电时消耗能量。如果电容器极板由有效为零电阻的良导体制成, 并且分隔两个极板的电介质的电导率可以忽略不计, 则没有实际电流可以流过电介质, 并且电容器中的任何地方都不会发生欧姆损耗。那么电容器充电所消耗的能量去了哪里呢? 能量最终以静电势能的形式存储在电介质中。储存的能量 W_e 与 Q 、 C 和 V 相关。

假设我们要通过将电容器两端的电压从 $v = 0$ 提升到 $v = V$ 来给电容器充电。在此过程中, 电荷 $+q$ 积聚在一个导体上, 而 $-q$ 积聚在另一导体上。实际上, 电荷 q 已从一个导体转移到另一个导体。电容器两端的电压 v 与 q 的关系为

$$v = \frac{q}{C}.$$
(4.128)

根据 v 的定义, 将额外增量电荷 dq 从一个导体转移到另一个导体所需的功 dW_e 为

$$dW_e = v dq = \frac{q}{C} dq.$$
(4.129)

如果我们在最初未充电的电容器的导体之间转移总电荷 Q , 则所做的总功为

$$W_e = \int_0^Q \frac{q}{C} dq = \frac{1}{2} \frac{Q^2}{C} \quad (\text{J}).$$
(4.130)

使用 $C = Q/V$, 其中 V 是最终电压, W_e 也可以表示为

$$W_e = \frac{1}{2} CV^2 \quad (\text{J}).$$
(4.131)

例 4-14 中讨论的平行板电容器的电容由公式 1 给出。(4.123) 为 $C = \epsilon A/d$, 其中 A 是每个板的表面积, d 是它们之间的间距。此外, 电容器两端的电压 V 与电介质中电场 E 的大小有关, 关系为 $V = Ed$ 。在等式中使用这两个表达式。(4.131) 给出

$$W_e = \frac{1}{2} \frac{\epsilon A}{d} (Ed)^2 = \frac{1}{2} \epsilon E^2 (Ad) = \frac{1}{2} \epsilon E^2 \mathcal{V},$$
(4.132)

其中 $\mathcal{V} = Ad$ 是电容器的体积。这个表达式证实了本节开头的断言, 即电容器充电所消耗的能量被存储在两个导体之间的介电材料中存在的电场中。

静电能量密度 w_e 定义为每单位体积的静电势能 W_e :

$$w_e = \frac{W_e}{\mathcal{V}} = \frac{1}{2} \epsilon E^2 \quad (\text{J/m}^3).$$
(4.133)

尽管这个表达式是针对平行板电容器导出的, 但它对于任何包含电场 $\mathbf{E}$ 的介电介质 (包括真空) 同样有效。此外, 对于任何体积 $\mathcal{V}$, 其中存储的总静电势能是

$$W_e = \frac{1}{2} \int_{\mathcal{V}} \epsilon E^2 d\mathcal{V} \quad (\text{J}).$$
(4.134)

回到平行板电容器, 带相反电荷的板通过电力 $\mathbf{F}_e$ 相互吸引。就图4-28的坐标系而言,

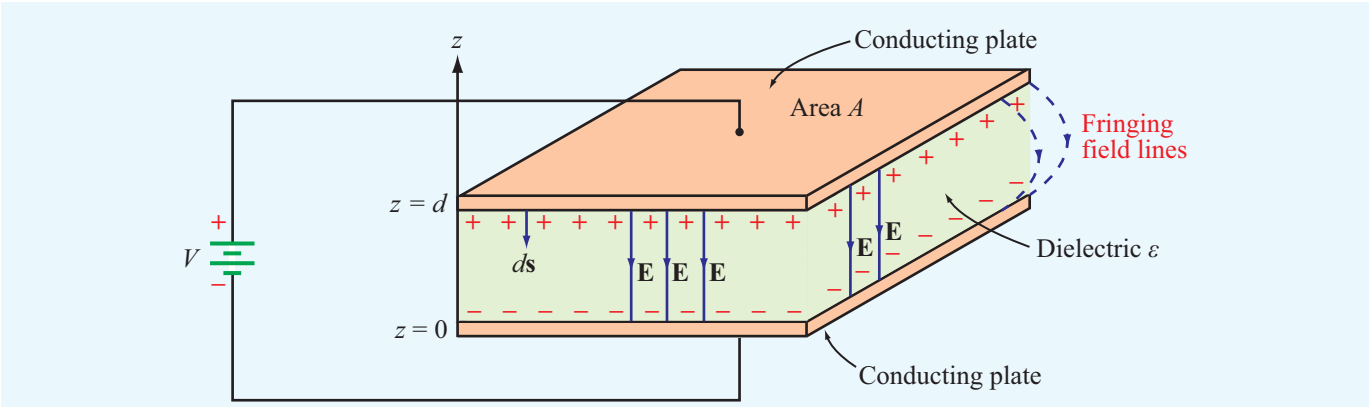


Figure 4-28 A dc voltage source connected to a parallel-plate capacitor.

electrical force acting on the upper plate is along $-\hat{z}$ (due to attraction by the lower plate). Hence, it is given by

$$\mathbf{F}_e = -\hat{z} F_e \quad (\text{force on upper plate}). \quad (4.135)$$

Our plan is to compute $\mathbf{F}_e$ from energy considerations. We start by converting the spacing d into a variable z and using $C = \epsilon A/z$ in Eq. (4.131):

$$W_e = \frac{1}{2} CV^2 = \frac{1}{2} \frac{\epsilon AV^2}{z}. \quad (4.136)$$

If V is maintained at a constant level, W_e decreases when increasing the separation z between the plates. If an external, upward-directed force $\mathbf{F} = -\mathbf{F}_e$ is applied to counter the electrostatic force $\mathbf{F}_e$ and used to move the upper plate upwards by a distance dz , the expended mechanical work is

$$dW = \mathbf{F} \cdot \hat{z} dz. \quad (4.137)$$

The work dW is equal to the loss in electrostatic energy stored in the capacitor. That is,

$$dW = -dW_e. \quad (4.138)$$

Also, $\mathbf{F}_e = -\mathbf{F}$, which leads to

$$dW_e = \mathbf{F}_e \cdot \hat{z} dz = -\hat{z} F_e \cdot \hat{z} dz = -F_e dz. \quad (4.139)$$

From Eq. (4.136),

$$dW_e = -\frac{1}{2} \epsilon \frac{AV^2}{z^2} dz. \quad (4.140)$$

Equating Eqs. (4.139) and (4.140) and replacing z with d leads to

$$F_e = \frac{1}{2} \epsilon \frac{AV^2}{d^2}, \quad (4.141a)$$

and

$$\mathbf{F}_e = -\hat{z} \frac{1}{2} \epsilon A \frac{V^2}{d^2} \quad (\text{N}). \quad (4.141b)$$

(parallel-plate capacitor)

This is the electrostatic force exerted on the upper plate. The force on the lower plate is identical in magnitude and opposite in direction.

The relationship given by Eq. (4.139) pertains to a capacitor with $d\mathbf{l} = \hat{z} dz$. We can generalize the result for $d\mathbf{l}$ along any direction as

$$\mathbf{F}_e = -\nabla W_e. \quad (4.142)$$

Example 4-16: Force on Sliding Dielectric

The two plates of the parallel-plate capacitor shown in Fig. 4-29 are each of length ℓ and width w , and the separation between them is d . The capacitor contains a dielectric block of dimensions $\ell \times w \times d$ and permittivity ϵ . The block can slide in and out of the capacitor cavity along its length dimension. Compute the force $\mathbf{F}_e$ acting on the dielectric block when it is partially outside of the cavity and the voltage across the capacitor is V .

Solution: From Eq. (4.122), the electric field inside the capacitor cavity is

$$E = \frac{V}{d}.$$

This is true in both the section containing the dielectric block and the section filled with air. The total electrostatic energy of

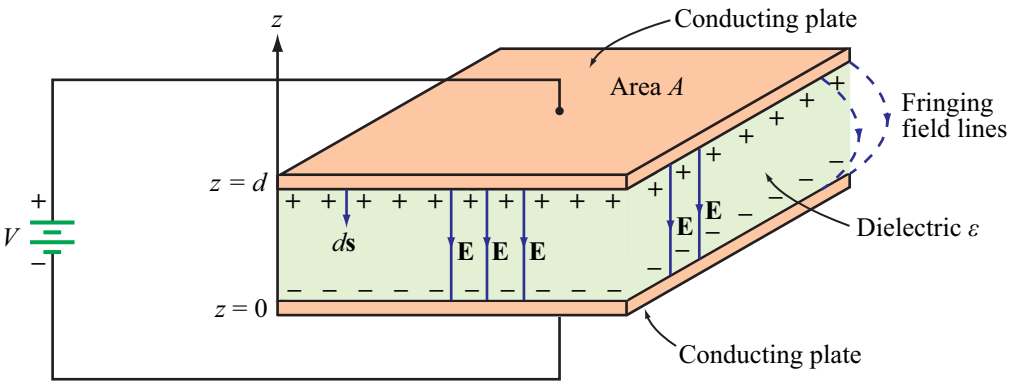


图 4-28 连接到平行板电容器的直流电压源。

作用在上板上的电力沿 $-\hat{z}$ （由于下板的吸引力）。因此，它由下式给出

$$\mathbf{F}_e = -\hat{z} F_e \quad (\text{force on upper plate}). \quad (4.135)$$

我们的计划是从能量考虑来计算 $\mathbf{F}_e$ 。我们首先将间距 d 转换为变量 z 并在等式中使用 $C = \epsilon A/z$ 。 (4.131)：

$$W_e = \frac{1}{2} CV^2 = \frac{1}{2} \frac{\epsilon AV^2}{z}. \quad (4.136)$$

如果 V 保持在恒定水平，则当增加板之间的间距 z 时， W_e 会减小。如果施加一个向上的外力 $\mathbf{F} = -\mathbf{F}_e$ 来抵消静电力 $\mathbf{F}_e$ 并将上板向上移动距离 dz ，则消耗的机械功为

$$dW = \mathbf{F} \cdot \hat{z} dz. \quad (4.137)$$

功 dW 等于电容器中存储的静电能的损失。那是，

$$dW = -dW_e. \quad (4.138)$$

此外， $\mathbf{F}_e = -\mathbf{F}$ ，这导致

$$dW_e = \mathbf{F}_e \cdot \hat{z} dz = -\hat{z} F_e \cdot \hat{z} dz = -F_e dz. \quad (4.139)$$

从方程。 (4.136),

$$dW_e = -\frac{1}{2} \epsilon \frac{AV^2}{z^2} dz. \quad (4.140)$$

等式(4.139) 和 (4.140) 并将 z 替换为 d 导致

$$F_e = \frac{1}{2} \epsilon \frac{AV^2}{d^2}, \quad (4.141a)$$

和

$$\mathbf{F}_e = -\hat{z} \frac{1}{2} \epsilon A \frac{V^2}{d^2} \quad (\text{N}). \quad (4.141b)$$

(平行板电容器)

这是施加在上板上的静电力。下板上的力大小相同，方向相反。

由方程式给出的关系(4.139) 涉及具有 $d\mathbf{l} = \hat{z} dz$ 的容器。我们可以将 $d\mathbf{l}$ 沿任意方向的结果概括为

$$\mathbf{F}_e = -\nabla W_e. \quad (4.142)$$

例 4-16：滑动电介质上的力

图4-29所示的平行板电容器的两个极板的长度为 ℓ ，宽度为 w ，它们之间的间距为 d 。电容器包含尺寸为 $\ell \times w \times d$ 和介电常数为 ϵ 的介电块。该块可以沿着其长度尺寸滑入和滑出电容器腔。计算当介电块部分位于腔体外部并且电容器两端的电压为 V 时作用在介电块上的力 $\mathbf{F}_e$ 。

解决方案：从方程。 (4.122)，电容器腔内的电场为

$$E = \frac{V}{d}.$$

这对于包含介电块的部分和充满空气的部分都是如此。总静电能量为

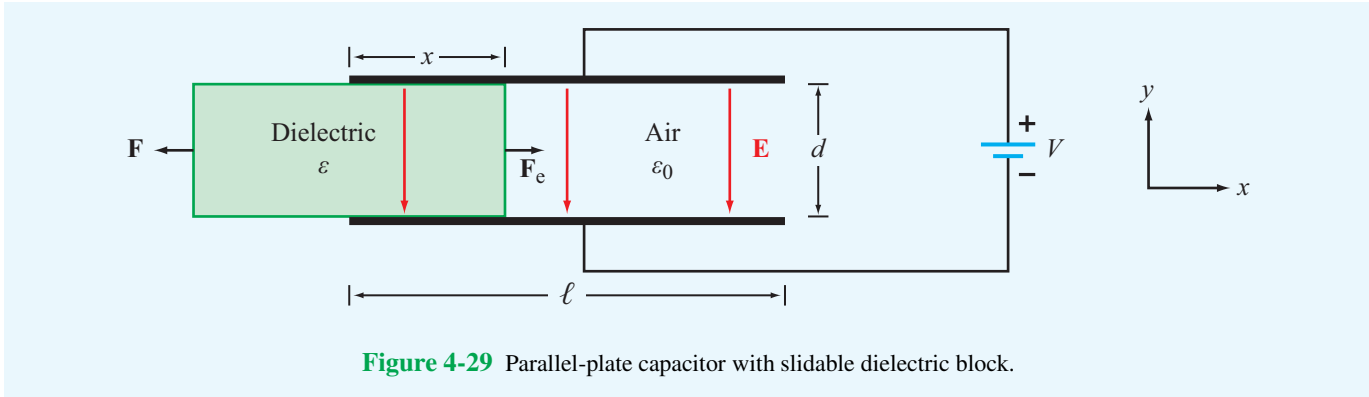


Figure 4-29 Parallel-plate capacitor with slidable dielectric block.

the capacitor consists of two components: one for the volume containing the dielectric block of permittivity ϵ and volume $\mathcal{V}_1 = xwd$, and another for the volume containing air with ϵ_0 and volume $\mathcal{V}_2 = (\ell - x)wd$. Hence,

$$\begin{aligned} W_e &= \frac{1}{2} \epsilon E^2 \mathcal{V}_1 + \frac{1}{2} \epsilon_0 E^2 \mathcal{V}_2 \\ &= \frac{1}{2} \epsilon \left(\frac{V}{d} \right)^2 xwd + \frac{1}{2} \epsilon_0 \left(\frac{V}{d} \right)^2 (\ell - x)wd \\ &= \frac{1}{2} \frac{V^2}{d} w [\epsilon x + \epsilon_0 (\ell - x)]. \end{aligned} \tag{4.143}$$

Since $\epsilon > \epsilon_0$, the electrostatic energy is maximum when $x = \ell$ (dielectric block fully inside the cavity). Sliding the dielectric block out of the capacitor requires exerting an external mechanical force $\mathbf{F}$ to oppose the electrostatic force $\mathbf{F}_e$, whose tendency is to oppose reduction in W_e . Thus, the direction of $\mathbf{F}_e$ is to pull the block back into the capacitor.

The magnitude of $\mathbf{F}_e$ can be obtained from

$$\begin{aligned} F_e &= \frac{dW_e}{dx} \\ &= \frac{d}{dx} \left[\frac{1}{2} \frac{V^2}{d} w [\epsilon x + \epsilon_0 (\ell - x)] \right] \\ &= \frac{1}{2} \frac{V^2}{d} w (\epsilon - \epsilon_0). \end{aligned} \tag{4.144}$$

Concept Question 4-26: To bring a charge q from infinity to a given point in space, a certain amount of work W is expended. Where does the energy corresponding to W go?

Concept Question 4-27: When a voltage source is connected across a capacitor, what is the direction of the electrical force acting on its two conducting surfaces?

Exercise 4-18: The radii of the inner and outer conductors of a coaxial cable are 2 cm and 5 cm, respectively, and the insulating material between them has a relative permittivity of 4. The charge density on the outer conductor is $\rho_\ell = 10^{-4}$ (C/m). Use the expression for $\mathbf{E}$ derived in Example 4-15 to calculate the total energy stored in a 20 cm length of the cable.

Answer: $W_e = 4.1$ J. (See [EM.1](#).)

4-11 Image Method

Consider a point charge Q at a distance d above a horizontally infinite, perfectly conducting plate (**Fig. 4-30(a)**). We want to determine V and $\mathbf{E}$ at any point in the space above the plate, as well as the surface charge distribution on the plate. Three different methods for finding $\mathbf{E}$ have been introduced in this chapter. The first method, based on Coulomb's law, requires knowledge of the magnitudes and locations of all the charges. In the present case, the charge Q induces an unknown and nonuniform distribution of charge on the plate. Hence, we cannot utilize Coulomb's method. The second method, based on Gauss's law, is equally difficult to use because it is not clear how to construct a Gaussian surface across which $\mathbf{E}$ is only tangential or only normal. The third method is based on evaluating the electric field using $\mathbf{E} = -\nabla V$ after solving Poisson's or Laplace's equation for V subject to the available boundary conditions, but it is mathematically involved.

Alternatively, the problem at hand can be solved using *image theory*.

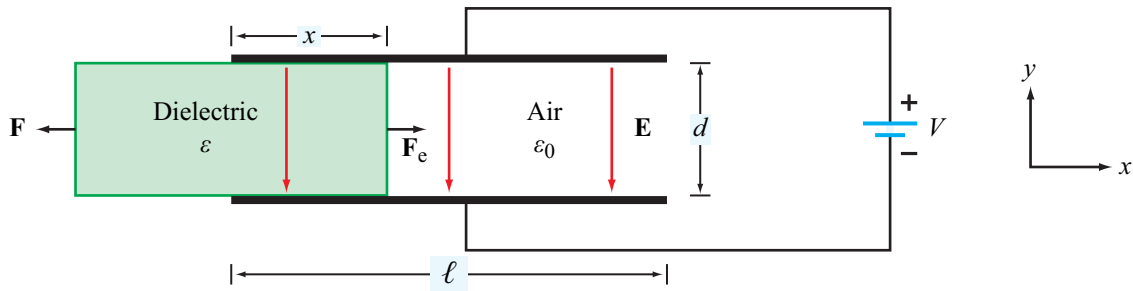


图 4-29 带有可滑动电介质块的平行板电容器。

该电容器由两部分组成：一个部分为包含介电常数为 ϵ 且体积为 $\mathcal{V}_1 = xwd$ 的介电块的体积，另一个部分为包含具有 ϵ_0 和体积 $\mathcal{V}_2 = (\ell - x)wd$ 的空氣的体积。因此，

$$\begin{aligned} W_e &= \frac{1}{2} \epsilon E^2 \mathcal{V}_1 + \frac{1}{2} \epsilon_0 E^2 \mathcal{V}_2 \\ &= \frac{1}{2} \epsilon \left(\frac{V}{d} \right)^2 xwd + \frac{1}{2} \epsilon_0 \left(\frac{V}{d} \right)^2 (\ell - x)wd \\ &= \frac{1}{2} \frac{V^2}{d} w [\epsilon x + \epsilon_0 (\ell - x)]. \end{aligned} \tag{4.143}$$

自从 $\epsilon > \epsilon_0$ ，当 x 时静电能最大 = ℓ (介电块完全位于空腔内)。将介电块滑出电容器需要施加外部机械力 F 来抵抗静电力 F_e ，其倾向是反对减少 W_e 。因此， F 的方向 e 是将块拉回到电容器中。

F_e 的大小可由下式获得

$$\begin{aligned} F_e &= \frac{dW_e}{dx} \\ &= \frac{d}{dx} \left[\frac{1}{2} \frac{V^2}{d} w [\epsilon x + \epsilon_0 (\ell - x)] \right] \\ &= \frac{1}{2} \frac{V^2}{d} w (\epsilon - \epsilon_0). \end{aligned} \tag{4.144}$$

概念问题 4-26: 为了将电荷 q 从无穷大带到空间中的给定点，需要消耗一定量的功 W 。 W 对应的能量去了哪里？

概念问题 4-27: 当电压源连接在电容器两端时，作用在其两个导电表面上的电力的方向是什么？

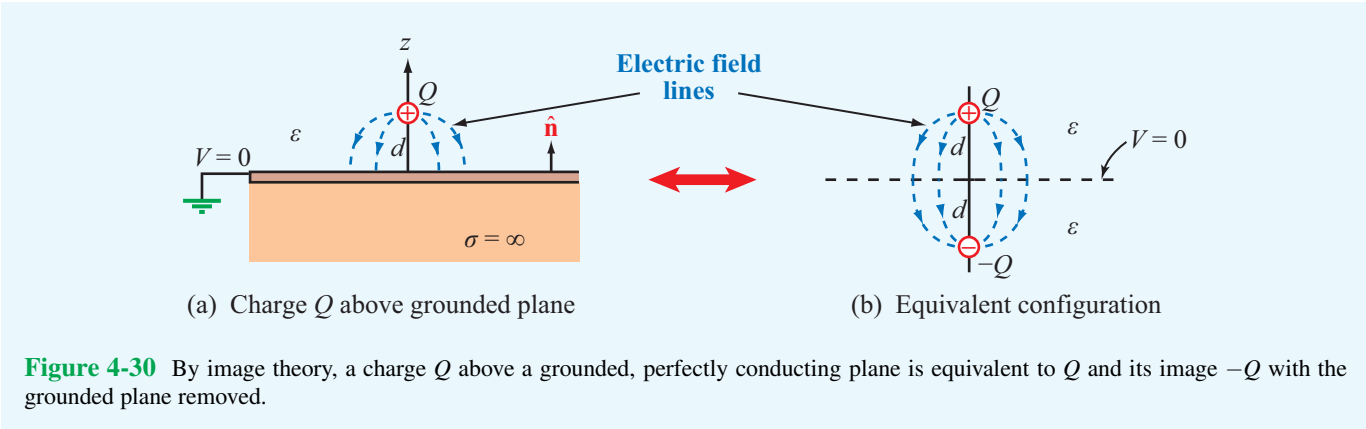
练习 4-18: 同轴电缆的内导体和外导体的半径分别为 2 cm 和 5 cm，它们之间的绝缘材料的相对介电常数为 4。外导体上的电荷密度为 $\rho_\ell = 10^{-4}$ (C/m)。使用例 4-15 中导出的 $\mathbf{E}$ 表达式来计算 20 cm 长度的电缆中存储的总能量。

答案: $W_e = 4.1$ J. (见 [EM.1](#)。)

4-11 镜像法

考虑在水平无限、理想导电的板上方距离 d 处的点电荷 Q (图 4-30(a))。我们想要确定板上方空间中任意点的 V 和 $\mathbf{E}$ ，以及板上的表面电荷分布。本章介绍了寻找 $\mathbf{E}$ 的三种不同方法。第一种方法基于库仑定律，需要了解所有电荷的大小和位置。在本例中，电荷 Q 在板上产生未知且不均匀的电荷分布。因此，我们不能使用库仑法。第二种方法基于高斯定律，同样难以使用，因为尚不清楚如何构造 $\mathbf{E}$ 仅为切线或法线的高斯曲面。第三种方法是在根据可用边界条件求解 V 的泊松或拉普拉斯方程后，使用 $\mathbf{E} = -\nabla V$ 评估电场，但它涉及数学。

或者，可以使用镜像理论来解决当前的问题。



► Any given charge configuration above an infinite, perfectly conducting plane is electrically equivalent to the combination of the given charge configuration and its image configuration with the conducting plane removed. ◀

The process is continued until the condition $V = 0$ is satisfied everywhere on all grounded planes. The image method applies not only to point charges but also to distributions of charge, such as the line and volume distributions depicted in Fig. 4-31. Once $\mathbf{E}$ has been determined, the charge induced on the plate can be found from

$$\rho_s = (\hat{\mathbf{n}} \cdot \mathbf{E})\epsilon_0, \quad (4.145)$$

where $\hat{\mathbf{n}}$ is the normal unit vector to the plate (Fig. 4-30(a)).

Example 4-17: Image Method for Charge above Conducting Plane

Use image theory to determine $\mathbf{E}$ at an arbitrary point $P = (x, y, z)$ in the region $z > 0$ due to a charge Q in free space at a distance d above a grounded conducting plate residing in the $z = 0$ plane.

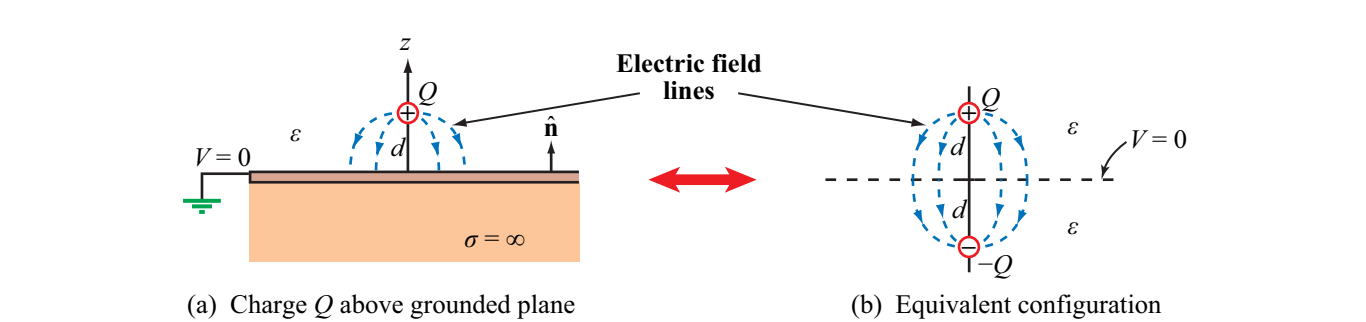
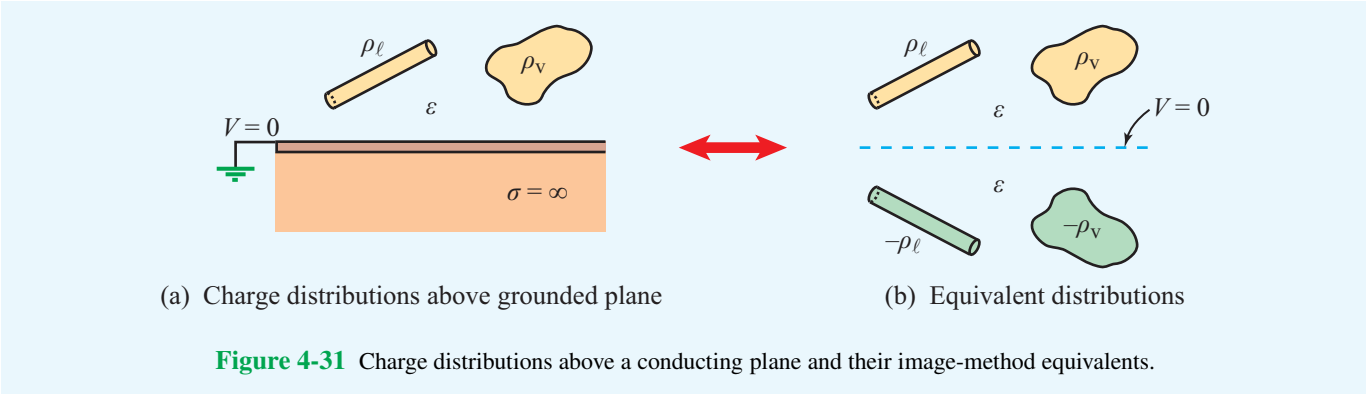


图 4-30 根据镜像理论，接地、理想导电平面上方的电荷 Q 等价于 Q 及其镜像 $-Q$ （除去接地平面）。

► 无限、理想导电平面上方的任何给定电荷配置在电学上等效于给定电荷配置及其去除导电平面后的镜像配置的组合。 ◀

继续该过程，直到所有接地平面上的所有位置都满足条件 $V = 0$ 。镜像法不仅适用于点电荷，也适用于电荷分布，如图4-31所示的线分布和体分布。一旦 $\mathbf{E}$ 确定，板上感应的电荷可以由下式找到：

$$\rho_s = (\hat{\mathbf{n}} \cdot \mathbf{E})\epsilon_0, \quad (4.145)$$

其中 $\hat{\mathbf{n}}$ 是板的法向单位矢量（图4-30(a)）。

例 4-17：导电平面上方电荷的镜像法

导电平面上方电荷 Q 的镜像法等效如图 4-30 的右侧部分所示。它由电荷 Q 本身和距 Q $2d$ 的镜像电荷 $-Q$ 组成，它们之间没有其他任何东西。现在，通过应用库仑法，可以轻松地在任意点 (x, y, z) 找到由两个隔离电荷产生的电场，如例 4-17 所示。根据对称性，两个电荷的组合总是在先前由导电表面占据的平面中的每个点产生电势 $V = 0$ 。如果电荷存在于多个接地平面中，则有必要建立其相对于每个平面的镜像，然后建立每个镜像相对于其余平面的镜像。

使用镜像理论来确定 $z > 0$ 区域中任意点 $P = (x, y, z)$ 处的 $\mathbf{E}$ ，这是由于位于 $z = 0$ 平面中的接地导电板上距离 d 处的自由空间中的电荷 Q 造成的。

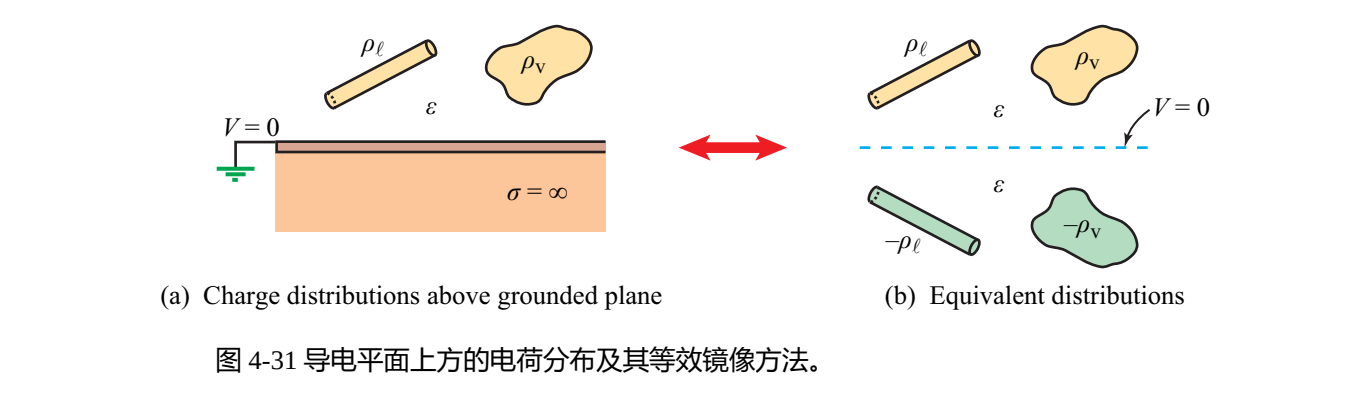


图 4-31 导电平面上方的电荷分布及其等效镜像方法。

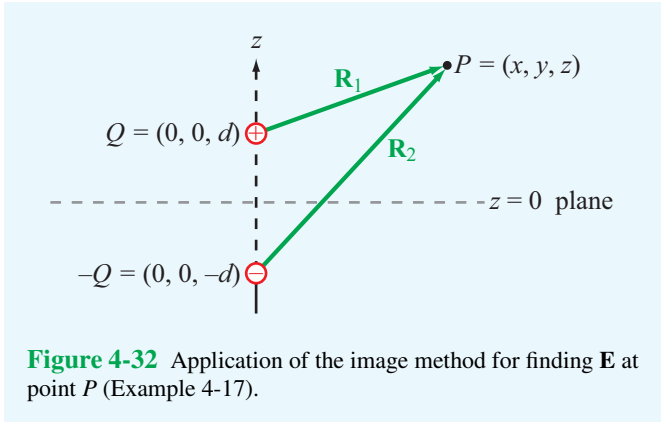


Figure 4-32 Application of the image method for finding $\mathbf{E}$ at point P (Example 4-17).

Solution: In Fig. 4-32, charge Q is at $(0,0,d)$, and its image $-Q$ is at $(0,0,-d)$. From Eq. (4.19), the electric field at point $P(x,y,z)$ due to the two charges is given by the following equation.

$$\begin{aligned}\mathbf{E} &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q\mathbf{R}_1}{R_1^3} + \frac{-Q\mathbf{R}_2}{R_2^3} \right) \\ &= \frac{Q}{4\pi\epsilon_0} \left[\frac{\hat{\mathbf{x}}x + \hat{\mathbf{y}}y + \hat{\mathbf{z}}(z-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} - \frac{\hat{\mathbf{x}}x + \hat{\mathbf{y}}y + \hat{\mathbf{z}}(z+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right]\end{aligned}$$

for $z \geq 0$.

Concept Question 4-28: What is the fundamental premise of the image method?

Concept Question 4-29: Given a charge distribution, what are the various approaches described in this chapter for computing the electric field $\mathbf{E}$ at a given point in space?

Exercise 4-19: Use the result of Example 4-17 to find the surface charge density ρ_s on the surface of the conducting plane.

Answer: $\rho_s = -Qd/[2\pi(x^2 + y^2 + d^2)^{3/2}]$. (See [EM](#).)

Chapter 4 Summary

Concepts

- Maxwell's equations are the fundamental tenets of electromagnetic theory.
- Under static conditions, Maxwell's equations separate into two uncoupled pairs with one pair pertaining to electrostatics and the other to magnetostatics.
- Coulomb's law provides an explicit expression for the electric field due to a specified charge distribution.
- Gauss's law states that the total electric field flux through a closed surface is equal to the net charge enclosed by the surface.
- The electrostatic field $\mathbf{E}$ at a point is related to the electric potential V at that point by $\mathbf{E} = -\nabla V$ with V often being referenced to zero at infinity.
- Because most metals have conductivities on the order of 10^6 (S/m), they are treated in practice as perfect conductors. By the same token, insulators with conduc-

- tivities smaller than 10^{-10} (S/m) often are treated as perfect dielectrics.
- Boundary conditions at the interface between two materials specify the relations between the normal and tangential components of $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{J}$ in one of the materials to the corresponding components in the other.
- The capacitance of a two-conductor body and resistance of the medium between them can be computed from knowledge of the electric field in that medium.
- The electrostatic energy density stored in a dielectric medium is $w_e = \frac{1}{2}\epsilon E^2$ (J/m³).
- When a charge configuration exists above an infinite, perfectly conducting plane, the induced field $\mathbf{E}$ is the same as that due to the configuration itself and its image with the conducting plane removed.

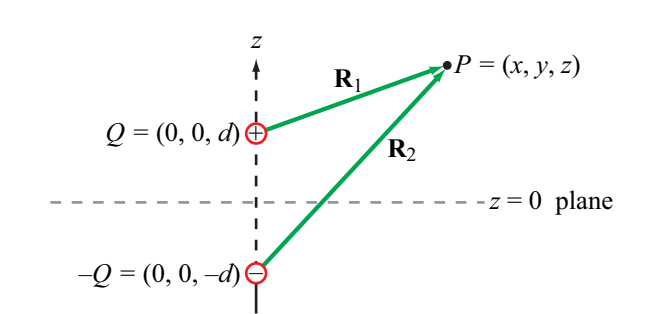


图 4-32 应用镜像法求 P 点 E (例 4-17)。

解：在图4-32中，电荷Q在(0,0,d)处，其像-Q在(0,0,-d)处。从方程。根据 (4.19) 式，两个电荷在点 P(x, y, z) 处的电场由以下方程给出。

$$\begin{aligned}\mathbf{E} &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q\mathbf{R}_1}{R_1^3} + \frac{-Q\mathbf{R}_2}{R_2^3} \right) \\ &= \frac{Q}{4\pi\epsilon_0} \left[\frac{\hat{\mathbf{x}}x + \hat{\mathbf{y}}y + \hat{\mathbf{z}}(z-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} - \frac{\hat{\mathbf{x}}x + \hat{\mathbf{y}}y + \hat{\mathbf{z}}(z+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right]\end{aligned}$$

对于 $z \geq 0$ 。

概念题4-28：镜像法的基本前提是什么？

概念问题 4-29：给定电荷分布，本章描述的计算空间给定点电场 $\mathbf{E}$ 的各种方法有哪些？

练习4-19：利用例4-17的结果求导电平面表面的表面电荷密度 ρ_s 。

答案： $\rho_s = -Qd/[2\pi(x^2 + y^2 + d^2)^{3/2}]$ 。（参见^{EM}。）

第四章总结

概念

- 麦克斯韦方程组是电磁理论的基本原理。
- 在静态条件下，麦克斯韦方程分为两对非耦合方程，一对涉及静电学，另一对涉及静磁学。
- 库仑定律为特定电荷分布产生的电场提供了明确的表达式。
- 高斯定律指出，通过封闭表面的总电通量 等于该表面所包围的净电荷。
- 某一点的静电场 $\mathbf{E}$ 与该点的电势 V 的关系为 $\mathbf{E} = -\nabla V$ ，其中 V 通常在无穷远处为零。
- 多数金属的电导率约为 10^6 S/m，在本章近似下通常视为理想导体。同理，电导率低于

- 10^{-10} S/m 的绝缘体通常近似视为理想电介质。
- 两种材料之间界面处的边界条件指定了一种材料中 $\mathbf{D}$ 、 $\mathbf{E}$ 和 $\mathbf{J}$ 的法向和切向分量与另一种材料中相应分量之间的关系。
- 双导体体的电容和它们之间的介质的电阻可以根据该介质中的电场的知识来计算。
- 存储在介电介质中的静电能量密度为 $w_e = \frac{1}{2}\epsilon E^2$ (J/m³)。
- 当电荷构型存在于无限的、理想的导电平面上方时，感应场 $\mathbf{E}$ 与构型本身及其镜像（除去导电平面后）产生的场相同。

Mathematical and Physical Models

Maxwell’s Equations for Electrostatics

Name	Differential Form	Integral Form
Gauss’s law	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$
Kirchhoff’s law	$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$

Electric Field

Current density	$\mathbf{J} = \rho_v \mathbf{u}$	Point charge	$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2}$
Poisson’s equation	$\nabla^2 V = -\frac{\rho_v}{\epsilon}$	Many point charges	$\mathbf{E} = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i(\mathbf{R} - \mathbf{R}_i)}{ \mathbf{R} - \mathbf{R}_i ^3}$
Laplace’s equation	$\nabla^2 V = 0$	Volume distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{v'} \hat{\mathbf{R}}' \frac{\rho_v d\mathbf{v}'}{R'^2}$
Resistance	$R = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}$	Surface distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{s'} \hat{\mathbf{R}}' \frac{\rho_s ds'}{R'^2}$
Boundary conditions	Table 4-3	Line distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_\ell dl'}{R'^2}$
Capacitance	$C = \frac{\int_S \epsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_l \mathbf{E} \cdot d\mathbf{l}}$	Infinite sheet of charge	$\mathbf{E} = \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0}$
RC relation	$RC = \frac{\epsilon}{\sigma}$	Infinite line of charge	$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \hat{\mathbf{r}} \frac{D_r}{\epsilon_0} = \hat{\mathbf{r}} \frac{\rho_\ell}{2\pi\epsilon_0 r}$
Energy density	$w_e = \frac{1}{2}\epsilon E^2$	Dipole	$\mathbf{E} = \frac{qd}{4\pi\epsilon_0 R^3} (\hat{\mathbf{R}} 2\cos\theta + \hat{\boldsymbol{\theta}} \sin\theta)$
		Relation to V	$\mathbf{E} = -\nabla V$

PROBLEMS

Sections 4-2: Charge and Current Distributions

*4.1 A cube 2 m on a side is located in the first octant in a Cartesian coordinate system with one of its corners at the origin. Find the total charge contained in the cube if the charge density is given by $\rho_v = xy^2e^{-2z}$ (mC/m³).

4.2 Find the total charge contained in a cylindrical volume defined by $r \leq 2$ m and $0 \leq z \leq 3$ m if $\rho_v = 20rz$ (mC/m³).

*4.3 Find the total charge contained in a round-top cone defined by $R \leq 2$ m and $0 \leq \theta \leq \pi/4$ given that $\rho_v = 10R^2\cos^2\theta$ (mC/m³).

4.4 If the line charge density is given by $\rho_l = 24y^2$ (mC/m), find the total charge distributed on the y axis from $y = -5$ to $y = 5$.

* Answer(s) available in Appendix E.

数学与物理模型

静电场的麦克斯韦方程组

名称	微分形式	积分形式
高斯定律	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$
基尔霍夫定律	$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$

电场

电流密度	$\mathbf{J} = \rho_v \mathbf{u}$	点电荷	$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2}$
泊松方程	$\nabla^2 V = -\frac{\rho_v}{\epsilon}$	多个点电荷	$\mathbf{E} = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i(\mathbf{R} - \mathbf{R}_i)}{ \mathbf{R} - \mathbf{R}_i ^3}$
拉普拉斯方程	$\nabla^2 V = 0$	体电荷分布	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{v'} \hat{\mathbf{R}}' \frac{\rho_v d\mathbf{v}'}{R'^2}$
电阻	$R = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}$	面电荷分布	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{s'} \hat{\mathbf{R}}' \frac{\rho_s ds'}{R'^2}$
边界条件	见表4-3	线电荷分布	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_\ell dl'}{R'^2}$
电容	$C = \frac{\int_S \epsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_l \mathbf{E} \cdot d\mathbf{l}}$	无限均匀面电荷	$\mathbf{E} = \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0}$
RC 关系	$RC = \frac{\epsilon}{\sigma}$	无限均匀线电荷	$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \hat{\mathbf{r}} \frac{D_r}{\epsilon_0} = \hat{\mathbf{r}} \frac{\rho_\ell}{2\pi\epsilon_0 r}$
电能密度	$w_e = \frac{1}{2}\epsilon E^2$	电偶极子	$\mathbf{E} = \frac{qd}{4\pi\epsilon_0 R^3} (\hat{\mathbf{R}} 2\cos\theta + \hat{\boldsymbol{\theta}} \sin\theta)$
说明：面电荷位于 $z=0$ ；表中方向对应 $z>0$, $z<0$ 时电场反向。电偶极子公式适用于 R 远大于电荷间距 d 。		与电势 V 的关系	$\mathbf{E} = -\nabla V$

问题

第 4-2 节：电荷和电流分布

*4.1 边长为 2 m 的正方体位于笛卡尔坐标系中的第一个八分圆内，其一个角位于原点。如果电荷密度由 $\rho_v = xy^2e^{-2z}$ (mC/m³) 给出，则求立方体中包含的总电荷。

4.2 求由 $r \leq 2$ m 和 $0 \leq z \leq 3$ m if $\rho_v = 20rz$ (mC/m³) 定义的圆柱形体积中包含的总电荷。

*4.3 给定 $\rho_v = 10R^2\cos^2\theta$ (mC/m³)，求 $R \leq 2$ m 和 $0 \leq \theta \leq \pi/4$ 定义的圆顶锥体中包含的总电荷。

4.4 如果线电荷密度由 $\rho_l = 24y^2$ (mC/m) 给出，求分布在 y 轴上从 $y = -5$ 到 $y = 5$ 的总电荷。

* 标记的习题在附录 E 提供答案。

Important Terms

Provide definitions or explain the meaning of the following terms:

boundary conditions	electric field intensity $\mathbf{E}$	Joule's law
capacitance C	electric flux density $\mathbf{D}$	Kirchhoff's voltage law
charge density	electric potential V	Laplace's equation
conductance G	electric susceptibility χ_e	linear material
conduction current	electron drift velocity $\mathbf{u}_e$	Ohm's law
conductivity σ	electron mobility μ_e	perfect conductor
conductor	electrostatic energy density w_e	perfect dielectric
conservative field	electrostatic potential energy W_e	permittivity ϵ
constitutive parameters	electrostatics	Poisson's equation
convection current	equipotential	polarization vector $\mathbf{P}$
Coulomb's law	Gaussian surface	relative permittivity ϵ_r
current density $\mathbf{J}$	Gauss's law	semiconductor
dielectric breakdown voltage V_{br}	hole drift velocity $\mathbf{u}_h$	static condition
dielectric material	hole mobility μ_h	superconductor
dielectric strength E_{ds}	homogeneous material	volume, surface, and line
dipole moment $\mathbf{p}$	image method	charge densities
electric dipole	isotropic material	

4.5 Find the total charge on a circular disk defined by $r \leq a$ and $z = 0$ if:

- (a) $\rho_s = \rho_{s0} \cos \phi$ (C/m²)
- (b) $\rho_s = \rho_{s0} \sin^2 \phi$ (C/m²)
- (c) $\rho_s = \rho_{s0} e^{-r}$ (C/m²)
- (d) $\rho_s = \rho_{s0} e^{-r} \sin^2 \phi$ (C/m²)

where ρ_{s0} is a constant.

4.6 If $\mathbf{J} = \hat{\mathbf{y}}4xz$ (A/m²), find the current I flowing through a square with corners at (0,0,0), (2,0,0), (2,0,2), and (0,0,2).

* 4.7 If $\mathbf{J} = \hat{\mathbf{R}}5/R$ (A/m²), find I through the surface $R = 5$ m.

4.8 An electron beam shaped like a circular cylinder of radius r_0 carries a charge density given by

$$\rho_v = \left(\frac{-\rho_0}{1+r^2} \right) \quad (\text{C/m}^3)$$

where ρ_0 is a positive constant and the beam's axis is coincident with the z axis.

- (a) Determine the total charge contained in length L of the beam.
- (b) If the electrons are moving in the $+z$ direction with uniform speed u , determine the magnitude and direction of the current crossing the z -plane.

4.9 A circular beam of charge of radius a consists of electrons moving with a constant speed u along the $+z$ direction.

The beam's axis is coincident with the z axis and the electron charge density is given by

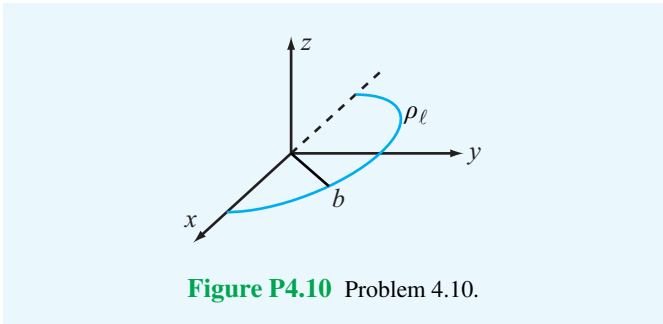
$$\rho_v = -cr^2 \quad (\text{C/m}^3)$$

where c is a constant and r is the radial distance from the axis of the beam.

* (a) Determine the charge density per unit length.

(b) Determine the current crossing the z -plane.

4.10 A line of charge of uniform density ρ_ℓ occupies a semicircle of radius b as shown in Fig. P4.10. Use the material presented in Example 4-4 to determine the electric field at the origin.



Important Terms

Provide definitions or explain the meaning of the following terms:

boundary conditions	electric field intensity $\mathbf{E}$	Joule's law
capacitance C	electric flux density $\mathbf{D}$	Kirchhoff's voltage law
charge density	electric potential V	Laplace's equation
conductance G	electric susceptibility χ_e	linear material
conduction current	electron drift velocity $\mathbf{u}_e$	Ohm's law
conductivity σ	electron mobility μ_e	perfect conductor
conductor	electrostatic energy density w_e	perfect dielectric
conservative field	electrostatic potential energy W_e	permittivity ϵ
constitutive parameters	electrostatics	Poisson's equation
convection current	equipotential	polarization vector $\mathbf{P}$
Coulomb's law	Gaussian surface	relative permittivity ϵ_r
current density $\mathbf{J}$	Gauss's law	semiconductor
dielectric breakdown voltage V_{br}	hole drift velocity $\mathbf{u}_h$	static condition
dielectric material	hole mobility μ_h	superconductor
dielectric strength E_{ds}	homogeneous material	volume, surface, and line
dipole moment $\mathbf{p}$	image method	charge densities
electric dipole	isotropic material	

4.5 求由 $r \leq a$ 和 $z = 0$ 定义的圆盘上的总电荷，如果： 电子束的轴与 z 轴重合，电子电荷密度由下式给出

- (a) $\rho_s = \rho_{s0} \cos \phi$ (C/m²) (b) $\rho_s = \rho_{s0} \sin^2 \phi$ (C/m²) (c) $\rho_s = \rho_{s0} e^{-r}$ (C/m²) (d) $\rho_s = \rho_{s0} e^{-r} \sin^2 \phi$ (C/m²)
- () () () $\rho_v = -cr^2$ (C/m³)

其中 c 是常数， r 是距束轴的径向距离。

其中 ρ_{s0} 是常数。

4.6 如果 $\mathbf{J} = \hat{\mathbf{y}}4xz$ (A/m²)，求流经角点为 (0, 0, 0)、(2, 0, 0)、(2, 0, 2) 和 (0, 0, 2) 的正方形的电流 I 。

* 4.7 如果 $\mathbf{J} = \hat{\mathbf{R}}5/R$ (A/m²)，求 I 穿过表面 $R = 5$ m。

4.8 形状像半径为 r_0 的圆柱体的电子束携带的电荷密度为：

$$\rho_v = \left(\frac{-\rho_0}{1+r^2} \right) \quad (\text{C/m}^3)$$

其中 ρ_0 是正常数，束轴与 z 轴重合。

- (a) 确定束流长度 L 中包含的总电荷。(b) 如果电子以均匀速度 u 沿 $+z$ 方向移动，请确定穿过 z 平面的电流的大小和方向。

4.9 半径为 a 的圆形电荷束由沿 $+z$ 方向以恒定速度 u 移动的电子组成。

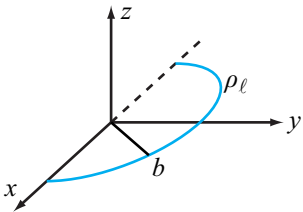


图 P4.10 问题 4.10。

Section 4-3: Coulomb’s Law

*4.11 A square with sides of 2 m has a charge of $40\text{ }\mu\text{C}$ at each of its four corners. Determine the electric field at a point 5 m above the center of the square.

4.12 Three point charges, each with $q = 3\text{ nC}$, are located at the corners of a triangle in the x - y plane, with one corner at the origin, another at (2 cm,0,0), and the third at (0,2 cm,0). Find the force acting on the charge located at the origin.

*4.13 Charge $q_1 = 6\text{ }\mu\text{C}$ is located at (1 cm,1 cm,0) and charge q_2 is located at (0,0,4 cm). What should q_2 be so that $\mathbf{E}$ at (0,2 cm,0) has no y component?

4.14 A line of charge with uniform density $\rho_\ell = 8\text{ }(\mu\text{C/m})$ exists in air along the z axis between $z = 0$ and $z = 5\text{ cm}$. Find $\mathbf{E}$ at (0,10 cm,0).

4.15 Electric charge is distributed along an arc located in the x - y plane and defined by $r = 2\text{ cm}$ and $0 \leq \phi \leq \pi/4$. If $\rho_\ell = 5\text{ }(\mu\text{C/m})$, find $\mathbf{E}$ at (0,0, z) and then evaluate it at

- * (a) the origin,
- (b) $z = 5\text{ cm}$, and
- (c) $z = -5\text{ cm}$.

4.16 A line of charge with uniform density ρ_l extends between $z = -L/2$ and $z = L/2$ along the z axis. Apply Coulomb’s law to obtain an expression for the electric field at any point $P(r,\phi,0)$ on the x - y plane. Show that your result reduces to the expression given by Eq. (4.33) as the length L is extended to infinity.

*4.17 Repeat Example 4-5 for the circular disk of charge of radius a , but in the present case, assume the surface charge density to vary with r as

$$\rho_s = \rho_{s0}r^2 \quad (\text{C/m}^2)$$

where ρ_{s0} is a constant.

4.18 Multiple charges at different locations are said to be in equilibrium if the force acting on any one of them is identical in magnitude and direction to the force acting on any of the others. Suppose we have two negative charges: one located at the origin and carrying charge $-9e$ and the other located on the positive x axis at a distance d from the first one and carrying charge $-36e$. Determine the location, polarity, and magnitude of a third charge whose placement would bring the entire system into equilibrium.

Section 4-4: Gauss’s Law

*4.19 Three infinite lines of charge, all parallel to the z axis, are located at the three corners of the kite-shaped arrangement

shown in Fig. P4.19. If the two right triangles are symmetrical and of equal corresponding sides, show that the electric field is zero at the origin.

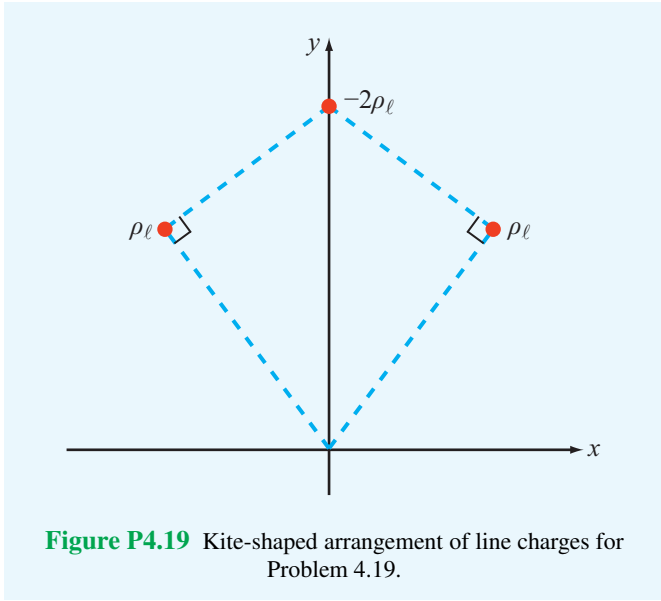


Figure P4.19 Kite-shaped arrangement of line charges for Problem 4.19.

4.20 Three infinite lines of charge, $\rho_{l1} = 3\text{ (nC/m)}$, $\rho_{l2} = -3\text{ (nC/m)}$, and $\rho_{l3} = 3\text{ (nC/m)}$ are all parallel to the z axis. If they pass through the respective points (0, $-b$), (0,0), and (0, b) in the x - y plane, find the electric field at (a ,0,0). Evaluate your result for $a = 2\text{ cm}$ and $b = 1\text{ cm}$.

4.21 A horizontal strip lying in the x - y plane is of width d in the y direction and infinitely long in the x direction. If the strip is in air and has a uniform charge distribution ρ_s , use Coulomb’s law to obtain an explicit expression for the electric field at a point P located at a distance h above the centerline of the strip. Extend your result to the special case where d is infinite and compare it with Eq. (4.25).

4.22 Given the electric flux density

$$\mathbf{D} = \hat{\mathbf{x}}2(x+y) + \hat{\mathbf{y}}(3x-2y) \quad (\text{C/m}^2),$$

determine

- (a) ρ_v by applying Eq. (4.26),
- (b) the total charge Q enclosed in a cube 2 m on a side, located in the first octant with three of its sides coincident with the x , y , and z axes and one of its corners at the origin, and
- (c) the total charge Q in the cube, obtained by applying Eq. (4.29).

第 4-3 节：库仑定律

*4.11 边长为 2 m 的正方形，其四个角的电荷均为 $40\text{ }\mu\text{C}$ 。确定正方形中心上方 5 m 处的电场。

4.12 三个点电荷，每个 $q = 3\text{ nC}$ ，位于 x - y 平面中三角形的角上，其中一个角位于原点，另一个角位于 (2 cm, 0, 0)，第三个角位于 (0, 2 cm, 0)。求作用在原点电荷上的力。

*4.13 电荷 $q_1 = 6\text{ }\mu\text{C}$ 位于 (1 cm, 1 cm, 0)，电荷 q_2 位于 (0, 0, 4 cm)。 q_2 应该是多少，以便 $\mathbf{E}$ 在 (0, 2 cm, 0) 处没有 y 分量？

4.14 空气中存在一条密度均匀的电荷线 $\rho_\ell = 8\text{ }(\mu\text{C/m})$ ，沿 z 轴位于 $z = 0$ 和 $z = 5\text{ cm}$ 之间。在 (0,10 cm,0) 处找到 $\mathbf{E}$ 。

4.15 电荷沿 x - y 平面内的弧分布，由 $r = 2\text{ cm}$ 和 $0 \leq \phi \leq \pi/4$ 定义。如果 $\rho_\ell = 5\text{ }(\mu\text{C/m})$ ，则在 (0, 0, z) 处找到 $\mathbf{E}$ ，然后在

- * (a) 原点,
- (b) $z = 5\text{ 厘米}$ ，以及
- (c) $z = -5\text{ 厘米}$ 。

4.16 具有均匀密度 ρ_l 的电荷线沿 z 轴在 $z = -L/2$ 和 $z = L/2$ 之间延伸。应用库仑定律获得 x - y 平面上任意点 $P(r,\phi,0)$ 处的电场表达式。证明你的结果可以简化为方程式给出的表达式。(4.33) 当长度 L 扩展到无穷大时。

*4.17 对半径为 a 的电荷圆盘重复例 4-5，但在本例中，假设表面电荷密度随 r 变化为

$$\rho_s = \rho_{s0}r^2 \quad (\text{C/m}^2)$$

其中 ρ_{s0} 是常数。

4.18 如果作用在其中任何一个电荷上的力与作用在其他任何一个电荷上的力的大小和方向相同，则称不同位置的多个电荷处于平衡状态。假设我们有两个负电荷：一个位于原点并携带电荷 $-9e$ ，另一个位于正 x 轴上，距第一个负电荷距离 d ，并携带电荷 $-36e$ 。确定第三个电荷的位置、极性和大小，该电荷的放置将使整个系统达到平衡。

第 4-4 节：高斯定律

*4.19 三个无限电荷线均平行于 z 轴，位于风筝形排列的三个角处

如图P4.19所示。如果两个直角三角形对称且对应边相等，则表明原点处电场为零。

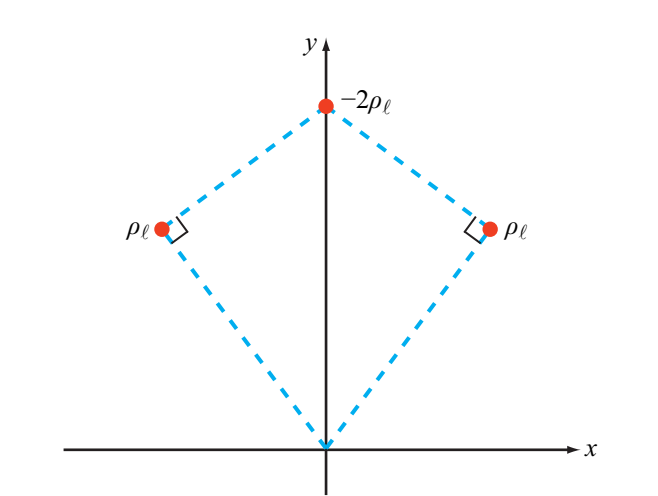


图 P4.19 问题 4.19 中线电荷的风筝形排列。

4.20 三个无限电荷线 $\rho_{l1} = 3\text{ (nC/m)}$ 、 $\rho_{l2} = -3\text{ (nC/m)}$ 和 $\rho_{l3} = 3\text{ (nC/m)}$ 均平行于 z 轴。如果它们穿过 x - y 平面上的相应点 (0, $-b$)、(0, 0) 和 (0, b)，则求 (a , 0, 0) 处的电场。评估 $a = 2\text{ 厘米}$ 和 $b = 1\text{ 厘米}$ 的结果。

4.21 位于 x - y 平面上的水平条带在 y 方向上的宽度为 d ，在 x 方向上的长度无限长。如果带材在空气中并且具有均匀的电荷分布 ρ_s ，则使用库仑定律获得位于带材中心线上方距离 h 处的点 P 处的电场的显式表达式。将你的结果扩展到 d 无穷大的特殊情况，并将其与方程进行比较。(4.25)。

4.22 给定电通量密度

$$\mathbf{D} = \hat{\mathbf{x}}2(x+y) + \hat{\mathbf{y}}(3x-2y) \quad (\text{C/m}^2),$$

决定

- (a) ρ_v 通过应用等式。(4.26)，(b) 包含在边长 2 m 的立方体中的总电荷 Q ，该立方体位于第一个八分圆内，其三个边与 x 、 y 和 z 轴重合，且其中一个角位于原点，以及
- (c) 通过应用式 (4.26) 获得的立方体中的总电荷 Q 。(4.29)。

*4.23 Repeat Problem 4.22 for $\mathbf{D} = \hat{x}xy^3z^3$ (C/m²).

4.24 Charge Q_1 is uniformly distributed over a thin spherical shell of radius a , and charge Q_2 is uniformly distributed over a second spherical shell of radius b with $b > a$. Apply Gauss's law to find $\mathbf{E}$ in the regions $R < a$, $a < R < b$, and $R > b$.

*4.25 The electric flux density inside a dielectric sphere of radius a centered at the origin is given by

$$\mathbf{D} = \hat{\mathbf{r}}\rho_0R \quad (\text{C/m}^2)$$

where ρ_0 is a constant. Find the total charge inside the sphere.

4.26 In a certain region of space, the charge density is given in cylindrical coordinates by the function

$$\rho_v = 5re^{-r} \quad (\text{C/m}^3).$$

Apply Gauss's law to find $\mathbf{D}$.

*4.27 An infinitely long cylindrical shell extending between $r = 1$ m and $r = 3$ m contains a uniform charge density ρ_{v0} . Apply Gauss's law to find $\mathbf{D}$ in all regions.

4.28 If the charge density increases linearly with distance from the origin such that $\rho_v = 0$ at the origin and $\rho_v = 4$ C/m³ at $R = 2$ m, find the corresponding variation of $\mathbf{D}$.

4.29 A spherical shell with outer radius b surrounds a charge-free cavity of radius $a < b$ (Fig. P4.29). If the shell contains a

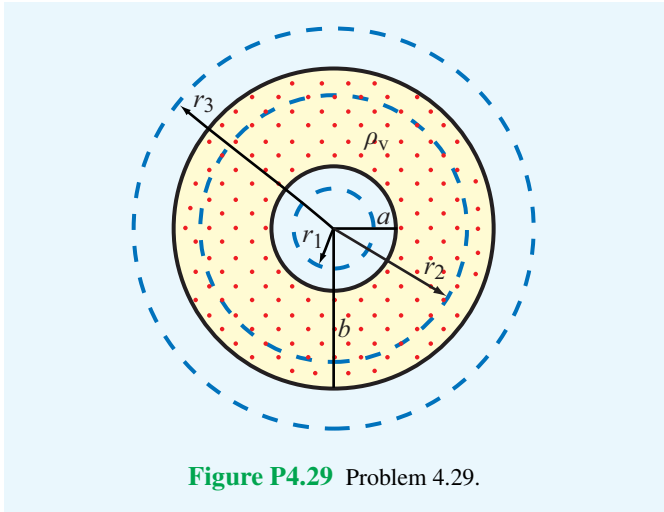


Figure P4.29 Problem 4.29.

charge density given by

$$\rho_v = -\frac{\rho_{v0}}{R^2}, \quad a \leq R \leq b,$$

where ρ_{v0} is a positive constant, determine $\mathbf{D}$ in all regions.

Section 4-5: Electric Potential

*4.30 A square in the x - y plane in free space has a point charge of $+Q$ at corner $(a/2, a/2)$, the same at corner $(a/2, -a/2)$, and a point charge of $-Q$ at each of the other two corners.

- (a) Find the electric potential at any point P along the x axis.
- (b) Evaluate V at $x = a/2$.

4.31 The circular disk of radius a shown in Fig. 4-7 has uniform charge density ρ_s across its surface.

- (a) Obtain an expression for the electric potential V at a point $P(0, 0, z)$ on the z axis.
- (b) Use your result to find $\mathbf{E}$ and then evaluate it for $z = h$. Compare your final expression with (4.24), which was obtained on the basis of Coulomb's law.

*4.32 A circular ring of charge of radius a lies in the x - y plane and is centered at the origin. Assume also that the ring is in air and carries a uniform density ρ_ℓ .

- (a) Show that the electrical potential at $(0, 0, z)$ is given by $V = \rho_\ell a / [2\epsilon_0(a^2 + z^2)^{1/2}]$.
- (b) Find the corresponding electric field $\mathbf{E}$.

4.33 Show that the electric potential difference V_{12} between two points in air at radial distances r_1 and r_2 from an infinite line of charge with density ρ_ℓ along the z axis is $V_{12} = (\rho_\ell / 2\pi\epsilon_0) \ln(r_2/r_1)$.

*4.34 Find the electric potential V at a location a distance b from the origin in the x - y plane due to a line charge with charge density ρ_ℓ and of length l . The line charge is coincident with the z axis and extends from $z = -l/2$ to $z = l/2$.

4.35 For the electric dipole shown in Fig. 4-15, $d = 1$ cm and $|\mathbf{E}| = 4$ (mV/m) at $R = 1$ m and $\theta = 0^\circ$. Find $\mathbf{E}$ at $R = 2$ m and $\theta = 90^\circ$.

*4.23 对 $\mathbf{D} = \hat{x}xy^3z^3$ (C/m²) 重复问题 4.22。

4.24 电荷 Q_1 均匀分布在半径为 a 的薄球壳上，电荷 Q_2 均匀分布在半径为 b 且 $b > a$ 的第二个球壳上。应用高斯定律求出区域 $R < a$ 、 $a < R < b$ 和 $R > b$ 中的 $\mathbf{E}$ 。

*4.25 以原点为中心、半径为 a 的电介质球内的电通量密度由下式给出

$$\mathbf{D} = \hat{\mathbf{r}}\rho_0R \quad (\text{C/m}^2)$$

其中 ρ_0 是常数。求球体内部的总电荷。

4.26 在空间的某个区域，电荷密度在柱坐标系中由函数给出

$$\rho_v = 5re^{-r} \quad (\text{C/m}^3).$$

应用高斯定律求 $\mathbf{D}$ 。

*4.27 在 $r = 1$ m 和 $r = 3$ m 之间延伸的无限长圆柱壳包含均匀电荷密度 ρ_{v0} 。应用高斯定律求出所有区域中的 $\mathbf{D}$ 。

4.28 如果电荷密度随着距原点的距离线性增加，使得原点处的 $\rho_v = 0$ 和 $R = 2$ m 处的 $\rho_v = 4$ C/m³，求 $\mathbf{D}$ 的相应变化。

4.29 外半径为 b 的球壳包围着半径为 $a < b$ 的无电荷空腔 (图 P4.29)。如果外壳包含

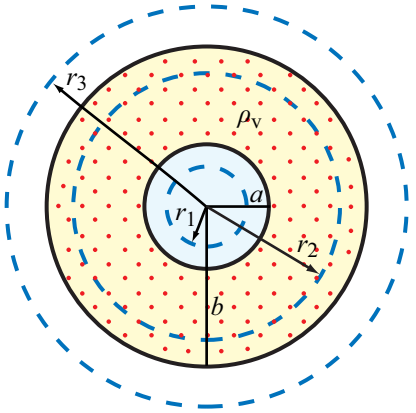


图 P4.29 问题 4.29。

电荷密度由下式给出

$$\rho_v = -\frac{\rho_{v0}}{R^2}, \quad a \leq R \leq b,$$

其中 ρ_{v0} 是正常数，确定所有区域中的 $\mathbf{D}$ 。

第 4-5 节：电势

*4.30 自由空间中 x - y 平面上的正方形在角点 $(a/2, a/2)$ 处具有 $+Q$ 的点电荷，在角点 $(a/2, -a/2)$ 处具有相同的点电荷，在其他两个角处各有一个点电荷 $-Q$ 。

- (a) 求沿 x 轴任意点 P 的电势。(b) 评估 $x = a/2$ 处的 V 。

4.31 图 4-7 所示的半径为 a 的圆盘在其表面上具有均匀的电荷密度 ρ_s 。

- (a) 求出 z 轴上点 $P(0, 0, z)$ 处的电势 V 的表达式。(b) 使用结果求出 $\mathbf{E}$ ，然后计算它的 $z = h$ 。将你的最终表达式与根据库仑定律获得的 (4.24) 进行比较。

*4.32 半径为 a 的电荷圆环位于 x - y 平面内，且中心位于原点。还假设环在空气中并且具有均匀的密度 ρ_ℓ 。

- (a) 证明 $(0, 0, z)$ 处的电势由 $V = \rho_\ell a / [2\epsilon_0(a^2 + z^2)^{1/2}]$ 给出。
- (b) 求相应的电场 $\mathbf{E}$ 。

4.33 证明沿 z 轴，距离密度为 ρ_ℓ 的无限电荷线，空气中距离 r_1 和 r_2 的两点之间的电势差 V_{12} 为 $V_{12} = (\rho_\ell / 2\pi\epsilon_0) \ln(r_2/r_1)$ 。

*4.34 求 x - y 平面中距原点距离 b 的位置处由于电荷密度为 ρ_ℓ 且长度为 l 的线电荷而产生的电势 V 。线电荷与 z 轴重合，并从 $z = -l/2$ 延伸到 $z = l/2$ 。

4.35 对于图 4-15 所示的电偶极子，在 $R = 1$ m 和 $\theta = 0^\circ$ 处， $d = 1$ cm 和 $|\mathbf{E}| = 4$ (mV/m)。在 $R = 2$ m 和 $\theta = 90^\circ$ 处找到 $\mathbf{E}$ 。

4.36 For each of the distributions of the electric potential V shown in **Fig. P4.36**, sketch the corresponding distribution of $\mathbf{E}$ (in all cases, the vertical axis is in volts and the horizontal axis is in meters).

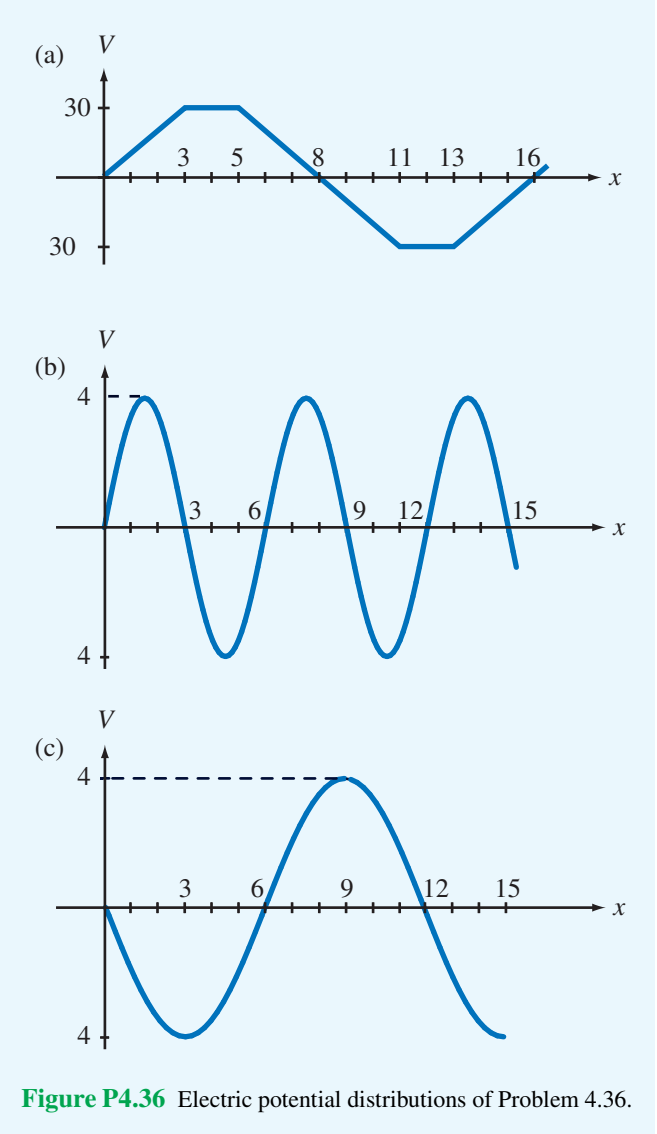


Figure P4.36 Electric potential distributions of Problem 4.36.

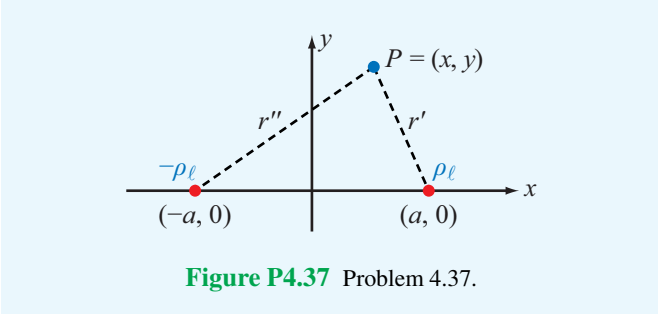


Figure P4.37 Problem 4.37.

4.38 Given the electric field

$$\mathbf{E} = \hat{\mathbf{R}} \frac{18}{R^2} \quad (\text{V/m})$$

find the electric potential of point A with respect to point B where A is at $+2$ m and B at -4 m (both on the z axis).

***4.39** An infinitely long line of charge with uniform density $\rho_l = 9$ (nC/m) lies in the x - y plane parallel to the y axis at $x = 2$ m. Find the potential V_{AB} at point $A(3 \text{ m}, 0, 4 \text{ m})$ in Cartesian coordinates with respect to point $B(0, 0, 0)$ by applying the result of Problem 4.33.

4.40 The x - y plane contains a uniform sheet of charge with $\rho_{s1} = 0.2$ (nC/m²). A second sheet with $\rho_{s2} = -0.2$ (nC/m²) occupies the plane $z = 6$ m. Find V_{AB} , V_{BC} , and V_{AC} for $A(0, 0, 6 \text{ m})$, $B(0, 0, 0)$, and $C(0, -2 \text{ m}, 2 \text{ m})$.

Section 4-6: Conductors

4.41 A cylindrical bar of silicon has a radius of 4 mm and a length of 8 cm. If a voltage of 5 V is applied between the ends of the bar and $\mu_e = 0.13$ (m²/V·s), $\mu_h = 0.05$ (m²/V·s), $N_e = 1.5 \times 10^{16}$ electrons/m³, and $N_h = N_e$, find the following:

- (a) the conductivity of silicon,
- (b) the current I flowing in the bar,
- * (c)** the drift velocities $\mathbf{u}_e$ and $\mathbf{u}_h$,
- (d) the resistance of the bar, and
- (e) the power dissipated in the bar.

4.42 Repeat Problem 4.41 for a bar of germanium with $\mu_e = 0.4$ (m²/V·s), $\mu_h = 0.2$ (m²/V·s), and $N_e = N_h = 2.4 \times 10^{19}$ electrons or holes/m³.

4.43 A 100-m long conductor of uniform cross-section has a voltage drop of 4 V between its ends. If the density of the current flowing through it is 1.4×10^6 (A/m²), identify the material of the conductor.

***4.37** As shown in **Fig. P4.37**, two infinite lines of charge, both parallel to the z axis, lie in the x - z plane: one with density ρ_ℓ and located at $x = a$ and the other with density $-\rho_\ell$ and located at $x = -a$. Obtain an expression for the electric potential $V(x, y)$ at a point $P = (x, y)$ relative to the potential at the origin.

4.36 对于图 P4.36 所示的每种电势 V 的分布，画出相应的 $\mathbf{E}$ 分布（在所有情况下，纵轴的单位是伏特，横轴的单位是米）。

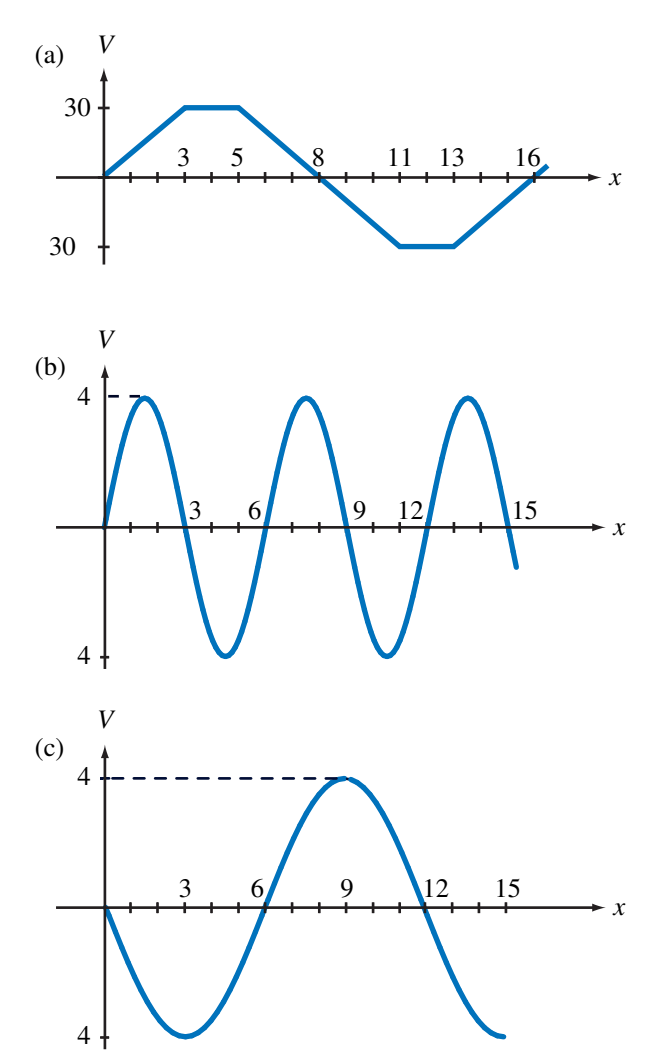


图 P4.36 问题 4.36 的电势分布。

***4.37** 如图 P4.37 所示，两条无限电荷线均平行于 z 轴，位于 x - z 平面内：一条密度为 ρ_ℓ ，位于 $x = a$ ，另一条密度为 $-\rho_\ell$ ，位于 $x = -a$ 。求出 $P = (x, y)$ 点处的电势 $V(x, y)$ 相对于原点电势的表达式。

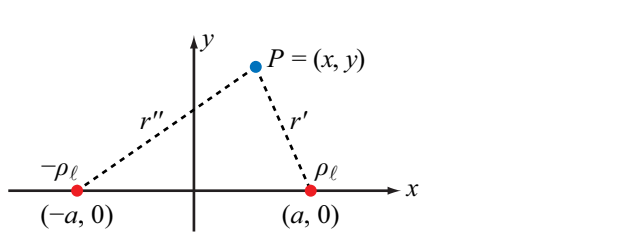


图 P4.37 问题 4.37。

4.38 给定电场

$$\mathbf{E} = \hat{\mathbf{R}} \frac{18}{R^2} \quad (\text{V/m})$$

求 A 点相对于 B 点的电势，其中 A 位于 $+2$ m， B 位于 -4 m（均在 z 轴上）。

***4.39** 一条具有均匀密度 $\rho_l = 9$ (nC/m) 的无限长电荷线位于 x - y 平面内，平行于 y 轴，位于 $x = 2$ m 处。应用问题 4.33 的结果，求直角坐标系中 $A(3 \text{ m}, 0, 4 \text{ m})$ 点相对于 $B(0, 0, 0)$ 点的势 V_{AB} 。

4.40 x - y 平面包含 $\rho_{s1} = 0.2$ (nC/m²) 的均匀电荷层。 $\rho_{s2} = -0.2$ (nC/m²) 的第二片占据平面 $z = 6$ m。求 $A(0, 0, 6 \text{ m})$ 、 $B(0, 0, 0)$ 和 $C(0, -2 \text{ m}, 2 \text{ m})$ 的 V_{AB} 、 V_{BC} 和 V_{AC} 。

第 4-6 节：导体

4.41 圆柱形硅棒的半径为 4 毫米，长度为 8 厘米。如果在棒两端之间施加 5 V 电压， $\mu_e = 0.13$ (m²/V·s)、 $\mu_h = 0.05$ (m²/V·s)、 $N_e = 1.5 \times 10^{16}$ 电子/m³ 和 $N_h = N_e$ ，找到以下内容：

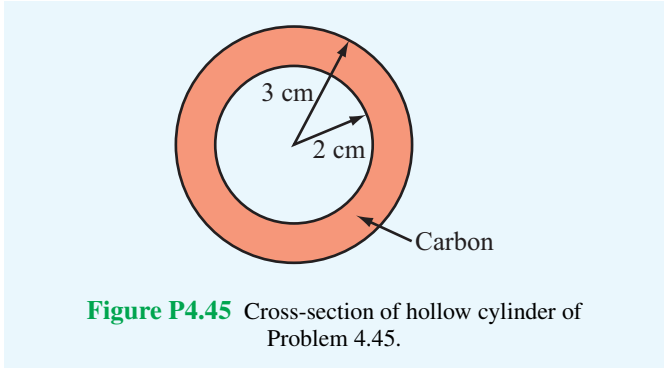
- (a) 硅的电导率，(b) 流过硅棒的电流 I ，
- * (c)** 漂移速度 u_e 和 u_h ，(d) 棒的电阻，以及 (e) 棒中消耗的功率。

4.42 对具有 $\mu_e = 0.4$ (m²/V·s)、 $\mu_h = 0.2$ (m²/V·s) 和 $N_e = N_h = 2.4 \times 10^{19}$ 个电子或空穴/m³ 的锗棒重复问题 4.41。

4.43 一根 100 米长的均匀横截面导体，其两端之间的电压降为 4 V。如果流过它的电流密度为 1.4×10^6 (A/m²)，则确定导体的材料。

4.44 A coaxial resistor of length l consists of two concentric cylinders. The inner cylinder has radius a and is made of a material with conductivity σ_1 , and the outer cylinder, extending between $r = a$ and $r = b$, is made of a material with conductivity σ_2 . If the two ends of the resistor are capped with conducting plates, show that the resistance between the two ends is $R = l / [\pi(\sigma_1 a^2 + \sigma_2(b^2 - a^2))]$.

***4.45** Apply the result of Problem 4.44 to find the resistance of a 20 cm long hollow cylinder (**Fig. P4.45**) made of carbon with $\sigma = 3 \times 10^4$ (S/m).



4.46 A 2×10^{-3} mm thick square sheet of aluminum has $5 \text{ cm} \times 5 \text{ cm}$ faces. Find the following:

- (a) The resistance between opposite edges on a square face.
- (b) The resistance between the two square faces. (See Appendix B for the electrical constants of materials.)

4.47 A cylinder-shaped carbon resistor is 8 cm in length, and its circular cross section has a diameter $d = 1 \text{ mm}$.

- (a) Determine the resistance R .
- (b) To reduce its resistance by 40%, the carbon resistor is coated with a layer of copper of thickness t . Use the result of Problem 4.44 to determine t .

Section 4-8: Boundary Conditions

***4.48** With reference to **Fig. 4-21**, find $\mathbf{E}_1$ if

$$\mathbf{E}_2 = \hat{x}3 - \hat{y}2 + \hat{z}2 \quad (\text{V/m}),$$

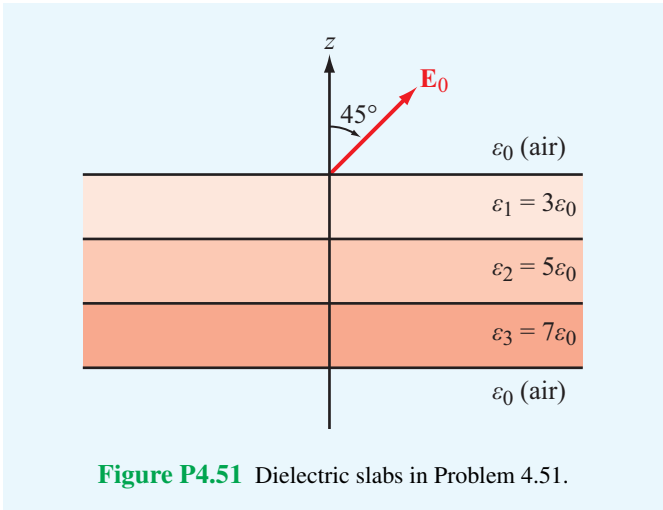
$\epsilon_1 = 2\epsilon_0$, $\epsilon_2 = 18\epsilon_0$, and the boundary has a surface charge density $\rho_s = 3.54 \times 10^{-11}$ (C/m²). What angle does $\mathbf{E}_2$ make with the z axis?

4.49 An infinitely long cylinder of radius a is surrounded by a dielectric medium that contains no free charges. If the

tangential component of the electric field in the region $r \geq a$ is given by $\mathbf{E}_t = -\hat{\phi} \cos \phi / r^2$, find $\mathbf{E}$ in that region.

***4.50** If $\mathbf{E} = \hat{\mathbf{R}}150$ (V/m) at the surface of a 5 cm conducting sphere centered at the origin, what is the total charge Q on the sphere's surface?

4.51 **Figure P4.51** shows three planar dielectric slabs of equal thickness but with different dielectric constants. If $\mathbf{E}_0$ in air makes an angle of 45° with respect to the z axis, find the angle of $\mathbf{E}$ in each of the other layers.



Sections 4-9 and 4-10: Capacitance and Electrical Energy

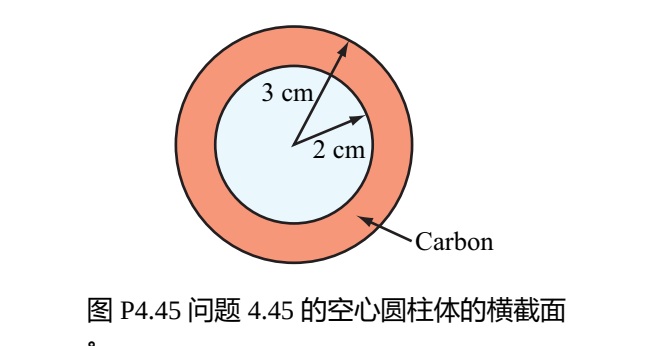
4.52 Determine the force of attraction in a parallel-plate capacitor with $A = 5 \text{ cm}^2$, $d = 2 \text{ cm}$, and $\epsilon_r = 4$ if the voltage across it is 50 V.

4.53 Dielectric breakdown occurs in a material whenever the magnitude of the field $\mathbf{E}$ exceeds the dielectric strength anywhere in that material. In the coaxial capacitor of Example 4-15,

- (a)** At what value of r is $|\mathbf{E}|$ maximum?
- (b)** What is the breakdown voltage if $a = 1 \text{ cm}$, $b = 2 \text{ cm}$, and the dielectric material is mica with $\epsilon_r = 6$?

4.44 长度为 l 的同轴电阻器由两个同心圆柱体组成。内圆柱体具有半径 a 并且由具有电导率 σ_1 的材料制成，并且在 $r = a$ 和 $r = b$ 之间延伸的外圆柱体由具有电导率 σ_2 的材料制成。如果电阻器的两端都盖有导电片，则表明两端之间的电阻为 $R = l / [\pi(\sigma_1 a^2 + \sigma_2(b^2 - a^2))]$ 。

***4.45** 应用问题 4.44 的结果求出 20 厘米长的碳制空心圆柱体 (**图 P4.45**) 的电阻，其 $\sigma = 3 \times 10^4$ (S/m)。



4.46 一块 2×10^{-3} 毫米厚的方形铝板具有 $5 \text{ cm} \times 5 \text{ cm}$ 的面。找到以下内容：

- (a) 方形面上相对边缘之间的阻力。
- (b) 两个正方形面之间的电阻。（有关材料的电常数，请参阅附录 B。）

4.47 圆柱形碳电阻器长 8 厘米，其圆形横截面直径 $d = 1 \text{ mm}$ 。

- (a) 确定电阻 R 。
- (b) 为了降低 40% 的电阻，碳电阻器上涂有一层厚度为 t 的铜。使用问题 4.44 的结果来确定 t 。

第 4-8 节：边界条件

***4.48** 参考图 4-21，求 $\mathbf{E}_1$ 如果

$$\mathbf{E}_2 = \hat{x}3 - \hat{y}2 + \hat{z}2 \quad (\text{V/m}),$$

$\epsilon_1 = 2\epsilon_0$ 、 $\epsilon_2 = 18\epsilon_0$ ，边界表面电荷密度为 $\rho_s = 3.54 \times 10^{-11}$ (C/m²)。 $\mathbf{E}_2$ 与 z 轴成什么角度？

4.49 半径为 a 的无限长圆柱体被不含自由电荷的介电介质包围。如果

$r \geq a$ 区域中电场的切向分量由 $E_t = -\hat{\phi} \cos \phi / r^2$ 给出，求该区域中的 $\mathbf{E}$ 。

***4.50** 如果 $\mathbf{E} = \hat{\mathbf{R}}150$ (V/m) 位于以原点为中心的 5 cm 导电球体表面，则该球体表面的总电荷 Q 是多少？

4.51 **图 P4.51** 显示了三个厚度相等但介电常数不同的平面电介质板。如果空气中的 $\mathbf{E}_0$ 与 z 轴成 45° 角，则求其他各层中 $\mathbf{E}$ 的角度。

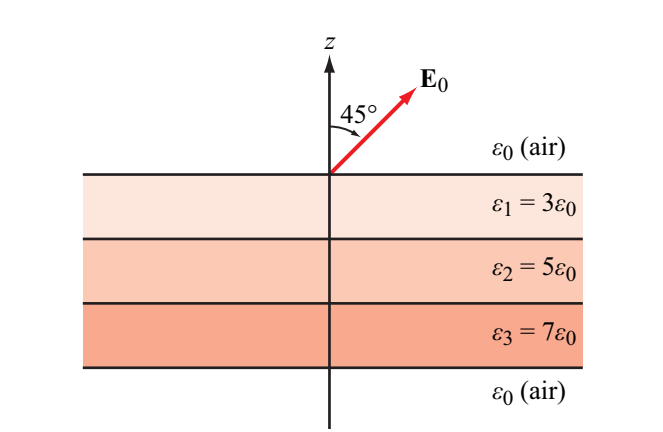


图 P4.51 问题 4.51 中的电介质板。

Sections 4-9 and 4-10: Capacitance and Electrical Energy

4.52 确定 $A = 5 \text{ cm}^2$ 、 $d = 2 \text{ cm}$ 和 $\epsilon_r = 4$ 的平行板电容器（如果其两端电压为 50 V）的吸引力。

4.53 当电场 $\mathbf{E}$ 的大小超过材料中任何地方的介电强度时，材料中就会发生介电击穿。在实施例 4-15 的同轴电容器中，

- (a)** r 的值是多少时 $|\mathbf{E}|$ 最大？
- (b)** 如果 $a = 1 \text{ cm}$ ， $b = 2 \text{ cm}$ ，介电材料为 $\epsilon_r = 6$ 云母，击穿电压是多少？

4.54 An electron with charge $Q_e = -1.6 \times 10^{-19}$ C and mass $m_e = 9.1 \times 10^{-31}$ kg is injected at a point adjacent to the negatively charged plate in the region between the plates of an air-filled, parallel-plate capacitor with separation of 1 cm and rectangular plates each 10 cm^2 in area (**Fig. P4.54**). If the voltage across the capacitor is 10 V, find the following:

(a) the force acting on the electron,

(b) the acceleration of the electron, and

(c) the time it takes the electron to reach the positively charged plate, assuming that it starts from rest.

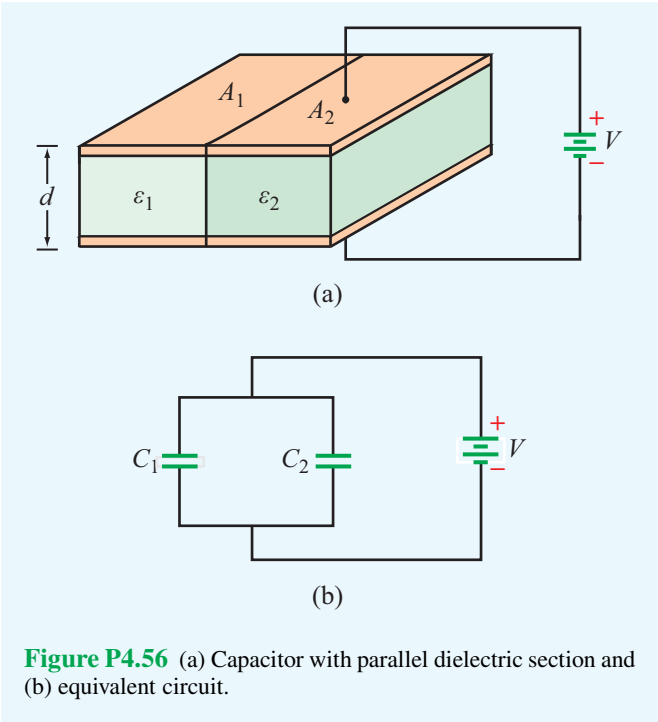
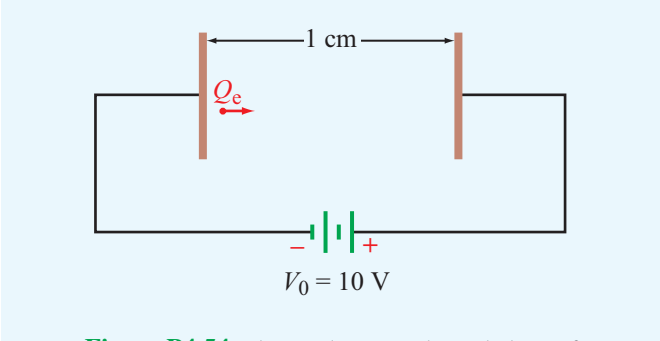


Figure P4.56 (a) Capacitor with parallel dielectric section and (b) equivalent circuit.

***4.55** In a dielectric medium with $\epsilon_r = 4$, the electric field is given by

$$\mathbf{E} = \hat{\mathbf{x}}(x^2 + 2z) + \hat{\mathbf{y}}x^2 - \hat{\mathbf{z}}(y + z) \quad (\text{V/m})$$

Calculate the electrostatic energy stored in the region $-1\text{ m} \leq x \leq 1\text{ m}$, $0 \leq y \leq 2\text{ m}$, and $0 \leq z \leq 3\text{ m}$.

4.56 **Figure P4.56(a)** depicts a capacitor consisting of two parallel, conducting plates separated by a distance d . The space between the plates contains two adjacent dielectrics: one with permittivity ϵ_1 and surface area A_1 and another with ϵ_2 and A_2 . The objective of this problem is to show that the capacitance C of the configuration shown in **Fig. P4.56(a)** is equivalent to two capacitances in parallel, as illustrated in **Fig. P4.56(b)**, with

$$C = C_1 + C_2, \quad (4.146)$$

where

$$C_1 = \frac{\epsilon_1 A_1}{d}, \quad (4.147)$$

$$C_2 = \frac{\epsilon_2 A_2}{d}. \quad (4.148)$$

To this end, proceed as follows:

- (a) Find the electric fields $\mathbf{E}_1$ and $\mathbf{E}_2$ in the two dielectric layers.
- (b) Calculate the energy stored in each section and use the result to calculate C_1 and C_2 .
- (c) Use the total energy stored in the capacitor to obtain an expression for C . Show that Eq. (4.146) is indeed a valid result.

4.57 Use the result of Problem 4.56 to determine the capacitance for each of the following configurations.

- * (a)** Conducting plates are on top and bottom faces of the rectangular structure in **Fig. P4.57(a)**.
- (b) Conducting plates are on front and back faces of the structure in **Fig. P4.57(a)**.
- (c) Conducting plates are on top and bottom faces of the cylindrical structure in **Fig. P4.57(b)**.

4.54 电荷为 $Q_e = -1.6 \times 10^{-19}$ C 且质量为 $m_e = 9.1 \times 10^{-31}$ kg 的电子被注入到空气填充的平行板电容器的板之间区域中与带负电的板相邻的点，该电容器的板间距为 1 cm，矩形板的面积各为 10 cm^2 (图 P4.54)。如果电容器两端的电压为 10 V，则得出：

(a) 作用在电子上的力，(b) 电子的加速度，以及 (c) 电子到达带正电的板所需的时间（假设它从静止开始）。

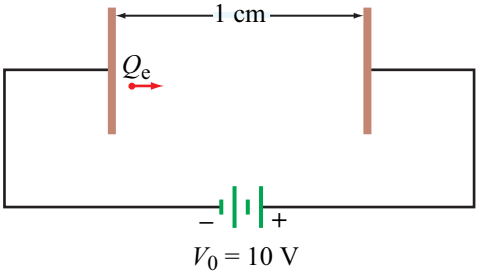


图 P4.54 问题 4.54 的带电板之间的电子。

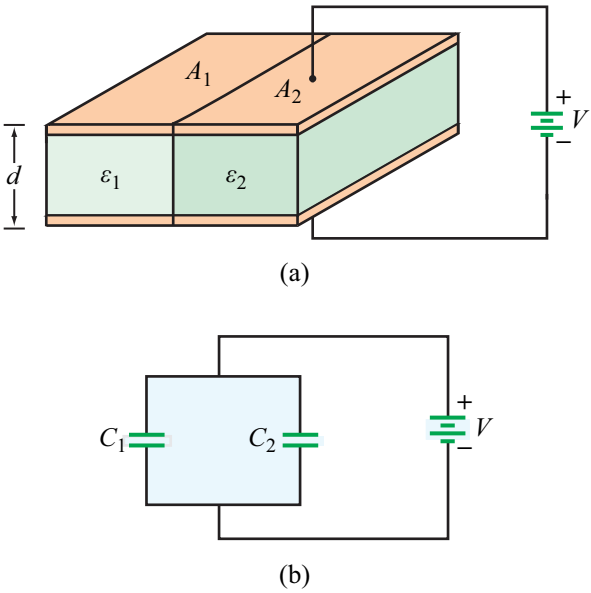


图 P4.56 (a) 具有平行介电部分的电容器和 (b) 等效电路。

***4.55** 在 $\epsilon_r = 4$ 的介电介质中，电场由下式给出

$$\mathbf{E} = \hat{\mathbf{x}}(x^2 + 2z) + \hat{\mathbf{y}}x^2 - \hat{\mathbf{z}}(y + z) \quad (\text{伏/米})$$

计算区域 $-1\text{ m} \leq x \leq 1\text{ m}$, $0 \leq y \leq 2\text{ m}$ 和 $0 \leq z \leq 3\text{ m}$ 中存储的静电能。

4.56 图 P4.56(a) 描绘了一个由两个平行的导电板组成的电容器，两个导电板之间的距离为 d 。板之间的空间包含两个相邻的电介质：一个具有介电常数 ϵ_1 和表面积 A_1 ，另一个具有 ϵ_2 和 A_2 。该问题的目的是证明图 P4.56(a) 所示配置的电容 C 相当于两个并联电容，如图 P4.56(b) 所示，其中

$$C = C_1 + C_2, \quad (4.146)$$

在哪里

$$C_1 = \frac{\epsilon_1 A_1}{d}, \quad (4.147)$$

$$C_2 = \frac{\epsilon_2 A_2}{d}. \quad (4.148)$$

为此，请执行以下操作：

(a) 求两个介电层中的电场 $\mathbf{E}_1$ 和 $\mathbf{E}_2$ 。(b) 计算每个部分存储的能量，并使用结果计算 C_1 和 C_2 。(c) 使用电容器中存储的总能量来获得 C 的表达式。(4.146) 确实是一个有效的结果。

4.57 使用问题 4.56 的结果来确定以下每个配置的电容。

*** (a)** 导电板位于图P4.57(a)中矩形结构的顶面和底面。

(b) 导电板位于图 P4.57(a) 结构的正面和背面。

(c) 导电板位于图 P4.57(b) 中圆柱形结构的顶面和底面。

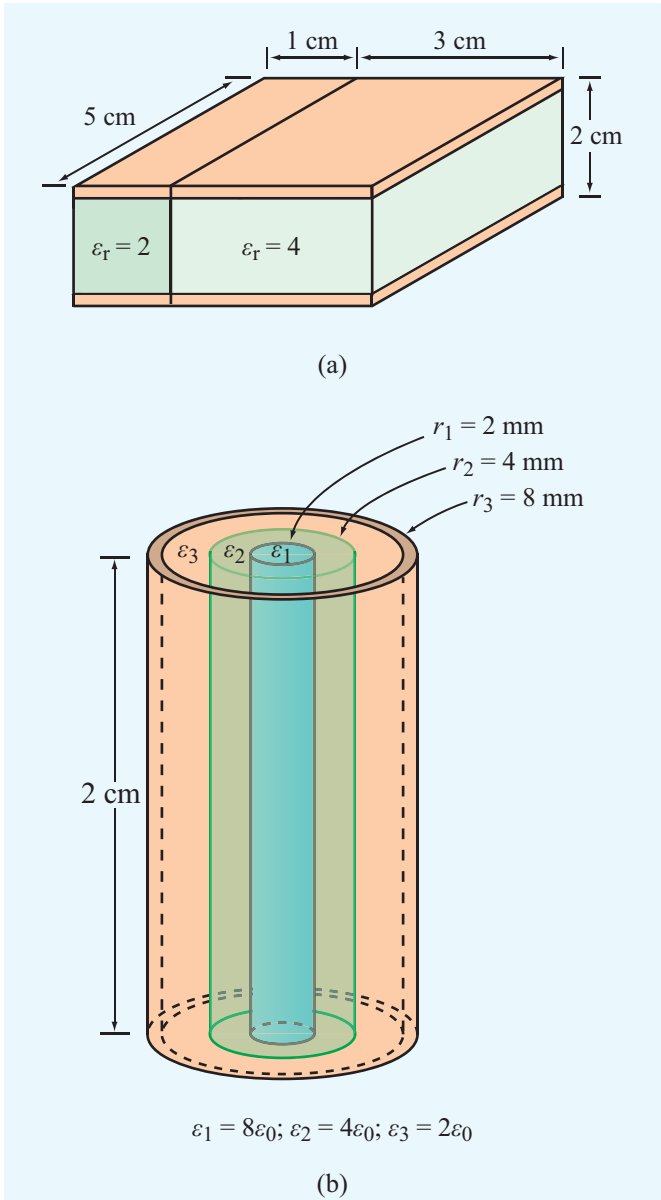


Figure P4.57 Dielectric sections for Problems 4.57 and 4.59.

4.58 The capacitor shown in Fig. P4.58 consists of two parallel dielectric layers. Use energy considerations to show that the equivalent capacitance of the overall capacitor, C , is equal to the series combination of the capacitances of the individual layers, C_1 and C_2 , namely

$$C = \frac{C_1 C_2}{C_1 + C_2}, \quad (4.149)$$

where

$$C_1 = \epsilon_1 \frac{A}{d_1}, \quad C_2 = \epsilon_2 \frac{A}{d_2}.$$

- (a) Let V_1 and V_2 be the electric potentials across the upper and lower dielectrics, respectively. What are the corresponding electric fields E_1 and E_2 ? By applying the appropriate boundary condition at the interface between the two dielectrics, obtain explicit expressions for E_1 and E_2 in terms of ϵ_1 , ϵ_2 , and V and the indicated dimensions of the capacitor.
- (b) Calculate the energy stored in each of the dielectric layers and then use the sum to obtain an expression for C .
- (c) Show that C is given by Eq. (4.149).

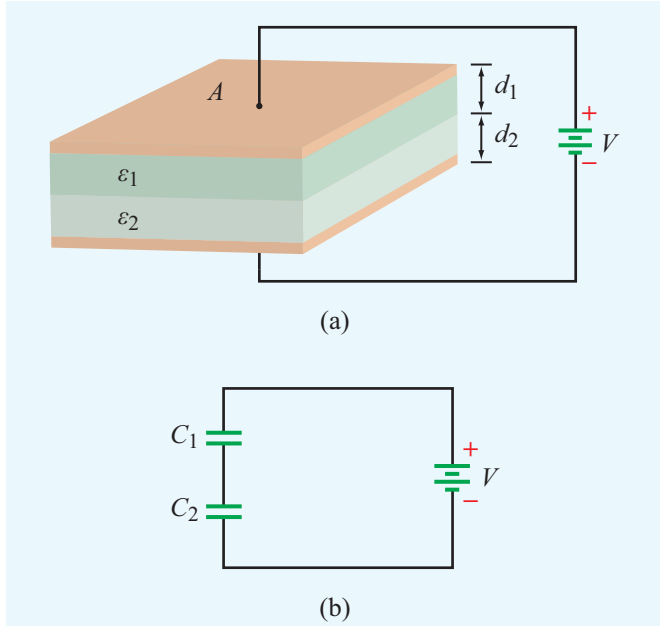


Figure P4.58 (a) Capacitor with parallel dielectric layers and (b) equivalent circuit (Problem 4.58).

4.59 Use the expressions given in Problem 4.58 to determine the capacitance for the configurations in Fig. P4.57(a) when the conducting plates are placed on the right and left faces of the structure.

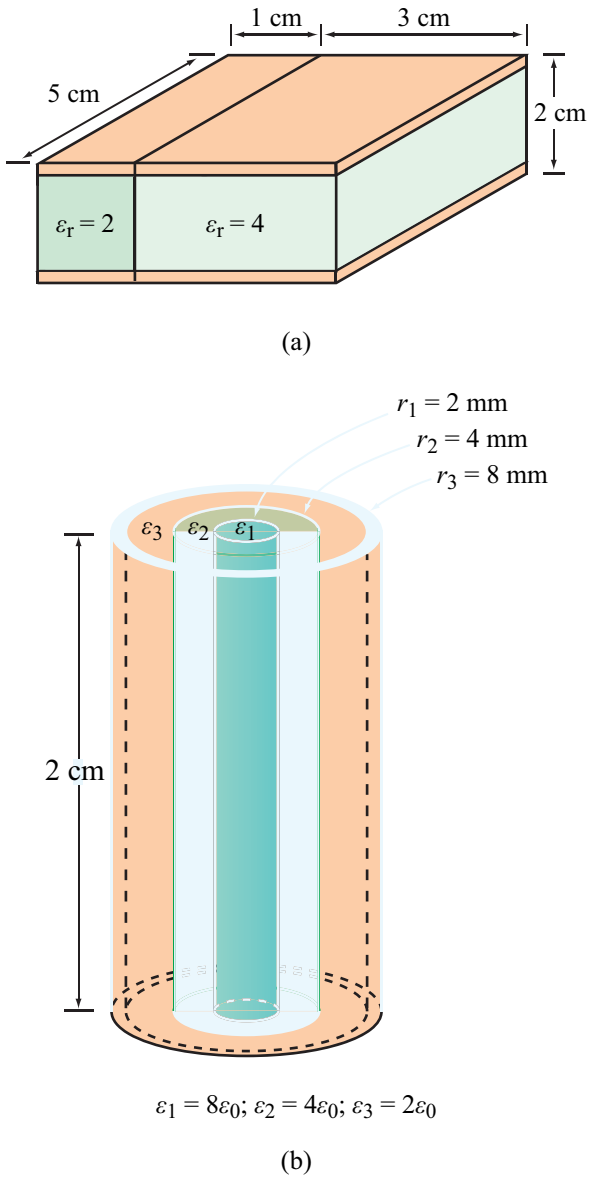


图 P4.57 问题 4.57 和 4.59 的电介质部分。

4.58 图 P4.58 所示的电容器由两个平行的介电层组成。使用能量考虑表明整个电容器的等效电容 C 等于各层电容 C_1 和 C_2 的串联组合，即

$$C = \frac{C_1 C_2}{C_1 + C_2}, \quad (4.149)$$

在哪里

$$C_1 = \epsilon_1 \frac{A}{d_1}, \quad C_2 = \epsilon_2 \frac{A}{d_2}.$$

- (a) 令 V_1 和 V_2 分别为上电介质和下电介质两端的电势。相应的电场 E_1 和 E_2 是什么？通过在两个电介质之间的界面处应用适当的边界条件，根据 ϵ_1 、 ϵ_2 和 V 以及电容器的指示尺寸获得 E_1 和 E_2 的显式表达式。
- (b) 计算每个介电层中存储的能量，然后使用总和获得 C 的表达式。
- (c) 证明 C 由方程式给出。 (4.149)。

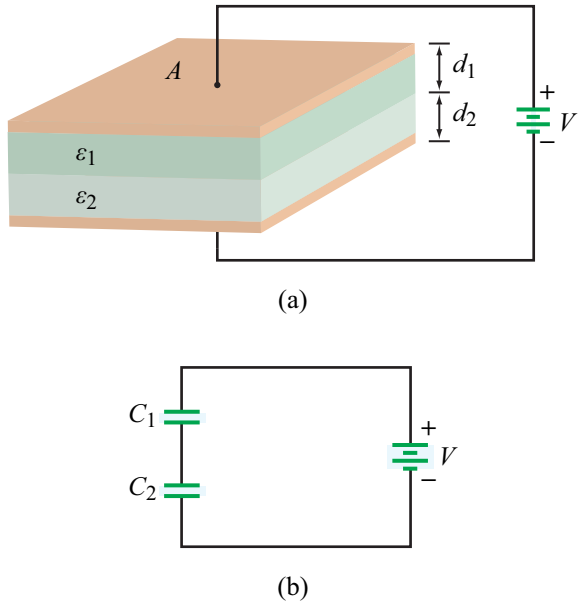


图 P4.58 (a) 具有平行介电层的电容器和 (b) 等效电路 (问题 4.58)。

4.59 当导电板放置在结构的左右面上时，使用问题 4.58 中给出的表达式来确定图 P4.57(a) 中的配置的电容。